

Modern Geometries

Andrew Misseldine

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Contents

1	Finite Geometry	1
1.1	Axioms	1
1.2	Models	4
1.3	Finite Geometries	7
1.4	Fano Geometry	11
1.5	Young Geometry	14
2	Incidence Geometry	17
2.1	The Incidence Axioms	17
2.2	The Parallel Axioms	21
2.3	Affine Geometry	25
2.4	Non-Euclidean Geometry	29
3	Ordered Geometry	33
3.1	The Betweenness Axioms	33
3.2	Linear Ordering	37
3.3	Angles	40
3.4	The Crossbar Theorem	43
4	Congruence Geometry	47
4.1	The Congruence Axioms of Segments	47
4.2	The Congruence Axioms of Angles	50
4.3	Euclid's Fourth Postulate	53
4.4	Interior and Exterior Angles	56
4.5	Triangle Congruence	59
4.6	Midpoints and Bisectors	62
4.7	The Triangle Inequality	65
4.8	Circles	68
5	Neutral Geometry	73
5.1	The Continuity Axioms	73
5.2	The Saccheri-Legendre Theorem	77
5.3	Quadrilaterals	80
5.4	Angle of Parallelism	83
6	Euclidean Geometry	87
6.1	Rectangles	87
6.2	Euclidean Geometry	92
6.3	Equivalents to the Euclidean Parallel Postulate	96
7	Hyperbolic Geometry	99
7.1	The Hyperbolic Parallel Postulate	99
7.2	Defect of Triangles	103
7.3	Axioms of Area	106
7.4	Hyperbolic Area and Distance	109

8	Elliptic Geometry	113
8.1	Projective Geometry	113
8.2	The Elliptic Parallel Postulate	117
8.3	Poles and Duality	120
8.4	Elliptic Polygons	123

Chapter 1

Finite Geometry

“Finally there are simple ideas of which no definition can be given; there are also axioms or postulates, or in a word primary principles, which cannot be proved and have no need of proof.” – Gottfried Leibniz

1.1 Axioms

Geometry is arguably the oldest branch of mathematics, whose applications were studied thousands of years ago by the ancient Egyptians, Babylonians, and Chinese and whose theory was studied by the ancient Greeks. The study of modern geometry often begins with the axiomatic method as this is the logical perspective that separates the modern study of geometry from the ancient. In fact, the axiomatic method is the bed-stone for which ALL of pure mathematics rests. The **axiomatic method** can be summarized in the following way:

Definition 1.1.1. The Axiomatic Method

1. Any axiomatic system must contain a set of technical terms that are deliberately chosen as **undefined terms** and are subject to the interpretation of the reader.
2. All other technical terms of the system are ultimately defined by means of the undefined terms. These terms are the **definitions** of the system.
3. The axiomatic system contains a set of statements, dealing with undefined terms and definitions, that are chosen to remain unproved. These are the **axioms** of the system.
4. All other statements of the system must be logical consequences of the axioms. These derived statements are called the **theorems** of the axiomatic system.

Axioms are considered to be the fundamentally true statements of any theory. Axioms are true because they are true, at least from a philosophical point of view. Essentially, “something” cannot be proven from “nothing,” that is, in order to prove a statement true one must already have some known true-statement to derive the new statement from. If Theorem 1 was proven from Theorem 2, then how was Theorem 2 proven? There must have existed some other theorem, say Theorem 3, which proved Theorem 2 true. What about Theorem 3 then? Continuing in this process, we see there must be an infinite line of logical dominoes that must already have be set up just to be able to prove Theorem 1. The only way to avoid this logical paradox is to accept some statements “fundamental” true, that is, a statement known to be true without proof. These are what axioms are in logic. Thus, the truth of these axioms is not at issue, just the reader’s willingness to accept them as true.¹

Axiomatic systems are ubiquitous in pure mathematics. In algebra, the axioms of a group determine group theory. An additional axiom is added to determine abelian group theory. Several more axioms are added to produce ring theory, and a few more are added to produce field theory. Returning to abelian groups, axioms different to those of rings, can be added to an abelian group to form vector spaces, normed vector spaces, inner product spaces, or Lie algebras, to name only a few. Axioms can be removed from group theory in order to study its generalization, such as semigroups or quasigroups. In topology, axioms

are introduced to determine what an open set is, how can points be separated, how far apart two points are, how connected is a set. In analysis, axioms are used to determine what is a natural number, what is a real number, and what is a set.

Example 1.1.2. We will next explore a specific, simple axiomatic system.

Undefined terms: Fe , Fo , and Fum , the last of which is a relation from Fe 's to Fo 's.

Definition: If a Fe Fum 's a Fo , we say that the Fo Fum 's the Fe .

Axiom 1: There exist exactly three distinct Fe 's in this system.

Axiom 2: Any two distinct Fe 's Fum exactly one Fo .

Axiom 3: Not all Fe 's Fum the same Fo .

Axiom 4: Any two distinct Fo 's Fum at least one Fe .

Using these four axioms, we prove Theorems 1.1.3, 1.1.4, 1.1.5, and 1.1.6.

Theorem 1.1.3. *Each Fo Fum 's exactly two Fe 's.*

Proof. Let F be a Fo . By Axiom 3, not all Fe 's Fum F , and by Axiom 1 there are exactly three Fe 's. We conclude that F does not Fum more than two Fe 's. Suppose that fewer than two Fe 's Fum F . Since there are three Fe 's, at least two Fe 's do not Fum F . Let G be the unique Fo for which these two Fe 's both Fum , guaranteed by Axiom 2. Then F and G are distinct (since G Fum 's Fe 's not Fum 'ing F). By Axiom 4, F and G Fum a Fe in common, which must be the third Fe not used to construct G . Then G Fum 's all three Fe 's, a contradiction to Axiom 3. Therefore, F Fum 's at least two Fe 's, which proves the theorem. $\square$

Theorem 1.1.4. *A Fo is completely characterized by the Fe 's it Fum 's, that is, if two Fo 's are distinct then they do not Fum the same Fe 's.*

Proof. Let F and G be distinct Fo 's. Both Fo 's Fum exactly two Fe 's by Theorem 1.1.3. If F and G Fum the same two Fe 's, then this would contradict Axiom 2 since F and G would be two distinct Fo 's Fum 'ing the same pair of Fe 's. Therefore, F and G Fum different Fe 's. $\square$

Theorem 1.1.5. *Two distinct Fo 's Fum exactly one Fe .*

Proof. Let F and G be two distinct Fo 's. By Axiom 4, we know that there exists a Fe , say x , that Fum 's F and G . By Theorem 1.1.3, if F and G Fum more than one Fe , then they Fum the same two Fe 's. By Theorem 1.1.4, they would be the same Fo . $\square$

Theorem 1.1.6. *There are exactly three distinct Fo 's.*

Proof. By Theorem 1.1.3, each Fo Fum 's exactly two Fe 's. Axiom 1 says that there are exactly three Fe 's. Therefore, there are $\binom{3}{2} = 3$ Fo 's. $\square$

Exercises

1. Consider the following axiomatic system with undefined terms: pi , ka , and chu , the last being a relation on pi and ka :

Definition. pi - ka - chu Axioms

- I. There exist exactly five pi 's.
- II. Any two distinct pi 's chu exactly one ka .
- III. Each ka chu 's exactly two pi 's.

- (i) How many ka 's are there in the system?
- (ii) Prove that any two ka 's chu at most one pi together.
- (iii) Prove that not all pi 's chu the same ka .

ⁱInterestingly enough, theists and atheists have used the above logical concern, and other philosophical technicalities, to both prove and disprove the existence of a deity. For example, theists argue that there must be a first in the long line of cause-and-effect going backward in time and that this first must be God. Atheists have argued similarly with the Big Bang. Why cannot they agree? Even when the argument is valid (devoid of logical error), which is nearly never the case in these kind of arguments, the two debaters are certainly arguing from two different sets of axioms, the union of which will certainly be inconsistent. These axiomatic differences lead many to believe differently in religion and politics. In the end, there are some truths that we must take on faith, which is a mathematically provable statement.

“It has long been an axiom of mine that the little things are infinitely the most important.”
 – Arthur Conan Doyle

1.2 Models

In addition to the previous logical concern of infinite descent, the purpose of the axioms is to provide a definition to the undefined terms, in a way of speaking. The axioms provide a level of rigidity to the undefined terms, that is, a reader can apply any interpretation to the undefined terms that they wish so long as all the axioms associated with these undefined terms are satisfied. By giving each undefined term in a system a particular meaning, we create an **interpretation** of that system. If, for a given interpretation of a system, all the axioms are true statements, we call the interpretation a **model**.

Example 1.2.1. Consider the axiomatic system established in Example 1.1.2. We will interpret the *Fe*’s as people, *Fo*’s as committees, and *Fum* to mean that the person is a member of the committee. Then the axioms would be interpreted as:

Axiom 1: There are exactly three people.

Axiom 2: Every pair of people belongs to exactly one committee.

Axiom 3: Not everyone belongs to the same committee.

Axiom 4: Every pair of committees has at least one person on both committees.

Let the people be Bob, Ted, and Carol, and the committees be Entertainment (consisting of Bob and Ted), Finance (Ted and Carol), and Refreshments (Bob and Carol). This is then a model of the axiomatic system from Example 1.1.2.

Example 1.2.2. Consider the axiomatic system established in Example 1.1.2. This time, we will interpret the *Fe*’s as books, *Fo*’s as shelves, and *Fum* to mean that the book is on the shelf. Then the axioms would be interpreted as:

Axiom 1: There are exactly three books.

Axiom 2: Every pair of books belongs to exactly one shelf.

Axiom 3: Not every book is on the same shelf.

Axiom 4: Every pair of shelves has at least one book on both shelves.

With this interpretation, it is not possible for a book to be contained on more than one shelf, but Axiom 4 would require that a book be on two shelves. Thus, this interpretation cannot lead to a model of the axiomatic system from Example 1.1.2.

Because theorems are logical consequences of the axioms and are independent of the interpretation, all theorems of an axiomatic system are true for every model. This is the real strength of the axiomatic method. For example, consider the model from Example 1.2.1. From the theorems proven previously, we know each committee has exactly two members, every pair of committees has exactly one person on both committees, and there are exactly three committees. Although these results seem trivial when seeing the model, for more complicated axiomatic systems such theorems will be valuable. Also, we know that any interpretation that violates any of these theorems cannot be a model. For example, any interpretation with people and committees with four committees cannot be a model since it violates a theorem derived from the axioms.

When working with models, it is possible to create two models which are essentially the same. We say that two models are **isomorphic** if there exists a one-to-one correspondence between all the undefined terms such that any relationship between the undefined terms is preserved. In terms of the axiomatic method, isomorphic models are the same.

Example 1.2.3. Consider again the axiomatic system from Example 1.1.2. We will interpret the *Fe*’s as elements, *Fo*’s as sets, and *Fum* to mean set containment. Let the elements be x , y , and z

and let the sets be $F = \{x, y\}$, $G = \{x, z\}$, and $H = \{y, z\}$. Then there are three elements, which satisfies Axiom 1. All possible pairs of elements are represented in the three sets, which satisfies Axiom 2. No set contains all the elements, which satisfies Axiom 3. Finally, all intersections of sets are nonempty, which satisfies Axiom 4. Therefore, this is a model of *Fe-Fo-Fum* axiomatic system.

An **isomorphism** of this axiomatic model means a one-to-one correspondence between the interpretation of the *Fe*'s and a one-to-one correspondence between the interpretation of the *Fo*'s such that all *Fum* relations are preserved by the correspondence. This set model is isomorphic to the model from Example 1.2.1, where the one-to-one correspondence is given as

$$\begin{aligned} \text{Bob} &\leftrightarrow x, & \text{Ted} &\leftrightarrow y, & \text{Carol} &\leftrightarrow z \\ \text{Entertainment} &\leftrightarrow F = \{x, y\}, & \text{Finance} &\leftrightarrow H = \{y, z\}, & \text{Refreshments} &\leftrightarrow G = \{x, z\} \end{aligned}$$

Definition 1.2.4. An individual axiom is said to be **independent** if it is not a theorem of the other axioms. We say a set of axioms is **independent** if each axiom is individually independent. An axiom which is not independent in a system is called **dependent**.

To show that an axiom is independent, one need only produce a model demonstrating the other axioms but not satisfying the axiom in question. To show dependence, a proof must be proved to show that the statement is a theorem of the other axioms.

Example 1.2.5. We will show that Axiom 1 is independent in the *Fe-Fo-Fum* axiomatic system. Consider the four elements a, b, c , and d (*Fe*'s) and the four sets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, and $\{b, c, d\}$ (*Fo*'s). Then each of the six pairs of elements are contained in only one set. Axiom 2 is satisfied. There is no set which contains all four elements. Axiom 3 is satisfied. Finally, the intersection of any two sets is a single element. Axiom 4 is satisfied. Thus, we have a model of *Fe-Fo-Fum* axioms 2-4. On the other hand, Axiom 1 fails since there are four *Fe*'s. Therefore, Axiom 1 is independent in the *Fe-Fo-Fum* system.

We present some other important logical definitions that are relevant to the study of axiomatics.

Definition 1.2.6. We say that an axiomatic system is **inconsistent** if a statement and its negation are both theorems of the axioms. We say that an axiomatic system is **consistent** if it is not inconsistent. We say that an axiomatic system is **complete** if every statement or its negation is a theorem of the axioms. We say that an axiomatic system is **incomplete** if it is not complete.

In an inconsistent system, all statements are both true and false, which makes the axiomatics completely useless. For this reason, it is important to determine whether a system is inconsistent or not. We, unfortunately, cannot expect complicated axiomatics to be complete. In a nutshell, the Gödel Incompleteness Theorem says that any axiomatic system that can perform basic arithmetic is incomplete. The axioms of geometry are included in this, but fortunately the only known holes in completeness are silly, self-referential statements that are only studied by logicians.

Exercises

1. Consider the following axiomatic system with undefined terms: pi , ka , and chu , the last being a relation on pi and ka :

Definition. pi - ka - chu Axioms

- I. There exist exactly five pi 's.
- II. Any two distinct pi 's chu exactly one ka .
- III. Each ka chu 's exactly two pi 's.

Devise a model for the pi - ka - chu axioms.

2. Consider the following axiomatic system with undefined terms: Fe , Fo , and Fum , the last being a relation on Fe and Fo :

Definition. Fe - Fo - Fum Axioms

- I. There exist exactly three distinct Fe 's in this system.
- II. Any two distinct Fe 's Fum exactly one Fo .
- III. Not all Fe 's Fum the same Fo .
- IV. Any two distinct Fo 's Fum at least one Fe .

- (i) Demonstrate the independenceⁱ of Axioms 2 in Fe - Fo - Fum Theory.
- (ii) Demonstrate the independence of Axioms 3 in Fe - Fo - Fum Theory.
- (iii) Demonstrate the independence of Axioms 4 in Fe - Fo - Fum Theory.

ⁱRecall that proving an axiom is independent is done by constructing a model for which all axioms of the theory are true except the one in question, in which case the model demonstrates the negation of said axiom.

“Our minds are finite, and yet even in these circumstances of finitude we are surrounded by possibilities that are infinite, and the purpose of life is to grasp as much as we can out of that infinitude.”

– Alfred North Whitehead

1.3 Finite Geometries

We now will begin our official study of geometry. For all geometric axiomatic systems we will use the following undefined terms.

Definition 1.3.1. A **geometry** is an axiomatic system which contains the following undefined terms: **points**, **lines**, and the relation **incidence** that is, “a point is incident to a line” or “a point is on a line.” Often times, a model of a geometry will also be referred to as a geometry.

The following are definition that apply to any geometry.

Definition 1.3.2. If a point is on a line, we say that the line **contains** the point. We say that two distinct lines **intersect** if there exists a point on both lines. We say that two distinct lines are **parallel** if they do not intersect. We say that a set of points is **collinear**, if there exists a single line incident to all points in this set. We say that a set of lines is **concurrent** if there exists a single point incident to all lines in the set.

We say that a geometry is **finite** if the number of points and lines is finite. A geometry is infinite otherwise.

For two lines ℓ and m , we will denote that ℓ is parallel to m as $\ell \parallel m$. Thus, $\ell \nparallel m$ implies that ℓ intersects m . Clearly, $\parallel$ is a symmetric relation, that is, $\ell \parallel m$ implies that $m \parallel \ell$. Although it seems somewhat awkward to do so now, we will also say that ℓ is *parallel* to ℓ , that is, $\ell \parallel \ell$. As parallel was defined originally for distinct lines, we may extend the definition of parallel in this fashion. This makes $\parallel$ into a reflexive relation. We will see later that under the right conditions, $\parallel$ can also be a transitive relation, hence an equivalence relation. But we must emphasize that this only happens in certain geometries, not in general.

Now we will recast our *Fe-Fo-Fum* axiomatic system in the language of geometry. In particular, we will exchange the term *Fe* with *point*, *Fo* with *line*, and *Fum* with *incident*. Although the theory does not change when we re-label the undefined terms, our intuition of the theory dramatically improves. Some of the below phrasing has been modified, but the axioms remain equivalent to their original forms.

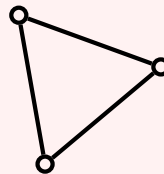
Definition 1.3.3. Three-Point Geometry

Axiom 1: There exists exactly three distinct points.

Axiom 2: For each two distinct points there exists
a unique line containing both of them.

Axiom 3: Not all points lie on the same line.

Axiom 4: All lines intersect.



Note that axiom 4 is mentioned above is much briefer than its original form. This is because the definition of “intersect” was used to abbreviate the statement. This is the reason that definitions are used, to simplify otherwise complicated statement. Notice that we could have also expressed it as “There exists no parallel lines,” which is equivalent to the axiom above given the definition of parallel.

A geometric model of the Three-Point Geometry is provided in the illustration above, where points are interpreted as dots on the page and lines as segments between dots. Based upon the theorems we have proven for *Fe-Fo-Fum* theory, up to isomorphism there is only one model for this axiomatic system. Thus,

the above diagram is essentially the only three-point geometry. The theorems of *Fe-Fo-Fum* theory translate immediately to the three-point geometry, as illustrated below, with the same proofs.

Theorem 1.3.4. *In three-point geometry, each line has exactly two points that belong to it.*

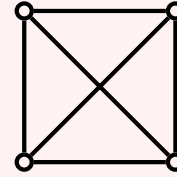
Theorem 1.3.5. *In three-point geometry, a line is completely characterized by the points on it, that is, if two lines are distinct then they do not contain the same points.*

Theorem 1.3.6. *In three-point geometry, two distinct lines intersect at exactly one point.*

Theorem 1.3.7. *In three-point geometry, there are exactly three distinct lines.*

Definition 1.3.8. Four-Point Geometry

- Axiom 1:* There exists exactly four distinct points.
Axiom 2: For each two distinct points there exists
 a unique line containing both of them.
Axiom 3: Each line is on exactly two points.



A model of the four-point geometry is illustrated above. The two “crossing” lines above are understood to actually be parallel, that is, they do not intersect since the only points in the geometry are the dots drawn. Of course, the lines do not need to be represented as straight lines, which can lead to non-“crossing” models if the above is too bothersome. Also, the points and lines can be drawn as vertices and edges of a tetrahedron, because no requirement to have the geometry embedded in $\mathbb{R}^2$ is provable. Up to isomorphism, the above model is the only model of four-point geometry, which we prove below.

Theorem 1.3.9. *In four-point geometry, if two distinct lines intersect, then the intersection is exactly one point.*

Proof. By assumptions, the two lines have at least one point of intersection. If there were more than one point in the intersection, then there would be at least two points of intersection. Then there would exist two distinct lines containing these two distinct points. This contradicts Axiom 2. Therefore, the intersection is exactly one point. $\square$

Theorem 1.3.10. *In four-point geometry, there are exactly six lines.*

Proof. By Axiom 1, we know there are exactly four points, by Axiom 2, each pair of points corresponds to exactly one line, and by Axiom 3, we know that each line is made up of exact two points. Therefore, the number of distinct lines is equal to the number of distinct (unordered) pairs of points, which is $\binom{4}{2} = 6$. $\square$

Theorem 1.3.11. *In four-point geometry, each point has exactly three lines incident to it.*

Proof. By Axiom 2, each point x has a line containing it with each of the other points in the geometry. By Axiom 1, there are three other points. If there were another line incident to x , then by Axiom 3 this line would have to contain a second point other than x . But this would contradict the uniqueness of the line given in Axiom 2 since we have already accounted for those three lines. Therefore, there are exactly three lines incident to x . $\square$

Theorem 1.3.12. *In four-point geometry, each distinct line has exactly one line parallel to it. In fact, if ℓ is a line and P is a point not on ℓ , then there exists exactly one parallel line to ℓ which contains P .*

Proof. Let ℓ be a line and P a point not on ℓ . Let x and y be the two points on ℓ , given by Axiom 3. By assumption, x and y are distinct from P . Let Q be the fourth point in the geometry, given by Axiom 1. Let m be the line containing P and Q by Axiom 2. Since m contains only two points, by Axiom 3, m does not intersect ℓ . Therefore, m and ℓ are parallel and m contains P . This proves the second statement. We can also see by construction, that m is the only parallel line to ℓ since there are only four points and each line is a set of two points. This proves the first statement. $\square$

Exercises

1. Prove that there exists a set of two lines in the four-point geometry that contain all the points of the geometry.

2. Replacing the undefined terms *pi*, *ka*, and *chu*, with point, line, and incident, respectively, the axioms of *pi-ka-chu* become:

Definition 1.3.13. 5-Point Handshake Geometry H_5

- I. (Point Existence): There exist exactly five points.
- II. (Line Determination): Any two distinct points have exactly one line on both of them.
- III. (Handshake): Each line is on exactly two points.

- (i) Rewrite the theorems and their proofs from the Exercises in Section 1.1 using now these geometric terms.
- (ii) Prove that there exist exactly four lines on each point.
- (iii) Prove that for any line ℓ and any point $P \notin \ell$ there exists exactly two other distinct lines on P that are parallel to ℓ .
- (iv) Demonstrate the independence of the 5-Point Handshake Axioms.

3. Consider the geometry where points are given as $P = \mathbb{Z}_2^2 = \{(x, y) \mid x, y \in \mathbb{Z}_2 = \{0, 1\}\}$ and lines are solution sets to linear equations of the form $ax + by = c$ where $a, b, c \in \mathbb{Z}_2$ and a, b are not simultaneously zero. A point (x_1, y_1) is incident to the line $ax + by = c$ if it is a solution to the equation.

- (i) How many points in the geometry?
- (ii) How many lines in this geometry?
- (iii) Which point is incident to which line?
- (iv) Construct a model of this geometry.
- (v) Is this geometry isomorphic to any geometry we have seen previously?

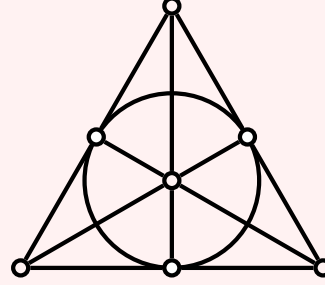
“We must accept finite disappointment, but never lose infinite hope.” – Martin Luther King, Jr.

1.4 Fano Geometry

We present another example of a finite geometry.

Definition 1.4.1. Fano Geometry

- Axiom 1:* There exists at least one line.
Axiom 2: There exists exactly three points on every line.
Axiom 3: Not all points lie on the same line.
Axiom 4: For each two distinct points there exists a unique line containing both of them.
Axiom 5: All lines intersect.
 (There are no parallel lines).



Theorem 1.4.2. *In Fano geometry, two distinct lines intersect in exactly one point.*

Proof. By Axiom 5, the two lines have at least one point of intersection. If there were more than one point in the intersection, then there would be at least two points, that is, there would exist two distinct lines containing these two distinct points. This contradicts Axiom 4. Therefore, the intersection is exactly one point. $\square$

Theorem 1.4.3. *In Fano geometry, there are exactly seven points and seven lines.*

Proof. There exists a line ℓ_1 by Axiom 1. By Axiom 2, there exists exactly three points A , B , and C on ℓ_1 . By Axiom 3, there exists a fourth point D not on ℓ_1 . By Axiom 4, there exists a unique line incident to D and A , D and B , and D and C , call them ℓ_2 , ℓ_3 , and ℓ_4 , respectively. Each of these three lines is distinct from ℓ_1 since they all contain D . Also, by Axiom 2, they each contain a third point.

Considering ℓ_2 , if it contained B or C , then this would violate Axiom 4 since ℓ_1 is already a line containing A and B or A and C . Thus, the third point on ℓ_2 given by Axiom 1 is a new point, call it E . Thus, ℓ_2 is distinct from ℓ_3 and ℓ_4 since they are incident to different points. Similarly, ℓ_3 and ℓ_4 contain a third point each different from the five points already mentioned, call them F and G respectively, otherwise Axiom 4 would be violated because there were two distinct lines incident to the same pair of points. This gives at least seven points and four lines. Also, all of the constructed lines intersect each other.

ℓ_1	A	B	C
ℓ_2	A	D	E
ℓ_3	B	D	F
ℓ_4	C	D	G
ℓ_5	A	F	G
ℓ_6	B	E	G
ℓ_7	C	E	F

Axiom 4 gives the existence of another line ℓ_5 containing A and F . This line contains a third point (Axiom 2). Also, ℓ_5 cannot be parallel to ℓ_4 . Of course, ℓ_5 cannot contain D or C as this would contradict Axiom 4 (there is already a line containing A and D and a line containing A and C). Thus, ℓ_5 must contain G . By similar reasoning, there is a line ℓ_6 containing B and E and a line ℓ_7 containing C and F , respectively. The only way to not contradict Axiom 5 is to have ℓ_6 contain G and ℓ_7 contain

E , respectively. This accounts for seven points and seven lines. It can be verified that these points and lines form a model of Fano geometry.

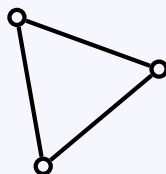
If there were another point H , then there would be a line ℓ_8 containing H and A . This line would contain a third point, but it cannot be any of the eight points considered since this would contradict Axiom 4. Call this third point I . But then ℓ_8 is parallel to ℓ_3 , a contradiction of Axiom 5. Therefore, there are exactly seven points. Also, if there were another line in this geometry, then either Axiom 4 is contradicted or another point is necessary. Therefore, there are only seven lines. $\square$

We note that in the previous proof, in addition to proving the statement of the theorem, we essentially have proven that all models of Fano geometry are isomorphic to the model constructed in the proof.

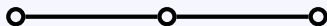
Theorem 1.4.4. *The five axioms of Fano geometry are independent.*

Proof. Consider as a model an empty set of points and an empty set of points. Then Axioms 2–5 are satisfied vacuously but Axiom 1 fails since no lines exist. Therefore, Axiom 1 is independent.

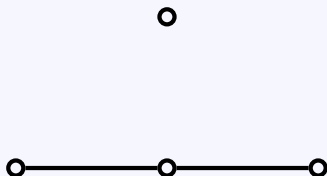
Three-point geometry provides a model of a geometry satisfying Axiom 1, 3, 4, 5 but fails Axiom 2. Therefore, Axiom 2 is independent.



Consider the geometric model illustrated below with the usual interpretation of points, lines, and incidence. Then Axioms 1, 2, and 4 are satisfied clearly, and Axiom 5 is satisfied vacuously. Clearly, Axiom 3 fails. Therefore, Axiom 3 is independent.



The model below clearly satisfies Axioms 1, 2, and 3, and Axiom 5 is satisfied vacuously. Also, Axiom 4 fails since there is no line containing the unique point off the single line and any of the three points on the unique line. Therefore, Axiom 4 is independent.



Young's geometry which we will discuss next time provides a model of Axioms 1–4 but fails Axiom 5. Thus, Axiom 5 is independent. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be Fano geometry.

1. Prove that each point is on exactly three lines.
2. Prove that the set of all lines on a point contains all the points of the geometry.
3. Prove that for any pair of points there exists exactly two lines containing neither point.
4. Prove that for any set of three nonconcurrent lines there exists exactly one point not on any of the three lines.

ⁱA 3D model of Fano geometry, coded by Alexis Borrell, Sidney Germer, can be viewed at <https://www.math3d.org/tUc5mutJM>.

“It takes a long time to become young.” – Pablo Picasso

1.5 Young Geometry

Axiom 5 of Fano geometry is a statement about parallel lines and that they do not exist. This statement was essential in constructing the unique model of Fano geometry. Young’s geometry is a modification of Fano’s geometry which this parallel axiom is replaced with one that allows the existence of parallel lines.

Definition 1.5.1. Young Geometry

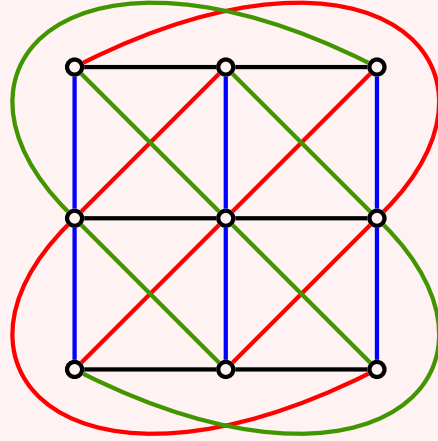
Axiom 1: There exists at least one line.

Axiom 2: There exists exactly three points on every line.

Axiom 3: Not all points lie on the same line.

Axiom 4: For each two distinct points there exists a unique line containing both of them.

Axiom 5: For each line ℓ and each point P not on ℓ , there exists exactly one parallel line to ℓ which contains P .



Theorem 1.5.2. *In Young geometry, every point is on exactly four lines. (Progressive Geometry Proofs Slides)*

Proof. Let P be a point in the geometry. We claim there exists a line ℓ such that $P \notin \ell$. Axiom 1 guarantees the existence of a line ℓ' . If $P \notin \ell'$, let $\ell = \ell'$. Otherwise, Axiom 2 gives us a point $A \in \ell'$ such that $A \neq P$. Axiom 3 also guarantees a point $B \notin \ell'$. Using Axiom 4, there exists a line ℓ which contains A and B . It must be that $\ell \neq \ell'$ since $B \in \ell \setminus \ell'$. On the other hand, $P \notin \ell$, otherwise there exists two lines containing A and P which contradicts Axiom 4.

Then for each point on ℓ , there is a unique line containing this point and P by Axiom 4. Likewise, these three lines are all distinct (otherwise the uniqueness of Axiom 4 is violated). Finally, there is a unique line containing P which is parallel to ℓ by Axiom 5. Thus, there are at least 4 lines incident to P .

Let m be another line incident to P that was not considered before. It cannot be parallel to ℓ , otherwise the uniqueness of parallel lines would be violated. Hence, m intersects ℓ . Call their intersection Q . Then m is the unique line determined by P and Q . Since Q is on ℓ , this line was already considered above, which leads to a contradiction. The result then follows. $\square$

Theorem 1.5.3 (Proclus’ Lemma). *In Young geometry, if lines ℓ and m intersect and if ℓ is parallel to a line n , then lines m and n intersect.*

Proof. Let P be a point of intersection of ℓ and m (although this point of intersection is unique, we do not need that here). By way of contradiction, suppose that m is parallel to n . Then lines ℓ and m are both parallel lines to n which both contain the point P . This contradicts Axiom 5. Therefore, m intersects n . $\square$

Theorem 1.5.4 (Transitivity of Parallelism). *In Young geometry, two lines parallel to the same line are parallel to each other.*

Proof. Let ℓ and m be lines which are both parallel to the line n . Suppose, for the sake of contradiction, that ℓ and m intersect. Then ℓ and m intersect and ℓ is parallel to n . Then Theorem 1.5.3 implies that m intersects n , which is a contradiction. Therefore, ℓ and m are parallel. $\square$

The previous theorem establishes that the relation of parallel is transitive.

Theorem 1.5.5. *In Young geometry, every line has exactly two lines that are parallel to it.*

Proof. Let ℓ be a line and let P be a point not on ℓ , which exists by Axiom 3. Let Q be a point on ℓ , which exists by Axiom 2. Now, there exists a unique parallel line to ℓ through the point P , by Axiom 5. Let m be the unique line containing P and Q , which exists by Axiom 4. Axiom 2 provides a third point on m , call this point R .

By Axiom 5, there exists a unique line containing P and parallel to ℓ . There exists a similar unique parallel line to ℓ through R . These two lines must be distinct, otherwise there would be multiple lines containing P and R , a violation of Axiom 4. This shows that there are at least two lines parallel to ℓ .

Suppose that m was a third line parallel to ℓ . By Theorem 1.5.3, m intersect the lines containing P , Q , and R . Since m and ℓ are parallel, the intersection cannot be Q . But if the intersection is P or R , then m would intersect a line parallel to ℓ , which violates the transitivity of parallel lines. Therefore, there are exactly two lines parallel to ℓ . $\square$

The last theorem will be proven in the homework, although in the next chapter we will prove this theorem more generally for affine geometry.

Theorem 1.5.6. *In Young geometry, there are exactly 9 points and 12 lines.*

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be Young geometry.

1. Prove in there exist exactly 9 points.
2. Prove in there exist exactly 12 lines.
3. Consider the geometry where points are given as $P = \mathbb{Z}_3^2 = \{(x, y) \mid x, y \in \mathbb{Z}_3 = \{0, 1, 2\}\}$ and lines are solution sets to linear equations of the form $ax + by = c$ where $a, b, c \in \mathbb{Z}_3$ and a, b are not simultaneously zero. A point (x_1, y_1) is incident to the line $ax + by = c$ if it is a solution to the equation.
 - (i) How many points in the geometry?
 - (ii) How many lines in this geometry?
 - (iii) Which point is incident to which line?
 - (iv) Construct of model of this geometry.
 - (v) Is this geometry isomorphic to any geometry we have seen previously?

ⁱA 3D model of Young geometry, coded by Alexis Borrell, Sidney Germer, can be viewed at <https://www.geogebra.org/m/n6hjdceu>.

ⁱⁱThe progressive slides in this section were created by Hayley John.

Chapter 2

Incidence Geometry

“What is character but the determination of incident? What is incident but the illustration of character?”
– Henry James

2.1 The Incidence Axioms

We have explored now four distinct models of finite geometry in our lectures, with a few others explored in the homework. Although these different geometries expressed very different properties, there are many theorems and properties that were true for all the geometries. These common properties give rise to the incidence axioms and their logical consequences.

Definition 2.1.1. Incidence Axioms

- Axiom I.1* (Line Determination): For each two distinct points there exists a unique line containing both of them.
- Axiom I.2* (Secancy): For every line there exist at least two distinct points on it.
- Axiom I.3* (Point Existence): There exists at least three distinct points.
- Axiom I.4* (Non-Collinearity): Not all points lie on the same line.

We say that a geometry is an **incidence geometry** (or a **linear space**) if it satisfies all four of the incidence axioms.

Example 2.1.2. The 3-point, 4-point, and 5-point, Fano, and Young geometries are all examples of finite incidence geometries.

Example 2.1.3. The Euclidean plane $\mathbb{R}^2$ is an example of an infinite incidence geometry,ⁱ where points are interpreted as ordered pairs (x, y) with $x, y \in \mathbb{R}$ and lines are interpreted as solution sets to linear equations of the form $Ax + By = C$ (where A , B , and C are real numbers such that A and B are not both zero).

Note that if we have any set X with “nicely” defined operations of addition and multiplication, such as $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{C}$, then this same scheme makes X^2 into an incidence geometry, which we call a **Cartesian geometry**.ⁱⁱ

Theorem 2.1.4. *In incidence geometry, if two distinct lines intersect, then the intersection is exactly one point.*

Proof. By assumptions, the two lines have at least one point of intersection. If there were more than one point in the intersection, then there would be at least two points of intersection. Then there would exist two distinct lines containing these two distinct points. This contradicts Line Determination (Axiom I.1). Therefore, the intersection is exactly one point. $\square$

Theorem 2.1.5. *In incidence geometry, if ℓ is any line, then there exists at least one point P such that P does not lie on ℓ .*

Proof. If no points existed off of ℓ , then all points would lie on a common line. This contradicts Non-Collinearity (Axiom 4). $\square$

Theorem 2.1.6. *In incidence geometry, if P and Q are two points such that $P \neq Q$, then there exists a point R such that P , Q , and R are non-collinear, that is, the points P , Q , and R do not all lie on the same line.*

Proof. The points P and Q determine a line ℓ (Axiom I.1). By Theorem 2.1.5, there exists a point R not on ℓ . Therefore, P , Q , and R are non-collinear. $\square$

Theorem 2.1.7. *In incidence geometry, if P is any point, then there exists points Q and R such that P , Q , and R are non-collinear, that is, the points P , Q , and R do not all lie on the same line.*

Proof. By Point Existence (Axiom I.3), there exists at least one other point Q such that $P \neq Q$. The result then follows from Theorem 2.1.6. $\square$

Theorem 2.1.8. *In incidence geometry, there exists three non-collinear points.*

Proof. By Point Existence (Axiom I.3), there exists a point P . The theorem then follows from Theorem 2.1.7. $\square$

Theorem 2.1.9. *In incidence geometry, if P is any point, then there exists at least one line ℓ such that P does not lie on ℓ .*

Proof. By Theorem 2.1.7, there exists points Q and R such that P , Q , and R are non-collinear. Let ℓ be the line determined by Q and R (Axiom I.1). By construction, P does not lie on ℓ . $\square$

Theorem 2.1.10. *In incidence geometry, for each point there exist at least two lines containing it.*

Proof. Let P be a point of the geometry. By Theorem 2.1.7, there exist points Q and R such that P , Q , and R are non-collinear. By Line Determination (Axiom I.1) there exists a line containing P and Q and a line containing P and R . Since P , Q , and R are non-collinear, these two lines must be distinct. $\square$

Of course, the incidence axioms say nothing of planes, hyper-planes, or other higher dimensional objects. Instead, the incidence axioms only provide a meaning about points, lines, and incidence. As such, higher dimensional Euclidean geometries $\mathbb{R}^n$ are also incidence geometries, where points are interpreted as vectors with n real entries and lines are interpreted as solution sets to matrix equations $A\mathbf{x} = \mathbf{b}$ (where A is an $m \times n$ matrix with nullity one and $\mathbf{b}$ is a real-valued vector with m entries).

Note that Exercises 1.3.3 and 1.5.3 provide examples of finite Cartesian incidence geometries.

Exercises

1. Prove that 5-point handshake geometryⁱ is an incidence geometry.
2. Prove that Fano geometry is an incidence geometry.
3. Prove that Young geometry is an incidence geometry.
4. Prove that the axioms of incidence are independent.
5. Consider the finite geometry with the following axioms:

Definition.

- I. There exist exactly three points.
- II. Each two distinct points share at least one common line.
- III. Each point is on exactly two distinct lines.

- (i) Prove that the geometry is consistent.
- (ii) Prove Axiom I is independent.
- (iii) Does this geometry have any parallel lines? Explain.
- (iv) Prove or disprove: Each two distinct points share exactly one common line.

ⁱSee Exercise 1.3.2.

“None but himself can be his parallel.” – Virgil

2.2 The Parallel Axioms

We continue to develop the theoryⁱ of incidence geometry.

Theorem 2.2.1. *In incidence geometry, there exists at least three lines.*

Proof. Let P , Q , and R be three non-collinear points, guaranteed by Theorem 2.1.8. By Line Determination (Axiom I.1), there exist at least three lines ℓ , m , and n determined by the pairs P and Q , P and R , and Q and R , respectively. Since these three points are non-collinear, we conclude that these three lines must be distinct because each line is missing exactly one point from this triple. $\square$

Theorem 2.2.2. *In incidence geometry, if ℓ is any line, then there exist lines m and n such that ℓ , m , and n are distinct and both m and n intersect ℓ . Furthermore, these lines are non-concurrent.*

Proof. By Secancy (Axiom I.2), there exists two distinct points P and Q on ℓ . By Theorem 2.1.6, there exists a third point R not on ℓ . By Line Determination (Axiom I.1), there exist lines m and n determined by the pairs P and R and Q and R , respectively. Since these three points are non-collinear, we conclude that these three lines must be distinct because each line is missing exactly one point from this triple. If there were a common point S incident to all three lines, then S would have to be distinct from P , Q , and R . Then the lines ℓ and m would have two distinct intersections P and S , a contradiction to Theorem 2.1.4. Therefore, there is no point common to all three lines. $\square$

Theorem 2.2.3. *In incidence geometry, there exists three non-concurrent lines.*

Proof. By Theorem 2.2.1, there exists a line ℓ . The result then follows from Theorem 2.2.2. $\square$

When looking at the axioms of Fano and Young geometry, we see they differ in only one axiom which states something about the existence of parallel lines. Because both are models of incidence geometry, we see that their respective parallel axioms are independent of incidence geometry, that is, the incidence axioms are not enough to determine the nature of parallel lines in a geometry.

We list three parallel *alternatives* meaning axioms which are mutually inconsistent, that is, any geometry with any pair of these postulates as axioms is inconsistent. As such, a geometry may have at most one of the following parallel axioms, or possibly none of them.

Definition 2.2.4. Parallel Axioms

Axiom P.1 (Elliptic Parallel Postulate): All lines intersect, that is, there exists no parallel lines.

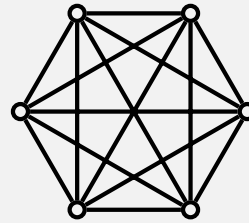
Axiom P.2 (Euclidean Parallel Postulate): For every line ℓ and every point P not on ℓ , there exists exactly one line parallel to ℓ which contains P .

Axiom P.3 (Hyperbolic Parallel Postulate): For every line ℓ and every point P not on ℓ , there exists more than one line parallel to ℓ which contains P .

The second parallel axiom gets its name because it is equivalent to Euclid’s famous fifth postulate of geometry, although the version we mentioned about is due to Playfair.

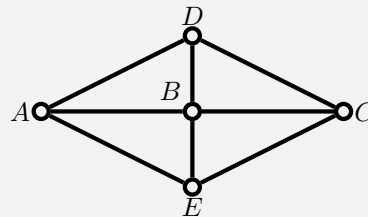
Example 2.2.5. Three-point geometry and Fano geometry satisfies the Elliptic Parallel Postulate (Axiom P.1). Four-point geometry and Young geometry satisfies the Euclidean Parallel Postulate (Axiom I.2). Five-point geometryⁱⁱ satisfies the Hyperbolic Parallel Postulate (Axiom I.3).

3-, 4-, and 5-point geometries are all examples of a larger infinite family of geometries, called **handshake geometry**. Let H_n be an incidence geometry with n vertices and all lines contain exactly two points, called a **handshake**. These three geometries are none other than H_3 , H_4 , and H_5 , respectively. The geometry H_6 is displayed below. For $n \geq 5$, H_n satisfies the hyperbolic parallel postulate.



Example 2.2.6. Consider the finite geometry illustrated to below. The points A , B , and C are considered to be collinear. Likewise, the points D , B , and E are considered to be collinear.

Consider the line determined by A and D , call it ℓ , and consider the point C . Then there exists a unique line on C which is parallel to ℓ , namely the line determined by C and E . This resembles the Euclidean Parallel Postulate. In particular, this geometry does not satisfy the Elliptic or Hyperbolic Parallel Postulates. On the other hand, consider the point B . The only lines incident to B are $\{A, B, C\}$ and $\{D, B, E\}$, both of which intersect ℓ . Thus, there is no line parallel to ℓ which contains B . This resembles the Elliptic Parallel Postulate. In particular, this geometry does not satisfy the Euclidean Parallel Postulate. This shows that an incidence geometry might not satisfy any of the parallel axioms.

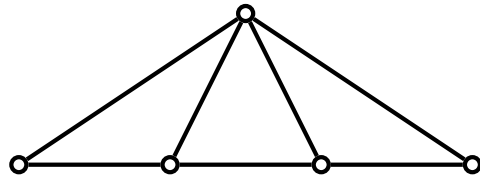
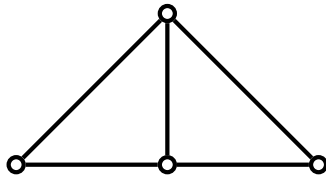


The above examples prove the following theorem.

Theorem 2.2.7. *The axioms of incidence together with a single parallel axiom is a consistent geometric theory. Also, each parallel axiom is independent of the four incidence axioms.*

Definition 2.2.8. We say that an incidence geometry is **affine** if it satisfies the Euclidean Parallel Postulate. We say that an incidence geometry is **projective** if it satisfies the Elliptic Parallel Postulate and all lines contain at least three points.

To justify why the extra condition when defining projective geometries about at least three points on a line, let us introduce **fan geometry**. Let F_n be an incidence geometry with n points and lines determined by the following rules: there is a single line that contains $n - 1$ of the points and for the one point off this line there exists $n - 1$ handshake lines between this point and the other $n - 1$ points. Three point geometry is just F_3 and F_4 and F_5 are displayed below left and below right, respectively. One can easily see that F_n is an incidence geometry satisfying the Elliptic Parallel Postulate, but it fails the extra condition that all lines have at least three points. This extra condition is to remove fan-like geometries from the family of projective geometries. The reasons behind this will be revealed in Section 8.1.



Elliptic Parallel Postulate Haiku

I am a point P
 I want a parallel please!
 Oh, there's none for me.

Hyperbolic Parallel Postulate Haiku

I am a point P
 There's so many parallels
 At least 2 for me!

Euclidean Parallel Postulate Haiku

I am a point P
 Elliptic? Hyperbolic?
 No, just 1 for me!

by Gwen Ellison

Exercises

1. Determine which of the following interpretations of the undefined terms are models of incidence geometry. In any case, indicate which parallel alternative is exhibited by the interpretation.

- (i) Points are points on a Euclidean plane, and lines are circles in the Euclidean plane.
- (ii) Points are points on a Euclidean plane, and lines are all those lines on the plane that pass through a given fixed point P .
- (iii) Points are points on a Euclidean plane, and lines are concentric circles all having the same fixed center.
- (iv) Points are Euclidean points in the interior of a fixed circle, and lines are the parts of Euclidean lines that intersect the interior of the circle.
- (v) Points are points on the surface of a Euclidean sphere, and lines are great circles on the surface of that sphere.
- (vi) The same as part (v) except that any two points that lie on opposite ends of a diameter are identified as the same point.
- (vii) Points are points on a Euclidean hemisphere (not including those points on the great circle that defines the hemisphere), and lines are great semicircles (i.e. those points on a great circle that intersect with the hemisphere).
- (viii) In Euclidean 3-space, points are interpreted as lines and lines are interpreted as Euclidean planes. Incidence means those lines which are immersed in the plane.
- (ix) Points are lines in Euclidean 3-space, and lines are Euclidean planes all of which contains the same fixed line.

2. Which of the geometries discussed in Chapter 1 are affine geometries?

3. Which of the geometries discussed in Chapter 1 are projective geometries?

4. Consider a finite geometry where the points are interpreted to be the vertices of a regular octahedron and the lines are the sets of exactly two points. Is this interpretation a model of an incidence geometry? How many lines does this geometry have? Which parallel axiom (if any) does it display?

5. Consider the following axiom set for an incidence geometry:

Definition. *Undefined terms:* point, line, incidence

Axioms:

- I. Given any two distinct points, there is a unique line incident with both.
- II. Given any two distinct lines, there is a unique point incident with both.
- III. There exists in this geometry a set of four points, no three of which are collinear.
- IV. Each line contains exactly n points.

- (i) Prove that $n > 2$.
- (ii) Verify that this axiom set establishes a projective geometry.
(Note that when $n = 3$ this geometry is isomorphic to Fano's geometry.)
- (iii) Prove that there exists a set of four lines no three of which are concurrent.
- (iv) Prove that each point is incident to exactly n lines.

ⁱⁱBy the word "theory," we mean the collection of theorems for this axiomatic system.

ⁱⁱSee Definition 1.3.13.

“You don’t accomplish a lot by changing people’s opinions by shoving facts down their throat. I think you change people’s opinions by opening your heart up and showing the parallels between you and another person. That’s how people’s ideas shift.” – Mary Lambert

2.3 Affine Geometry

We can formalize affine geometry into an axiomatic system. We will see that affine geometry generalizing the properties of Young geometry.

Definition 2.3.1. Affine Geometry

Axiom I.1 (Line Determination): For each two distinct points there exists a unique line containing both of them.

Axiom I.2 (Secancy): For every line there exist at least two distinct points on it.

Axiom I.3 (Point Existence): There exists at least three distinct points.

Axiom I.4 (Non-Collinearity): Not all points lie on the same line.

Axiom P.2 (Euclidean Parallel Postulate): For every line ℓ and every point P not on ℓ , there exists exactly one line parallel to ℓ which contains P .

Theorem 2.3.2 (Proclus’ Lemma). *In an affine geometry, suppose that ℓ , m , and n are lines such that ℓ is parallel to m and m intersects n . Then ℓ intersects n also.*

Proof. Let P be the point of intersection of m and n , which is unique in incidence geometry (Theorem 2.1.4). By way of contradiction, suppose that ℓ is parallel to n . Then lines m and n are both parallel lines to ℓ which contain the point P . This contradicts the Euclidean parallel postulate (Axiom P.2). Therefore, ℓ intersects n . $\square$

Theorem 2.3.3 (Transitivity of Parallelism). *In affine geometry, two lines parallel to the same line are parallel to each other.*

Proof. Let m and n be lines which are both parallel to the line ℓ . Suppose, for the sake of contradiction, that m and n intersect. This ℓ and m are parallel and m intersects n , Theorem 2.3.2 implies that ℓ intersects n , which is a contradiction. Therefore, m and n are parallel. $\square$

The previous theorem establishes that the relation of parallel is transitive.

Theorem 2.3.4. *In affine geometry, all lines contain the same number of points.*

Proof. Let ℓ and m be two lines. Let P be a point on ℓ which is not on m . This is possible because intersections of lines are at most one point and every line has at least two points, by secancy. Likewise, let Q be a point on m which is not on ℓ . Let n be the line determined by P and Q . Then for each point X on ℓ there is a unique line n_X which is parallel to n and contains the point X , by the Euclidean parallel postulate. This point of intersection with ℓ is unique. Also, by Theorem 2.3.2, n_X intersects m at a unique point Y_X . Since the existence of these parallel lines n_X is unique, this construction determines an unambiguous correspondence between the points X on ℓ and the points Y_X on m . As intersections are unique, this is a one-to-one correspondence. Therefore, there is a bijection between the points on ℓ and the points on m , which proves the theorem. $\square$

Definition 2.3.5. This constant number of points on an affine line is called the **order** of the geometry.

Theorem 2.3.6. In an affine geometry of order n , each point has $n + 1$ lines incident to it.

Proof. Let P be a point and let ℓ be a line not on P . By line determination, we may construct n distinct lines from the n points on ℓ to P . Also, by the Euclidean parallel postulate, there exists a unique line parallel to ℓ incident to P . This accounts $n + 1$ distinct lines incident to P .

Let m be another line incident to P that was not considered before. It cannot be parallel to ℓ , otherwise the uniqueness of parallel lines would be violated. Hence, m intersects ℓ . Call their intersection Q . Then m is the unique line determined by P and Q . Since Q is on ℓ , this line was already considered above, which leads to a contradiction. The result then follows. $\square$

Theorem 2.3.7. In an affine geometry of order n , then there exist n^2 points and $n(n + 1)$ lines in the geometry.

Proof. Let ℓ and m be two intersecting lines in the geometry. For each point P on m , consider the unique line ℓ_P parallel to ℓ which contains the point P on m . Each line ℓ_P contains exactly n points. As each line ℓ_P is parallel to ℓ , Theorem 2.3.3 states that each pair ℓ_P and ℓ_Q is pairwise parallel if and only if $P \neq Q$. Thus, no points are shared by the ℓ_P . There are n many distinct parallel lines ℓ_P for each of the n points P on m . This accounts for n^2 many points in the geometry. For any point X in the geometry, there is a unique line containing X and is parallel to ℓ , call this line ℓ' . Since ℓ intersects m and ℓ is parallel to ℓ' , ℓ' must intersect m too. By uniqueness, ℓ' must then be of the form ℓ_P for some point P on m . Therefore, there are exactly n^2 points.

Now, the product $n^2(n + 1)$ counts the number of times any line is incident to any point, that is, the cardinality of the incidence relation viewed as a subset of the Cartesian product of points and lines. If we partition the incidence pairs into classes based upon which line is used in the pair, then the number of classes will be the number of lines. Since there are n points on each line, each class contains n pairs, which gives the number of lines as

$$\frac{n^2(n + 1)}{n} = n(n + 1). \quad \square$$

Example 2.3.8. Four-point geometry is affine and has $n = 2$ points on each line. It has $n^2 = 2^2 = 4$ points, $n(n + 1) = 2(3) = 6$ lines, and each point is incident to $n + 1 = 3$ lines.

Young geometry is affine and has $n = 3$ points on each line. It has $n^2 = 3^2 = 9$ points, $n(n + 1) = 3(4) = 12$ lines, and each point is incident to $n + 1 = 4$ lines. This then provides a proof to Theorem 1.3.18.

The above theorems are extremely useful in classifying finite affine geometries up to isomorphism. The above theorems and their proofs are also valid for infinite affine geometries, where addition is understood to be disjoint unions of cardinals, multiplication is understood to be Cartesian products of cardinals, and division is understood to be partitions of cardinals.

A last question to address here is the following: even though we know how many points, lines, etc. will follow from the given order n , what orders are possible? Are some orders not allowed? Previous homework suggested that order 2 and order 3 affine geometry can be coordinatized using finite fields. This works in general, that is, let $\mathbb{F}$ be the unique finite field of order p^k . Then there exists a finite affine geometry of order

$n = p^k$ by defining points as elements of $\mathbb{F}^2$ and lines as linear equations over $\mathbb{F}$. This shows that the order of an affine geometry could be any power of a prime. It is conjectured and widely believed that no other order is possible, that is, the order of an affine geometry is a power of a prime.

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be an affine geometry of order n .

1. Prove every line has exactly n lines that are parallel to it (including the line itself).

Definition 2.3.9. We say that two distinct points **intersect** if there exists a line incident to both points. We say that two distinct points are **parallel**ⁱ if they do not intersect.

2. We explore here a geometry which has parallel lines AND parallel points. Consider the following axioms for **Pappus Geometry**:

Definition 2.3.10. Pappus Geometry

- I. (Line Existence) There exists at least one line.
- II. (Line Congruence) Every line has exactly three points.
- III. (Non-collinearity) Not all points are on the same line.
- IV. (Euclidean Parallel Postulate) If a point P is not on a given line ℓ , then there is exactly one line m that contains P and is parallel to ℓ .
- V. (Pappus Line Determination) If a line ℓ is not on a given point P , then there is exactly one point Q that contains ℓ and is parallel to P .
- VI. (Uniqueness of Intersection) If two lines intersect, then the intersection is unique.

Because of the presense of the Euclidean Parallel Postulate, this geometry has Proclus' Lemma and Transitivity of Paralell Lines. You may take those for granted. Similar results also hold for parallelism of points in Pappus geometry, which we prove now.

- (i) (Proclus' Lemma for Points) In a Pappus geometry, suppose that P , Q , and R are points such that P is parallel to Q and Q intersects R . Prove that P intersects R also.
- (ii) (Transitivity of Parallelism of Points) Prove two points parallel to the same points are parallel to each other.
- (iii) Prove not all lines are incident to the same point.
- (iv) If two points intersect, prove the intersection is unique.
- (v) Prove that every point is on exactly three lines.
- (vi) How many points are there in this geometry?
- (vii) How many lines are there in this geometry?
- (viii) For each point in this geometry, how many other points are parallel to it?

ⁱNote that any geometry with parallel points fails Line Determination, which means it is not an incidence geometry.

“There is no such thing as defeat in non-violence.” – Cesar Chavez

2.4 Non-Euclidean Geometry

We have seen from the previous section that the Euclidean parallel postulate is a very powerful axiom and the inclusion of it, even into a very broad axiomatic system such as incidence geometry, can lead to very powerful results. This fact is why many geometers have been fascinated with Euclid’s fifth postulate for centuries.

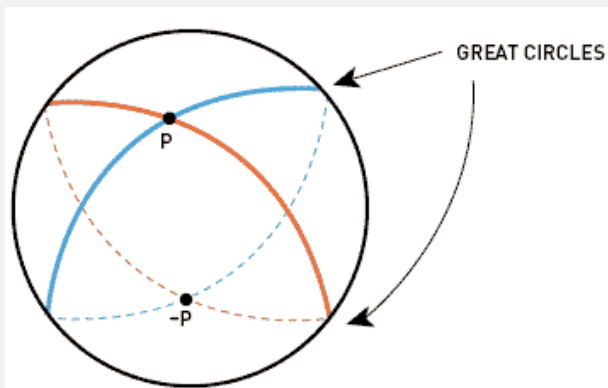
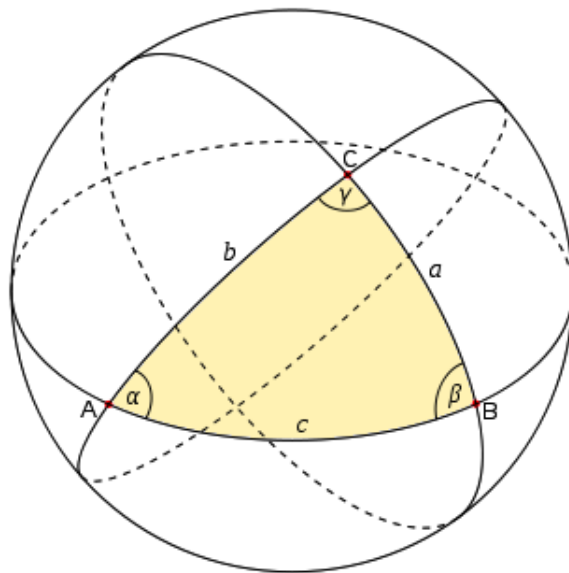
This is section we will explore non-Euclidean geometries, that is, geometries where the Euclidean parallel postulate fails, maybe because the elliptic or hyperbolic parallel postulates hold instead.

Example 2.4.1. Spherical geometry, denoted S^2 , is a geometry whose points are the set of points on the unit sphere in $\mathbb{R}^3$, whose lines are great circles on the sphere (circles whose diameter is the same as the diameter of the sphere), and incidence of points on line has its usual interpretation as points on curves.

A **geodesic**, the shortest pathⁱ between two distinct points, is illustrated to the right for two points A and B . Geodesics generalize the notion of “straight line” in a geometry, which is why lines in spherical geometry are defined as great circles, the geodesics of S^2 .

Spherical geometry satisfies the elliptic parallel postulate since all great circles intersect each other. In fact, spherical geometry is the best understood and most studied elliptic geometry, historically, particularly because of our residence on planet Earth which has spherical geometry. In fact, the word *geometry* derives from the Greek word γεωμετρία which means “earth-measure,” because ancient mathematicians developed many geometric ideas to better understand the geometry of Earth.

Although a very natural elliptic geometry, unfortunately spherical geometry is not a model of incidence geometry since it violates Axiom I.1 (Line Determination). To see a violation of Line Determination, we say that two points on a sphere are called **antipodes** if they are endpoints of the same diameter of the sphere. There does not exist a unique line determined by antipodes. In fact, there exist infinitely many great circles containing a pair of antipodal points. Furthermore, if a point P is on a great circle, then its antipode $-P$ will also be contained on that same great circle. Antipodes are, in fact, the only violation of line determination. Spherical geometry satisfying the other three axioms of incidence.



Example 2.4.2. Because spherical geometry is close to being an incidence geometry, there are two patchworks we will discuss. One patch is to instead define points on the sphere as antipodal pairs, as opposed to individual points. Lines are still great circles and an antipodal pair is incident to a great circle if the two points are on the circle. This does form an infinite incidence geometry additionally satisfying the elliptic parallel postulate, known as **the real projective plane** $\mathbb{RP}^2$.

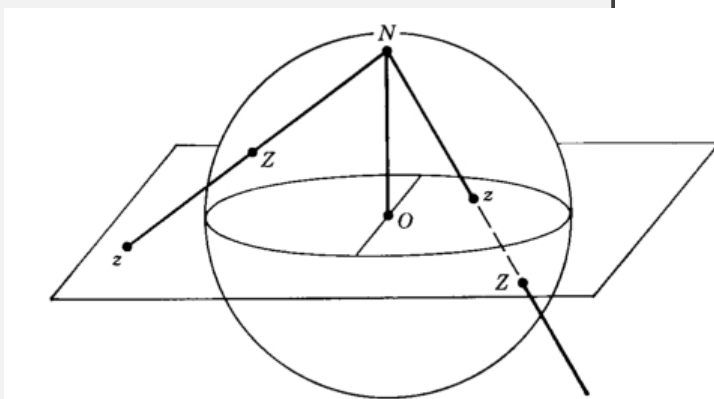
As geometry is a very visual branch of mathematics, many people are interested in “seeing” $\mathbb{RP}^2$. Unfortunately, $\mathbb{RP}^2$ cannot be embedded into $\mathbb{R}^3$ without self-intersection, which make visualizing it very difficult. The interested student is invited to search the web for visual models of the real projective plane.

This spherical model is just one example of $\mathbb{RP}^2$. We will explore another model in Chapter 7 as the Euclidean plane $\mathbb{R}^2$ adjoined a *line at infinity*. The interested reader is invited to search other models of the real projective plane, as there are many.

Example 2.4.3. Another patch to spherical geometry is to consider topological manifolds. A *topological space*, very loosely speaking, is a generalization of the notion of geometry which axiomatizes the ideas of point and closeness. While distance is one way to describe closeness, topology uses the undefined term open set to axiomatize closeness, a very broad notion that is sufficient to define the meaning of convergence of sequences and limits. In other words, a topology is the weakest form of geometry where Calculus can be done. In topology, a *continuous function* is a function between topological spaces which preserve limits.

A **n -manifold** M is a topological space which “locally looks like” n -dimensional Euclidean space $\mathbb{R}^n$, that is, for each point $p \in M$ there exists an open set U containing p , called a **neighborhood**, such that U and $\mathbb{R}^n$ are isomorphic as topological spaces and overlapping neighborhoods transition “well.” A **smooth n -manifold** is a manifold such that each neighborhood is isomorphic to $\mathbb{R}^n$ as a geometry. The isomorphism between U and $\mathbb{R}^n$ is called a **chart**.

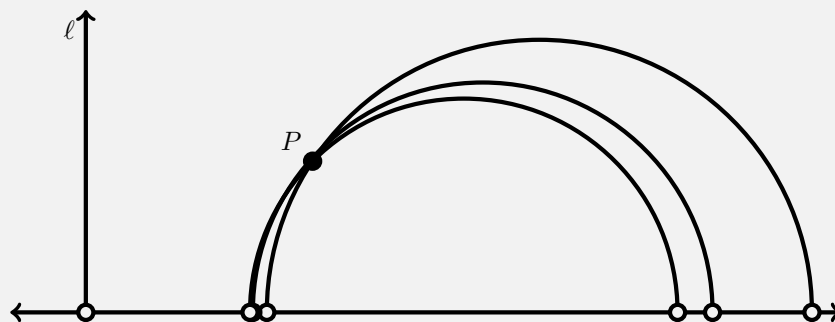
The sphere S^2 is a smooth 2-manifold since locally a sphere resembles the Euclidean plane. Stereographic projection (which interested students are invited to learn more about) is an example of a chart for this manifold. Recall that ancient mankind believed the earth to be flat. This is just the observation that S^2 is a 2-manifold. Therefore, S^2 is locally Euclidean (incidence) geometry but globally elliptic and violates line determination.



The real projective plane is another example of a 2-manifold. In the next example, we look at an example of a hyperbolic 2-manifold.

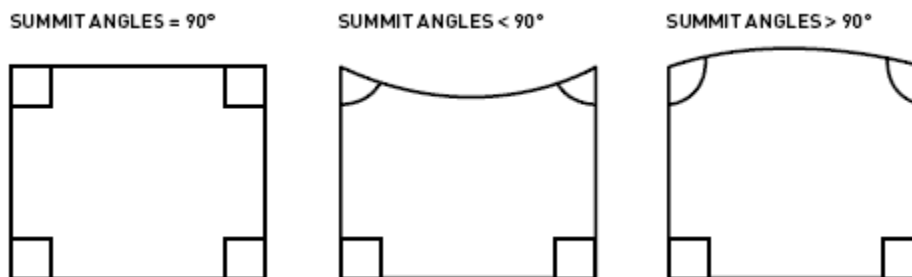
Example 2.4.4. Let $\mathbb{H}^2$ denote the set of points of the upper half-plane in $\mathbb{R}^2$, that is, $\mathbb{H}^2 = \{(x, y) \in \mathbb{R}^2 \mid y > 0\}$. In $\mathbb{H}^2$, lines are defined to be vertical lines and semicircles whose centers rest on the x -axis. The geometry $\mathbb{H}^2$ is often called the **real hyperbolic plane**, and it is a hyperbolic

2-manifold. It is also a model of the incidence axioms which satisfies the hyperbolic parallel postulate, as illustrated below.

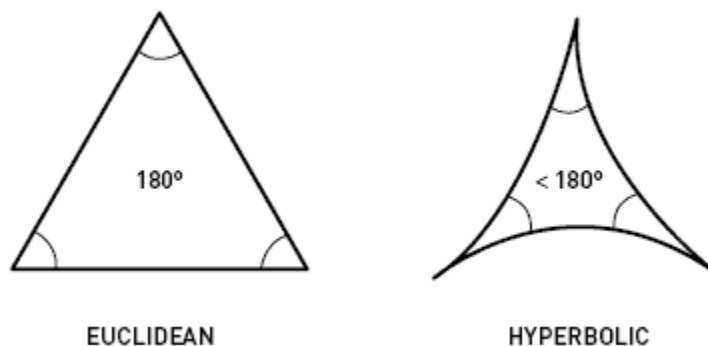


Much like the real projective plane, the real hyperbolic plane has many different models. The one presented above is known as the Poincaré half-plane model.

The existence of parallel lines is not the only consequence of the parallel postulates. For example, the existence of rectangles is equivalent to the Euclidean parallel postulate. For example, the below diagram illustrates what quadrilaterals with two right angles look like in Euclidean, hyperbolic, and elliptic geometry, respectively.



Perhaps this fact is not too surprising since rectangles are just special types of parallelograms. Perhaps a more surprising fact is that in non-Euclidean geometry the angle sum of a triangle does not add up to 180° . In fact, the angle sum equaling 180° is another equivalent statement to the Euclidean parallel postulate. The reader should try to visualize a triangle in spherical geometry (or the real projective plane!) which has three right angles! We will study many other “obvious” geometric facts that seem to fall apart in non-Euclidean geometry in future lectures.



A manifold, while locally flat, generally have some global *curvature*, a notion that can be made explicit with geometric axioms. In relativity theory, physicists believe that gravity, and perhaps other forces, curves space. Thus, space may look like $\mathbb{R}^3$, it is more likely a 3-manifold with curvature, probably hyperbolic. As such, the study of non-Euclidean geometry, especially higher geometry, has many important implications in our understanding of the universe and many other branches of science.

Exercises

1. Which geometries (Euclidean, Elliptic, Hyperbolic) do you think would exhibit each of the following properties?

- (a) Pairs of lines enclose an area.
- (b) Similar triangles are congruent.
- (c) Rectangles exist.
- (d) The sum of measures of the interior angles of a pentagon is less than 540° .
- (e) All lines cross at least once.
- (f) Lines parallel to the same line are parallel to each other.
- (g) Two lines cannot have more than one common perpendicular.
- (h) The sum of the angles of a quadrilateral is less than two flat angles.
- (i) If A , B , and C are points on a line, then one must be between the other two.
- (j) The distance between parallel lines changes if we choose different points on the lines.

2. Is Line Determination valid in the Poincaré half-plane model of hyperbolic geometry? Explain why or why not.

3. Consider the Poincaré disc model of hyperbolic geometry. Sketch a triangle within this model. Does it appear that the triangle has an angle sum greater than, less than, or equal to 180° . Justify your response.

In Euclidean geometry, we often make use of the relation involving three points called *betweenness*.ⁱⁱⁱ

4. If A , B , and C are three collinear points in a hyperbolic plane, do you think you can determine which is between the other two? Explain how or why not.

5. If A , B , and C are three collinear points in an elliptic plane, do you think you can determine which is between the other two? Explain how or why not.

ⁱSeveral undefined terms have been used in the definition of geodesic. A **path** is a continuous map from the unit interval $[0, 1]$ into a topology X . The word “shortest” implies a notion of distance in the geometry. Although distance (or metric) has not yet been axiomatized for geometry in general, we mean here the usual notion of distance in $\mathbb{R}^3$.

ⁱⁱA 3D model of the real projective plane, coded by Alexis Borrell, Sidney Germer, can be viewed at <https://www.geogebra.org/calculator/xpfmhrms>.

ⁱⁱⁱWe will discuss *betweenness* in depth in the next chapter.

Chapter 3

Ordered Geometry

“The difference between stupidity and genius is that genius has its limits.” – Albert Einstein

3.1 The Betweenness Axioms

Let A and B be two points in an incidence geometry. Line determination guarantees that there is a unique line containing A and B . We will denote this line as $\overleftrightarrow{AB}$.

We introduce another undefined term for geometry, that of *between*, which naively expresses the idea of location and is a ternary relation between three distinct points. Let A, B, C be three distinct points in a geometry. We will say that B is **between** A and C , denoted as $A - B - C$. We will also use the notation $A - B - C - D$ to denote

$$A - B - C, \quad A - B - D, \quad A - C - D, \quad B - C - D.$$

Using the notion of betweenness, we can generalize the idea of $\overleftrightarrow{AB}$.

Definition 3.1.1. Let A and B be points in a geometry. Then the **segment** $\overline{AB}$ is defined to be the set of points containing A and B and all point between A and B , that is,

$$\overline{AB} = \{C \mid A - C - B\} \cup \{A, B\}.$$

Likewise, the **ray** $\overrightarrow{AB}$ is defined as the following set of points:

$$\overrightarrow{AB} = \overline{AB} \cup \{C \mid A - B - C\}.$$

Definition 3.1.2. Let A and B be two points not on a line ℓ . We say that A and B are on the **same side** of ℓ if $\overline{AB}$ does not intersect ℓ . We say that A and B are on the **opposite side** of ℓ if $\overline{AB}$ intersects ℓ .

The following four betweenness axioms (or the order axioms) will try to give meaning to the notion of between.

Definition 3.1.3. Betweenness Axioms

Axiom B.1 (Collinearity and Symmetrization): If $A - B - C$, then A, B, C are three distinct collinear points and $C - B - A$.

Axiom B.2 (Extension): Let A and B be two points on a line ℓ . Then there exist a point C on ℓ such that $A - B - C$.

Axiom B.3 (Trichotomy): Let A, B , and C be three distinct collinear points. Then one and only one of the three points is between the other two.

Axiom B.4 (Pasch's Axiom):¹ Let A , B , and C be distinct points. Let ℓ be a line not containing A , B , or C . If ℓ intersects $\overline{AB}$, then ℓ intersects $\overline{AC}$ or $\overline{BC}$ but not both.

We say that incidence geometry which satisfies the four betweenness axioms is an **ordered geometry**.

Intuitively, Pasch's axiom says that if a line goes into a triangle through one side then it must come out through another side. A consequence of extension (Axiom B.2) is that all ordered geometries are infinite.

Example 3.1.4. The real Euclidean plane $\mathbb{R}^2$ and real hyperbolic plane $\mathbb{H}^2$ are ordered geometries with between taking on its usual meaning. The real projective plane $\mathbb{RP}^2$ is not an ordered geometry. If lines are great circles in this geometry, how can between be well defined?

A Cartesian geometry X^2 can an ordered geometry if its arithmetic is likewise ordered. For addition to be ordered, we mean that for all $x, y, z \in X$, if $x < y$ then $x + z < y + z$. For multiplication to be ordered, we mean for all $x, y, z \in X$, if $x < y$ and $z > 0$ then $xz < yz$, and if $x < y$ and $z < 0$ then $xz > yz$. Thus, $\mathbb{Q}^2$ is an ordered Cartesian geometry. But this does have limits. The arithmetic in $\mathbb{Z}$ is ordered, but not dense, meaning there exists $x, y \in \mathbb{Z}$ such that $x < y$ but there exists no $z \in \mathbb{Z}$ such that $x < z < y$. This defect makes $\mathbb{Z}^2$ an incidence geometry that is not ordered.

Theorem 3.1.5. Let A and B be two distinct points in an ordered geometry. Then

$$\overline{AB} = \overrightarrow{AB} \cap \overrightarrow{BA}, \quad \overleftarrow{AB} = \overrightarrow{AB} \cup \overrightarrow{BA}.$$

Proof. By definition, $\overline{AB} \subseteq \overrightarrow{AB}, \overrightarrow{BA}$. So, $\overline{AB} \subseteq \overrightarrow{AB} \cap \overrightarrow{BA}$. Let $P \in \overrightarrow{AB} \cap \overrightarrow{BA}$. Since $P \in \overrightarrow{AB}$, we have $P \in \overline{AB}$ or $A - B - P$. Since $P \in \overrightarrow{BA}$, we have $P \in \overline{BA}$ or $B - A - P$. Of course, by Symmetrization (Axiom B.1) $\overline{BA} = \overline{AB}$. Also, by Trichotomy of Betweenness (Axiom B.3) it does not hold that $A - B - P$ and $B - A - P$. Thus, $P \in \overline{AB}$. Therefore, $\overrightarrow{AB} \cap \overrightarrow{BA} \subseteq \overline{AB}$. The remaining equation is left as an exercise of the reader. $\square$

Theorem 3.1.6 (Plane Separation). Let A , B , and C be distinct points not on line ℓ in an ordered geometry.

- (i) If A and B are on the same side of ℓ and B and C are on the same side of ℓ , then A and C are on the same side of ℓ .
- (ii) If A and B are on opposite sides of ℓ and B and C are on opposite sides of ℓ , then A and C are on the same side of ℓ .
- (iii) If A and B are on opposite sides of ℓ and B and C are on the same side of ℓ , then A and C are on opposite sides of ℓ .

Proof. We first note that if A , B , and C were collinear, then there exists a betweenness relation on them by the Collinearity Axiom. We will begin by proving (i). Suppose for the sake of contradiction that A and C are on opposite sides of ℓ . Then ℓ intersects the segment $\overline{AC}$. By Pasch's Axiom (Axiom B.4), since A , B , and C are not on ℓ , ℓ must intersect either $\overline{AB}$ or $\overline{BC}$. If ℓ intersects $\overline{AB}$, then this would imply that A and B are on opposite sides of ℓ , a contradiction. Similarly, if ℓ intersects $\overline{BC}$ then B and C would be on opposite sides of ℓ , another contradiction. Therefore, A and C are on the same side of ℓ . The remaining two parts are left as an exercise of the reader. $\square$

Definition 3.1.7. Let P be a point and ℓ a line not on P in an ordered geometry. Then we define the **open half plane** $H^\circ(\ell, P)$ to be the set of all points on the same side of ℓ as P . Define the **(closed) half plane** $H(\ell, P) = H^\circ(\ell, P) \cup \ell$.

It can be shown that being on the same side of ℓ is an equivalence relation. Then the equivalence classes of this relation are the open half planes. Such would be hoped, there are only two half planes per line.

Theorem 3.1.8. *Let ℓ be a line in an ordered geometry. Then there are exactly two open half planes for ℓ .*

Proof. Let A be a point not on ℓ . Let P be a point on ℓ . Then by Extension (Axiom B.2) there exists a point B such that $A - P - B$. Then $\overline{AB}$ intersects ℓ at P . This shows that A and B are on opposite side of ℓ . This gives two distinct open half planes $H^\circ(\ell, A)$ and $H^\circ(\ell, B)$. Let C be a point not on ℓ . If C is on the same side as A , then $H^\circ(\ell, C) = H^\circ(\ell, A)$. If C and A are on opposite sides of ℓ , then C and B are on the same side by Plane Separation (ii). Thus, $H^\circ(\ell, C) = H^\circ(\ell, B)$. Therefore, there are exactly two open half planes. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be ordered geometries.

1. Let A and B be two distinct points. Prove $\overleftrightarrow{AB} = \overrightarrow{AB} \cup \overrightarrow{BA}$.
2. Prove that being on the same side of a line is an equivalence relation.
3. Prove Theorem 3.1.6 (ii) and (iii).
4. Prove that the Cartesian geometry $\mathbb{Z}^2$ violates Pasch's axiom.ⁱⁱ

ⁱWe are using a variation of Hilbert's version of Pasch's Axiom, named after Moritz Pasch who in 1882 recognized this was an assumption made by Euclid but could not be proven from Euclid's postulates. This version of Pasch's Axiom is logically equivalent to Hilbert's original version, but it will allow us to develop the theory of betweenness a little bit faster than Hilbert's original approach.

ⁱⁱNote, since $\mathbb{Z}^2$ is an affine geometry which satisfies axioms B.1-B.3, that Pasch's axioms are independent of the axioms of incidence and (other) axioms of betweenness.

“Innovation distinguishes between a leader and a follower.” – Steve Jobs

3.2 Linear Ordering

Theorem 3.2.1 (Linear Density). *Given two distinct points B and D on a line ℓ in an ordered geometry, there exists points A , C , and E lying on ℓ such that $A - B - D$, $B - C - D$, and $B - D - E$. (Progressive Geometry Proofs Slides¹)*

Proof. The existence of point E is immediate from the Extension Axiom. Likewise, the existence of A is also immediate from extension with the point B and D being switched.

Let F be a point not on ℓ and consider the line $m = \overleftrightarrow{BF}$. By Extension, there exists a point G on m such that $B - F - G$. Let $n = \overleftrightarrow{DG}$. By Extension, there exists a point H on n such that $G - D - H$. Finally, consider the line $\overleftrightarrow{FH}$. This line intersects m at the unique point F . Thus, $B, G \notin \overleftrightarrow{FH}$. Also, $\overleftrightarrow{FH}$ intersects n at the unique point H . Thus, $D \notin \overleftrightarrow{FH}$. But as was mentioned earlier, $\overleftrightarrow{FH}$ intersects $\overleftrightarrow{BG}$ at F . Thus, Pasch’s Axiom applies so that $\overleftrightarrow{FH}$ intersects $\overleftrightarrow{BD}$ or $\overleftrightarrow{GD}$. Again, $\overleftrightarrow{FH}$ intersects n at the unique point H , but $G - D - H$. So, $\overleftrightarrow{FH}$ does not intersect $\overleftrightarrow{DG}$. We conclude that $\overleftrightarrow{FH}$ intersects $\overleftrightarrow{BD}$ at a point called C . Therefore, $B - C - D$. $\square$

Theorem 3.2.2 (Linearity Rules). *Let A , B , C , and D be distinct points on a line ℓ in an ordered geometry. Then*

- (i) $A - B - C, A - C - D \implies A - B - C - D$
- (ii) $B - C - D, A - B - D \implies A - B - C - D$
- (iii) $A - B - C, B - C - D \implies A - B - C - D$

Proof. Note that to prove (i), we need to show that $A - B - D$ and $B - C - D$. Let E be a point not on ℓ and consider the line $\overleftrightarrow{EC}$. Then C is the unique intersection between these two lines. Since $A - B - C$, we know that $A - C - B$ does not hold by Trichotomy. Thus, $\overleftrightarrow{AB}$ does not intersect $\overleftrightarrow{EC}$, that is, A and B are on the same side as $\overleftrightarrow{EC}$. Since $A - C - D$, the segment $\overleftrightarrow{AD}$ intersects $\overleftrightarrow{EC}$ at C . Thus, A and D are on opposite sides. By Plane Separation, B and D are on opposite sides of $\overleftrightarrow{EC}$, that is, $\overleftrightarrow{BD}$ intersects $\overleftrightarrow{EC}$. But the unique intersection point is C . Thus, $C \in \overleftrightarrow{BD}$ from which we conclude that $B - C - D$.

We now consider the line $\overleftrightarrow{EB}$ which intersects ℓ at the unique point B . By a similar argument, C and D are on the same side of $\overleftrightarrow{EB}$ by $B - C - D$, A and C are on opposite sides by $A - B - C$, and then A and D are on opposite sides. From this we conclude that $A - B - D$. Therefore, $A - B - C - D$.

The remaining two parts are left as an exercise for the reader. $\square$

Definition 3.2.3. A **strict order** $\prec$ on a set X is a relation $\prec \subseteq X \times X$ such that

- 1) (Irreflexive property) $x \not\prec x$ for all $x \in X$;
- 2) (Transitive property) If $x \prec y$ and $y \prec z$, then $x \prec z$.

For any strict order $\prec$, we define $\succ$ to be the relation $x \succ y$ if and only if $y \prec x$. Additionally, we can define another relationⁱⁱ $\preceq$ on X by the rule: $x \preceq y$ if and only if $x = y$ or $x \prec y$. The relation $\succeq$ is defined analogously.

We say that a strict is **totally ordered** if the additional axiom is satisfied:

- 3) (Totality property) For all $x, y \in X$, either $x \prec y$, $x = y$, or $y \prec x$.

Example 3.2.4. We say that $a \leq b$ for $a, b \in \mathbb{R}$ (or $\mathbb{Q}, \mathbb{Z}, \mathbb{N}$) if $b - a$ is nonnegative. This provides the usual ordering of $\mathbb{R}$ which is a total ordering. In an ordered geometry, we can generalize the ordering of the real line.

Theorem 3.2.5 (Ordering of a Line). *For each line ℓ with two distinct points A and B there exists a unique total order on ℓ such that $A \prec B$ and if $C - D - E$ then either*

$$C \prec D \prec E \quad \text{or} \quad C \succ D \succ E.$$

Proof. We set $A \prec B$. Let $P \in \ell$ other than A or B . Then we say that

- (i) $P \prec A$ and $P \prec B$ if $P-A-B$, (ii) $A \prec P$ and $P \prec B$ if $A-P-B$, (iii) $A \prec P$ and $B \prec P$ if $A-B-P$

Let $P, Q \in \ell$ be distinct points other than A or B . Then we say that $P \prec Q$ if and only if one of the following holds:

- (i) $P-Q-A-B$, (ii) $P-A-Q-B$, (iii) $P-A-B-Q$, (iv) $A-P-Q-B$, (v) $A-P-B-Q$, (vi) $A-B-P-Q$.

As we never defined the relation $P \prec P$, $\prec$ is irreflexive by construction. By Trichotomy and Linearity Rules, all pairs of points are comparable, that is, $\prec$ is total. For transitivity, let $P \prec Q$ and $Q \prec R$. If $\{A, B\} \subseteq \{P, Q, R\}$, then transitivity for these three possibilities follows trivially from the above definitions. Consider next the case that $|\{P, Q, R\} \cap \{A, B\}| = 1$. There are six possibilities here, each of which leads to transitivity by the definition above. For example, if $Q \prec A$ and $Q \prec B$, then $Q - A - B$. The only interpretation of $P \prec Q$ that is consistent to this is $P - Q - A - B$. Thus, $P \prec A$. The remaining cases are proven similarly. If $\{P, Q, R\} \cap \{A, B\} = \emptyset$, then there are ten possible interpretations of $P \prec Q$ and $Q \prec R$. In any case, we can deduce $P \prec R$ from the Linearity Rules. For example, if $P - Q - A - B$ and $Q - R - A - B$, then $P - A - B$ and $R - A - B$. By Linearity Rules, we conclude that $P - R - A - B$. This shows that $P \prec R$. The remaining cases are proven similarly. Therefore, $\prec$ is transitive, which shows that $\prec$ is a total order. $\square$

Definition 3.2.6. Let ℓ be a line in an ordered geometry with order $\preceq$. Let $A, B \in \ell$. Then we define the following four **intervals**:

$$[A, B] = \{C \in \ell \mid A \preceq C \preceq B\}, \quad [A, B) = \{C \in \ell \mid A \preceq C \prec B\},$$

$$(A, B] = \{C \in \ell \mid A \prec C \preceq B\}, \quad (A, B) = \{C \in \ell \mid A \prec C \prec B\}.$$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be ordered geometries.

Definition 3.2.7. Let $\preceq$ be a linear order on a line ℓ . Let $P \in \ell$. Then define the two **intervals**

$$(-\infty, P] = \{Q \in \ell \mid Q \preceq P\}, \quad [P, \infty) = \{Q \in \ell \mid P \preceq Q\},$$

called the two **opposite sides** of P on ℓ .

1. (Line Separation) Let ℓ and m be two distinct lines which intersect at the point P . Let $\preceq$ be a linear order on ℓ . Prove that $(-\infty, P]$ and $[P, \infty)$ belong to different half planes of m .
2. Let A, B , and C be three distinct points such that $A-B-C$. Prove $\overrightarrow{BA} \cap \overrightarrow{BC} = \{B\}$ and $\overrightarrow{BA} \cup \overrightarrow{BC} = \overleftarrow{AC}$.
3. Prove Theorem 3.2.2 (ii) and (iii).
4. The following fill in some of the gaps from Theorem 3.2.5.
 - (a) Show that $P-Q-A-B$ and $A-B-Q-P$ are mutually exclusive. This proves another case for the antisymmetry of $\preceq$.
 - (b) Show that $A-B-P-Q$ and $A-B-Q-R$ implies $A-B-P-R$. This proves another case for the transitivity of $\preceq$.

ⁱⁱThe progressive slides in this section were created by Hayley John.

ⁱⁱOne can define the notion of a **partial order** by the axioms:

- 1) (Reflexive property) $x \preceq x$ for all $x \in X$;
- 2) (Antisymmetric property) if $x \preceq y$ and $y \preceq x$ then implies $x = y$;
- 3) (Transitive property) If $x \preceq y$ and $y \preceq z$, then $x \preceq z$.

These axioms highly resemble the axioms of an equivalence relation. Given any partial order $\preceq$, one can define a new relation $\prec$ on X by the rule: $x \prec y$ if and only if $x \preceq y$ and $x \neq y$. It is then true that $\preceq$ is a partial order if and only if $\prec$ is a strict order.

“The difference between something good and something great is attention to detail.” – Charles R. Swindoll

3.3 Angles

Definition 3.3.1. Let A , B , and C be three distinct non-collinear points in an ordered geometry. Then we define that **angle**, denoted as $\angle ABC$, by the formula

$$\angle ABC = H(\overrightarrow{AB}, C) \cap H(\overrightarrow{BC}, A).$$

The **interior** of an angle is defined as $\angle^\circ ABC = H^\circ(\overrightarrow{AB}, C) \cap H^\circ(\overrightarrow{BC}, A)$.

A closed half plane is called a **flat angle**, denoted $\angle ABC$ where A , B , and C are points on the line and $A - B - C$. We say that a ray is a **null angle**, denoted $\angle ABC$ where $\overrightarrow{BA} = \overrightarrow{BC}$.

Lemma 3.3.2 (Between-Cross Lemma). *Given an angle $\angle ABC$ and a point P on $\overrightarrow{AC}$ in an ordered geometry, P belongs to $\angle^\circ ABC$ if and only if $A - P - C$.*

Proof. Let $P \in \angle^\circ ABC$. By definition, P and A are on the same side of $\overrightarrow{BC}$. This implies that $P - A - C$ or $A - P - C$. Likewise, P and C are on the same side of $\overrightarrow{AB}$, which implies that $P - C - A$ or $C - P - A$. By Trichotomy, we conclude that $A - P - C$. Thus, $P \in \overrightarrow{AC}$.

Let $P \in \overrightarrow{AC}$, that is, $A - P - C$. We note that $\overrightarrow{AC}$ intersects $\overrightarrow{BC}$ at the unique point C . Thus, $\overrightarrow{AP}$ does not intersect $\overrightarrow{BC}$, that is, A and P are on the same side of $\overrightarrow{BC}$. Thus, $P \in H^\circ(\overrightarrow{BC}, A)$. By similar reasoning, we can conclude also that $P \in H^\circ(\overrightarrow{AB}, C)$. Therefore, $P \in \angle^\circ ABC$. $\square$

Proposition 3.3.3. *Let P be a point in $\angle^\circ ABC$ in an ordered geometry. Then*

- (a) *The open ray $\overrightarrow{BP}^\circ := \overrightarrow{BP} \setminus \{B\}$ is contained in $\angle^\circ ABC$.*
- (b) *The opposite ray to $\overrightarrow{BP}$, denoted $-\overrightarrow{BP}$, intersects $\angle ABC$ only at B .*
- (c) *If $C - B - D$, then A belongs to $\angle^\circ PBD$.*

Proof. For (a), let $Q \in \overrightarrow{BP}^\circ$. Then $B - Q - P$, $Q = P$, or $B - P - Q$. If $Q = P$, then we know by assumption that $Q \in \angle^\circ ABC$. In the other two cases, we can conclude that $B \notin \overrightarrow{QP}$, where B is the unique point on $\overrightarrow{QP}$ which intersects $\overrightarrow{AB}$. Thus, Q and P are on the same side of $\overrightarrow{AB}$. So, $Q \in H^\circ(\overrightarrow{AB}, P)$. By similar reasoning, $Q \in H^\circ(\overrightarrow{BC}, P)$. Therefore, $Q \in \angle^\circ ABC$. This proves (a).

For the proof of (b), let P' be a point such that $P - B - P'$. Thus, $\overrightarrow{BP'}$ is the opposite ray to $\overrightarrow{BP}$, that is, $\overrightarrow{BP'} = -\overrightarrow{BP}$. Let $Q \in \angle ABC \cap -\overrightarrow{BP}$. Since $\overrightarrow{QP}$ intersects $\ell = \overrightarrow{BC}$ at B , Q and P are on opposite sides of ℓ . Since P is an interior point to $\angle ABC$, we know that A and P are on the same side of ℓ . By Plane Separation, A and Q are on opposite sides of ℓ . Thus, Q is not an interior point of $\angle ABC$. Thus, Q is on the ray $\overrightarrow{BA}$ or $\overrightarrow{BC}$. Without the loss of generality, we may assume $Q \in \overrightarrow{BA}$. Then $Q \in \overrightarrow{BA} \cap -\overrightarrow{BP} = \{B\}$, that is, $B = Q$. This proves (b).

For the proof of (c), note that A and P are on the same side of $\overrightarrow{BD}$ since $\overrightarrow{BD} = \overrightarrow{BC}$ and P is an interior point of $\angle ABC$. So, $A \in H^\circ(\overrightarrow{BD}, P)$. Of course, if A and D are on the same side of $\overrightarrow{PB}$, then $A \in H^\circ(\overrightarrow{PB}, D)$ from which we can conclude that $A \in \angle^\circ PBD$.

Suppose for the sake of contradiction that A and D are on opposite sides of $\overleftrightarrow{PB}$. Then there exists a point E on the intersection of $\overleftrightarrow{PB}$ and $\overline{AD}$. In particular, $A - E - D$. By the Between-Cross Lemma, $E \in \angle^\circ ABD$. Thus, A and E are on the same side of $\overleftrightarrow{BD} = \overleftrightarrow{BC}$. As A and P are also on the same side of $\overleftrightarrow{BC}$, we can conclude that P and E are on the same side of $\overleftrightarrow{BC}$ by Plane Separation. Thus, B is not between P and E (otherwise, $\overline{PE}$ would intersect $\overleftrightarrow{BC}$ at B). Thus, $E \in \overrightarrow{BP}^\circ$. Then $P \in \overrightarrow{BE}^\circ \subseteq \angle^\circ ABD$ by part one of this proposition. Hence, P and D are on the same side of $\overleftrightarrow{AB}$. Also, P and C are on the same side of $\overleftrightarrow{AB}$ since $P \in \angle^\circ ABC$. We conclude, by Plane Separation, that C and D are on the same side of $\overleftrightarrow{AB}$, which is a contradiction since $\overline{CD}$ intersects $\overleftrightarrow{AB}$ at B . $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be ordered geometries.

Definition 3.3.4. We say that a set $\mathcal{S}$ of points in an ordered geometry is **convex** if $A, B \in \mathcal{S}$ implies $\overline{AB} \subseteq \mathcal{S}$.

1. Prove that a half plane (open or closed) is convex.
2. Prove that the intersection of two convex sets is convex.
3. Prove that lines, rays, and segments are convex.

We can express lines, rays, and segments as intersections of known convex sets. For example, let A, B be two points on opposite sides of a line ℓ . Then $\ell = H(\ell, A) \cap H(\ell, B)$.

4. Prove that $\angle ABC$ and $\angle^\circ ABC$ are convex for any non-collinear points A, B , and C .

“You can’t cross the sea merely by standing and staring at the water.” – Rabindranath Tagore

3.4 The Crossbar Theorem

Definition 3.4.1. A ray $\overrightarrow{BC}$ in an ordered geometry is **between** two non-opposite rays $\overrightarrow{BA}$ and $\overrightarrow{BD}$ if C is in the interior of $\angle ABD$ (which is independent of the choice of C). We will denote this as $\overrightarrow{BA} - \overrightarrow{BC} - \overrightarrow{BD}$.

Theorem 3.4.2 (Crossbar Theorem). *If a ray $\overrightarrow{BP}$ in an ordered geometry is between two rays $\overrightarrow{BA}$ and $\overrightarrow{BC}$, then $\overrightarrow{BP}$ intersects $\overline{AC}$ at D between A and C . (Progressive Geometry Proof Slides)*

Proof. By Extension, there exists a point C' such that $C - B - C'$. By construction, C and C' are on opposite sides of $\ell = \overleftrightarrow{BP}$. By the previous proposition, $A \in \angle PBC'$. Therefore, A and C' are on the same side of ℓ . By Plane Separation, we conclude that A and C are on opposite sides of ℓ . Thus $\overline{AC}$ intersects ℓ at a point D . Since A , B , and C are non-collinear, we know that B is not between A and C . In particular, $B \neq D$. The opposite ray to $\overrightarrow{BP}$ only intersects $\angle ABC$ at B , which means that it does not intersect D . Since ℓ is the union of $\overrightarrow{BP}$ and its opposite ray, we conclude that $\overrightarrow{BP}$ intersects $\overline{AC}$ at D . $\square$

Corollary 3.4.3 (Correspondence of Betweenness). *Let $\overrightarrow{BA}$, $\overrightarrow{BC}$, and $\overrightarrow{BD}$ be rays in an ordered geometry. Then $\overrightarrow{BA} - \overrightarrow{BC} - \overrightarrow{BD}$ if and only if there exists points $A' \in \overrightarrow{BA}$, $C' \in \overrightarrow{BC}$, and $D' \in \overrightarrow{BD}$ such that $A' - C' - D'$.*

Proof. Necessity ($\Rightarrow$) is an immediate consequence of the Crossbar Theorem. Sufficiency ($\Leftarrow$) is an immediate consequence of the Between-Cross Lemma. $\square$

Because of the correspondence of betweenness of points and rays, betweenness of rays inherits many of the same properties of betweenness of points, for example symmetrization, extension, density, and linearity rules. It is also true that trichotomy holds for three concurrent rays, well a weaker version of trichotomy holds, and its proof is not an immediate consequence of the above corollary.

Theorem 3.4.4 (Trichotomy of Rays). *Let ℓ be a line containing the point B in an ordered geometry. Let $\overrightarrow{BA}$, $\overrightarrow{BC}$, and $\overrightarrow{BD}$ be rays all contained in the same half-plane of ℓ . Then one and only one of the three rays is between the other two.*

Proof. First, suppose that A and D are on the opposite sides of $\overleftrightarrow{BC}$. Then $\overline{AD}$ intersects $\overleftrightarrow{BC}$ at some point C' . In particular, $A - C' - D$. By way of contradiction, suppose that $C' \in -\overrightarrow{BC}$. Then $C' \notin H(\ell, C)$ since $\overrightarrow{BC}$ and $-\overrightarrow{BC}$ are in opposite sides of ℓ . By Plane Separation, A and C' are on opposite sides of ℓ . Let $P \in \overline{AC'} \cap \ell$. Likewise, D and C' are on opposite sides of ℓ , and let $Q \in \overline{DC'} \cap \ell$. Thus, $P, Q \in \overline{AD} \cap \ell$, but $P \neq Q$ since $P - C' - Q$ (a conclusion of the linearity rules using $A - C' - D$, $A - P - C'$, and $C' - Q - D$). This is a contradiction, which implies $C' \in \overrightarrow{BC}$. Thus, $\overrightarrow{BA} - \overrightarrow{BC} - \overrightarrow{BD}$, by betweenness correspondence.

Of course, if A and C are on the opposite sides of $\overleftrightarrow{BD}$, then we will get a similar betweenness statement. Thus, we will next consider the case that A and D are on the same side of $\overleftrightarrow{BC}$ and A and C are on the same side of $\overleftrightarrow{BD}$. Then $A \in H^\circ(\overrightarrow{BC}, D) \cap H^\circ(\overrightarrow{BD}, C) = \angle CBD$. Therefore, $\overrightarrow{BC} - \overrightarrow{BA} - \overrightarrow{BD}$.

The uniqueness of the betweenness statements comes from the correspondence between betweenness of points and rays, since betweenness of points has trichotomy. $\square$

Definition 3.4.5. Let A , B , and C be three non-collinear points in an ordered geometry. Then we define the **triangle** $\triangle ABC$ by the rule:

$$\triangle ABC = \angle ABC \cap \angle BCA \cap \angle CAB.$$

The **interior** of the triangle $\triangle^\circ ABC$ is defined by the rule:

$$\triangle^\circ ABC = \angle^\circ ABC \cap \angle^\circ BCA \cap \angle^\circ CAB.$$

Finally, we define the **boundary** of the triangle $\partial\triangle ABC$ by the rule:

$$\partial\triangle ABC = \overline{AB} \cup \overline{BC} \cup \overline{CA}.$$

Proposition 3.4.6. *Given a triangle $\triangle ABC$ and $P \in \triangle^\circ ABC$ in an ordered geometry. Let $\ell = \overleftrightarrow{AB}$, $m = \overleftrightarrow{AC}$, and $n = \overleftrightarrow{BC}$. Then*

- (a) $\triangle ABC = H(\ell, P) \cap H(m, P) \cap H(n, P)$.
- (b) Any ray $\overrightarrow{PQ}$ meets $\partial\triangle ABC$ at a unique point.

Proof. It holds that $\angle ABC = H(\ell, C) \cap H(m, A) = H(\ell, P) \cap H(m, P)$ since P and A are on the same side of ℓ and P and C are on the same side of m . Similar statements hold for the other two angles determining $\triangle ABC$. The first result then follows.

The second part is an exercise for the reader. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be ordered geometries.

1. Prove Proposition 3.4.6 (b).

Construct the ray $\overrightarrow{AP}$ and apply the Crossbar Theorem to find some point D . Then consider two cases: (i) $\overrightarrow{AP} = \overrightarrow{PQ}$ and (ii) $\overrightarrow{AP} \neq \overrightarrow{PQ}$. In the second case, consider the two triangles $\triangle ADB$ and $\triangle ADC$ and apply Pasch's Axiom.

ⁱThe progressive slides in this section were created by Hayley John.

Chapter 4

Congruence Geometry

“All animals are [congruent], but some animals are more [congruent] than others.” – George Orwell

4.1 The Congruence Axioms of Segments

We introduce another undefined term this time for line segments. We say that two line segments $\overline{AB}$ and $\overline{CD}$ are **congruent** and denote this as $\overline{AB} \cong \overline{CD}$.

Below we include the first three of Hilbert’s six congruence axioms to give meaning to this undefined term.

Definition 4.1.1. Congruence Axioms

Axiom C.1 (Segment Translation): Given two distinct points A and B and a ray r emanating from the point A' , then there exists a unique point B' on r such that $\overline{AB} \cong \overline{A'B'}$. In particular, on the ray $r = \overrightarrow{AB}$, that unique point is $B' = B$.

Axiom C.2 (Transitivity of Segment Congruence): If $\overline{AB} \cong \overline{A'B'}$ and $\overline{A'B'} \cong \overline{A''B''}$, then $\overline{AB} \cong \overline{A''B''}$.

Axiom C.3 (Segment Addition): Let A, B , and C be points on a line ℓ such that $A - B - C$ and let A', B' , and C' be points on a line ℓ' such that $A' - B' - C'$. If $\overline{AB} \cong \overline{A'B'}$ and $\overline{BC} \cong \overline{B'C'}$, then $\overline{AC} \cong \overline{A'C'}$.

Axioms C.4–C.6 all relate to congruence of angles and will be listed in Definition 4.2.2.

An ordered geometry that satisfies the six congruence axioms is called a **congruence geometry**.

Proposition 4.1.2. *Let A, B, C, D be points in a congruence geometry. If $\overline{AB} \cong \overline{CD}$, then $\overline{CD} \cong \overline{AB}$.*

Proof. By Segment Translation (Axiom C.1), there exists a unique point B' on the ray $r = \overrightarrow{AB}$ such that $\overline{CD} \cong \overline{AB'}$. Then $\overline{AB} \cong \overline{AB'}$ by the transitivity of congruence (Axiom C.2). Again using the uniqueness of the point provided by Segment Translation, we have that $B' = B$. Thus, $\overline{CD} \cong \overline{AB}$. $\square$

The previous proposition in combination of Axioms C.1 and C.2 shows that congruence of line segments is an equivalence relation.

Proposition 4.1.3 (Betweenness Preservation). *Given $\overline{AC} \cong \overline{A'C'}$ and $A - B - C$, then there exists a unique point B' between A' and C' such that $\overline{AB} \cong \overline{A'B'}$ and $\overline{BC} \cong \overline{B'C'}$.*

Proof. We begin by using Segment Translation (Axiom C.1) to create a unique point B' on the ray $\overrightarrow{A'C'}$ such that $\overline{AB} \cong \overline{A'B'}$. Again, there is a unique point P such that $\overline{BC} \cong \overline{B'P}$ and $A' - B' - P$ by Segment Translation. Since $\overline{AB} \cong \overline{A'B'}$ and $\overline{BC} \cong \overline{B'P}$, we conclude that $\overline{AC} \cong \overline{A'P}$ by Segment Addition (Axiom C.3). By Transitivity of Congruence (Axiom C.2), we also have that $\overline{A'C'} \cong \overline{A'P}$. The uniqueness condition from Segment Translation guarantees that $C' = P$. Therefore, $\overline{AB} \cong \overline{A'B'}$, $\overline{BC} \cong \overline{B'C'}$, and $A' - B' - C'$. $\square$

Definition 4.1.4. For segments $\overline{AB}$ and $\overline{CD}$ if there exists a point E such that $C - E - D$ and $\overline{AB} \cong \overline{CE}$, then we write $\overline{AB} < \overline{CD}$. The relation $\overline{AB} \leq \overline{CD}$ means that either $\overline{AB} \cong \overline{CD}$ or $\overline{AB} < \overline{CD}$.

It is an exercise of the reader to prove that $<$ provides a total¹ order on the set of segments in an congruence geometry.

Theorem 4.1.5 (Congruence of Lines). *For any two lines ℓ and ℓ' , there exists a one-to-one correspondence $f : \ell \rightarrow \ell'$ such that $\overline{AB} \cong \overline{f(A)f(B)}$ for distinct points $A, B \in \ell$ and if $A - B - C$ then $f(A) - f(B) - f(C)$.*

Proof. Place an ordering on each of the lines and let $P \in \ell$ and $P' \in \ell'$. Assign $f(P) = P'$. For each $Q \in [P, \infty)$, there is a unique point $Q' \in [P', \infty)$ such that $\overline{PQ} \cong \overline{P'Q'}$ by Segment Translation. Assign $f(Q) = Q'$. Similarly, for each $R \in (-\infty, P]$ there is a unique point $R' \in (-\infty, P']$ such that $\overline{PR} \cong \overline{P'R'}$ by Segment Translation. Assign $f(R) = R'$. This defines a function $f : \ell \rightarrow \ell'$. Define $f' : \ell' \rightarrow \ell$ analogously such that $f'(P') = P$ and the other assignments are given by Segment Translation. Then we have $f' \circ f : \ell \rightarrow \ell$ is the identity mapping by the uniqueness of Segment Translation. Therefore, f is a bijective function.

Let $A, B \in \ell$ be distinct points. Note that $\overline{AP} \cong \overline{A'P'} = \overline{f(A)f(P)}$ and $\overline{PB} \cong \overline{P'B'} = \overline{f(P)f(B)}$ by construction of f . If $A - P - B$, then $\overline{AB} \cong \overline{A'B'} = \overline{f(A)f(B)}$ by Segment Addition (Axiom C.3). If $A - B - P$, then $\overline{AB} \cong \overline{A'B'} = \overline{f(A)f(B)}$ by Segment Subtraction (an exercise for the reader). Therefore, f preserves congruence of segments.

Let $A - B - C$ on ℓ . Then there exists a unique point B'' on ℓ' between $A' = f(A)$ and $C' = f(C)$ such that $\overline{AB} \cong \overline{A'B''}$ and $\overline{BC} \cong \overline{B''C'}$ by Betweenness Preservation. Since $\overline{AB} \cong \overline{f(A)f(B)} = \overline{A'B'}$ as shown above, the uniqueness of Segment Translation shows that $B' = B''$. Hence, $f(A) - f(B) - f(C)$. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be congruence geometries.

1. (Segment Subtraction) Let A , B , and C be points on a line ℓ such that $A - B - C$ and let A' , B' , and C' be points on a line ℓ' such that $A' - B' - C'$. Prove if $\overline{AB} \cong \overline{A'B'}$ and $\overline{AC} \cong \overline{A'C'}$, then $\overline{BC} \cong \overline{B'C'}$.
2. (Linear Order of Segments) Prove that $<$ forms a total strict order on the set of line segments.

ⁱFor the totality axiom, if $\overline{AB} \not\cong \overline{CD}$ then either $\overline{AB} < \overline{CD}$ or $\overline{AB} > \overline{CD}$.

“All mankind... being all [congruent] and independent, no one ought to harm another in his life, health, liberty or possessions.” – John Locke

4.2 The Congruence Axioms of Angles

We introduce a second meaning of congruence, this time with angles. We may say that two angles $\angle ABC$ and $\angle DEF$ are **congruent** and denote this as $\angle ABC \cong \angle DEF$.

Definition 4.2.1. Let $\triangle ABC$ and $\triangle A'B'C'$ be two triangles. We say $\triangle ABC$ is **congruent** to $\triangle A'B'C'$, denoted $\triangle ABC \cong \triangle A'B'C'$, if $\overline{AB} \cong \overline{A'B'}$, $\overline{BC} \cong \overline{B'C'}$, $\overline{AC} \cong \overline{A'C'}$, $\angle ABC \cong \angle A'B'C'$, $\angle BCA \cong \angle B'C'A'$, and $\angle CAB \cong \angle C'A'B'$.

Definition 4.2.2. Congruence Axioms

Axiom C.4 (Angle Translation): Given an angle $\angle ABC$ and a ray $\overrightarrow{B'A'}$, then there exists a unique ray $\overrightarrow{B'C'}$ on each side of $\overrightarrow{A'B'}$ such that $\angle ABC \cong \angle A'B'C'$. In particular, on the ray $\overrightarrow{BA}$ and side containing C , that unique ray is $\overrightarrow{BC}$.

Axiom C.5 (Transitivity of Angle Congruence): If $\angle ABC \cong \angle A'B'C'$ and $\angle A'B'C' \cong \angle A''B''C''$, then $\angle ABC \cong \angle A''B''C''$.

Axiom C.6 (Side-Angle-Side): Let A, B, C, A', B' , and C' be points. Suppose $\overline{AB} \cong \overline{A'B'}$, $\angle ABC \cong \angle A'B'C'$, and $\overline{BC} \cong \overline{B'C'}$. Then $\triangle ABC \cong \triangle A'B'C'$.

Axiom C.4 provides that angle congruence is reflexive and Axiom C.5 provides that angle congruence is transitive. Angle congruence is also symmetric (an exercise left for the reader). This means that angle congruence is also an equivalence relation, just like segment congruence.

Definition 4.2.3. Let A, B, C , and D be points in an ordered geometry. We say that angles $\angle ABC$ and $\angle CBD$ are **supplementary** if $A - B - D$. We also say the two angles are **supplements** of each other.

Intuitively, supplementary angles are two angles which when combined are a flat angle, a line.

We note that the supplement of an angle $\angle ABC$ always exists. Let D be a point such that $A - B - D$, by the Extension Axiom. Then $\angle CBD$ is the supplement of $\angle ABC$.

Proposition 4.2.4. *Supplements of congruent angles are congruent.*

Proof. Let $\angle ABC \cong \angle A'B'C'$ be two congruent angles with supplements $\angle CBD$ and $\angle C'B'D'$. Since these angles are supplements, we know that $A - B - D$ and $A' - B' - D'$. By Segment Translation, we may assume that the points A', C' , and D' can be adjusted so that $\overline{AB} \cong \overline{A'B'}$, $\overline{BC} \cong \overline{B'C'}$, and $\overline{BD} \cong \overline{B'D'}$.

Since $\overline{AB} \cong \overline{A'B'}$, $\angle ABC \cong \angle A'B'C'$, and $\overline{BC} \cong \overline{B'C'}$, we conclude by SAS that $\triangle ABC \cong \triangle A'B'C'$. From this, we conclude that $\overline{AC} \cong \overline{A'C'}$ and $\angle BAC \cong \angle B'A'C'$.

By Segment Addition, we have $\overline{AD} \cong \overline{A'D'}$ since $\overline{AB} \cong \overline{A'B'}$, $\overline{BD} \cong \overline{B'D'}$, $A - B - D$, and $A' - B' - D'$. Using SAS again, we conclude that $\triangle ACD \cong \triangle A'C'D'$ since $\overline{AD} \cong \overline{A'D'}$, $\angle DAC = \angle BAC \cong \angle B'A'C' = \angle D'A'C'$, and $\overline{AC} \cong \overline{A'C'}$. Since $\triangle ACD \cong \triangle A'C'D'$, we can conclude that $\overline{DC} \cong \overline{D'C'}$ and $\angle ADC \cong \angle A'D'C'$.

Finally, we have one final application of SAS. This time we can conclude by SAS that $\triangle BCD \cong \triangle B'C'D'$

since $\overline{BD} \cong \overline{B'D'}$, $\angle BDC = \angle ADC \cong \angle A'D'C' = \angle B'D'C'$, and $\overline{DC} \cong \overline{D'C'}$. Because of the congruence of triangles, we conclude that $\angle CBD \cong \angle C'B'D'$, which proves the result. $\square$

Definition 4.2.5. We say that an angle is **right** if it is congruent to its supplement.

Proposition 4.2.6. Any angle congruent to a right angle is itself a right angle.

Proof. Suppose that $\angle ABC \cong \angle A'B'C'$ and $\angle ABC$ is a right angle. We need to show that $\angle A'B'C'$ is right. Let D be a point on the opposite ray to $\overrightarrow{BA}$ so that $A - B - D$ and $\angle CBD$ is a supplement of $\angle ABC$. Since $\angle ABC$ is right, we know that $\angle ABC \cong \angle CBD$. By Transitivity of Angle Congruence, we know that $\angle A'B'C' \cong \angle CBD$. Similarly, choose the point D' such that $\angle C'B'D'$ is a supplement of $\angle A'B'C'$. Since supplements of congruent angles are congruent, we know that $\angle CBD \cong \angle C'B'D'$. By Transitivity of Angle Congruence, we conclude $\angle A'B'C' \cong \angle C'B'D'$. Therefore, $\angle A'B'C'$ is right. $\square$

Definition 4.2.7. Let ℓ and m be two lines of a congruence geometry which intersect at a unique point B . We say ℓ is **perpendicular** to m , denoted $\ell \perp m$ if the angle $\angle ABC$ is right where A is any point on ℓ not equal to B and C is any point on m not equal to B .

Note that if $\ell \perp m$ then $m \perp \ell$ follows from the definition.

Theorem 4.2.8 (Dropping a Perpendicular to a Line). For each line ℓ and each point P not on ℓ , there exists a line m which contains P and $m \perp \ell$.

Proof. Let $A, B \in \ell$ such that $A \neq B$. By Angle Translation, there exists a point C on the opposite side of ℓ from P such that $\angle BAP \cong \angle BAC$. By Segment Translation, there exists a unique point on the ray $\overrightarrow{AC}$, call it P' , such that $\overline{AP} \cong \overline{AP'}$. Let $m = \overleftrightarrow{PP'}$. We claim that $m \perp \ell$. If A, P , and P' are collinear, then we can conclude that $P - A - P'$ since P and P' are on opposite sides of ℓ by the Plane Separation Theorem. Thus, the angles $\angle BAP$ and $\angle BAP'$ are supplementary angles and by construction $\angle BAP \cong \angle BAP'$. Therefore, $\angle BAP$ is a right angle and $m \perp \ell$.

If A, P , and P' are not collinear, let Q be the point of intersection with m and ℓ . We note that $P - Q - P'$. Since $Q \in \ell$ by construction, then we know that A, B , and Q are all collinear. This implies that either $\angle QAP = \angle BAP$ or they are supplementary angles. In either case, since $\angle BAP \cong \angle BAP'$, we can conclude that $\angle QAP \cong \angle QAP'$. Since $\overline{AQ} \cong \overline{AQ}$, $\angle QAP \cong \angle QAP'$, and $\overline{AP} \cong \overline{AP'}$, SAS applies and we have $\triangle QAP \cong \triangle QAP'$. In particular, $\angle AQP \cong \angle AQP'$, which are supplementary angles. Therefore, $\angle AQP$ is right and $m \perp \ell$. $\square$

The point of intersection between a line and the perpendicular dropped onto it is called a **foot**.

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be congruence geometries.

1. Prove if $\angle ABC \cong \angle DEF$ then $\angle DEF \cong \angle ABC$.
2. (Triangle Translation) Given a triangle $\triangle ABC$ and a segment $\overline{A'B'}$ such that $\overline{AB} \cong \overline{A'B'}$, prove there exists a unique point C' for each side of the line $\overleftrightarrow{A'B'}$ such that $\triangle ABC \cong \triangle A'B'C'$.
3. (Isosceles Triangle Theorem) Given a triangle $\triangle ABC$ such that $\overline{AB} \cong \overline{AC}$, prove that $\angle ACB \cong \angle ABC$.

Definition 4.2.9. Let ℓ and m be two lines which intersect at a unique point B . Let A and D be points on ℓ on opposite sides of m . Let C and E be points on m on opposite sides of ℓ . Then we call the angles $\angle ABC$ and $\angle DBE$ **vertical angles**. Intuitively, vertical angles are on opposite sides of the common point B .

4. (Vertical Angles) Prove that vertical angles are congruent.

“Adopting the right attitude can convert a negative stress into a positive one.” – Hans Selye

4.3 Euclid's Fourth Postulate

Proposition 4.3.1 (Angle Addition). *Let $\overrightarrow{BA}$, $\overrightarrow{BC}$, and $\overrightarrow{BD}$ be concurrent rays such that $\overrightarrow{BA} - \overrightarrow{BC} - \overrightarrow{BD}$ and let $\overrightarrow{B'A'}$, $\overrightarrow{B'C'}$, and $\overrightarrow{B'D'}$ be concurrent rays such that $\overrightarrow{B'A'} - \overrightarrow{B'C'} - \overrightarrow{B'D'}$. If $\angle ABC \cong \angle A'B'C'$ and $\angle CBD \cong \angle C'B'D'$, then $\angle ABD \cong \angle A'B'D'$.*

Proof. Because of Segment Translation, we may assume that $\overline{BA} \cong \overline{B'A'}$, $\overline{BC} \cong \overline{B'C'}$, and $\overline{BD} \cong \overline{B'D'}$. By the Crossbar Theorem, we may also assume that $A - C - D$. Be aware we are not assuming that $A' - C' - D'$. Therefore, $\triangle ABC \cong \triangle A'B'C'$ and $\triangle CBD \cong \triangle C'B'D'$ by SAS. This implies that $\angle ACB \cong \angle A'C'B'$ and $\angle DCB \cong \angle D'C'B'$. It also implies that $\overline{AC} \cong \overline{A'C'}$, $\overline{CD} \cong \overline{C'D'}$, and $\angle BAC \cong \angle B'A'C'$.

We note that since $A - C - D$ that $\angle ACB$ and $\angle DCB$ are supplements. Let D'' be a point such that $A' - C' - D''$. Then $\angle D''C'B'$ is a supplement to $\angle A'C'B'$. Since supplement of congruent angles are congruent, we can conclude that $\angle DCB \cong \angle D''C'B'$. By Angle Transitivity of Congruence (Axiom C.5), we conclude that $\angle D'C'B' \cong \angle D''C'B'$. Since D' and D'' are on the same side of $\overrightarrow{B'C'}$, the uniqueness of Angle Translation gives that $\overline{C'D'} = \overline{C'D''}$. Thus, A' , C' , D' , and D'' are all collinear with $A' - C' - D'$. By Segment Addition, we then have that $\overline{AD} \cong \overline{A'D'}$. Finally, since $\overline{AB} \cong \overline{A'B'}$, $\angle BAC \cong \angle B'A'C'$, and $\overline{AD} \cong \overline{A'D'}$, we have that $\triangle ABD \cong \triangle A'B'D'$ by SAS. In particular, $\angle ABD \cong \angle A'B'D'$. $\square$

Proposition 4.3.2 (Betweenness Preservation). *Given $\angle ABD \cong \angle A'B'D'$ and $\overrightarrow{BA} - \overrightarrow{BC} - \overrightarrow{BD}$, then there exists a unique ray $\overrightarrow{B'C'}$ between $\overrightarrow{B'A'}$ and $\overrightarrow{B'D'}$ such that $\angle ABC \cong \angle A'B'C'$ and $\angle CBD \cong \angle C'B'D'$.*

Proof. We begin by using Angle Translation (Axiom C.4) to create a unique ray $\overrightarrow{B'C'}$ in the angle interior $\angle A'B'D'$ such that $\angle ABC \cong \angle A'B'C'$. Again, there is a unique ray $\overrightarrow{B'P}$ such that $\angle CBD \cong \angle C'B'P$ and $\overrightarrow{B'A'} - \overrightarrow{B'C'} - \overrightarrow{B'P}$ by Angle Translation. Since $\angle ABC \cong \angle A'B'C'$ and $\angle CBD \cong \angle C'B'P$, we conclude that $\angle ABD \cong \angle A'B'P$ by Angle Addition. By Transitivity of Congruence, we also have that $\angle A'B'D' \cong \angle A'B'P$. The uniqueness condition from Angle Translation guarantees that $\overrightarrow{B'D'} = \overrightarrow{B'P}$. Therefore, $\angle ABC \cong \angle A'B'C'$, $\angle CBD \cong \angle C'B'D'$, and $\overrightarrow{B'A'} - \overrightarrow{B'C'} - \overrightarrow{B'D'}$. $\square$

Because of we can “add” angles and angle congruence forms an equivalence relation, we see that Betweenness Preservation of Angles can be proven using the same proof as Betweenness Preservation of Segments. In the homework, the reader will see that other properties of congruence hold for angles as they do for segments because of the same proofs. This is the Axiomatic Method at work!

Definition 4.3.3. For angles $\angle ABC$ and $\angle DEF$ if there exists a ray $\overrightarrow{EP}$ such that $P \in \angle DEF$ and $\angle ABC \cong \angle DEP$, then we write $\angle ABC < \angle DEF$. The relation $\angle ABC \leq \angle DEF$ means that either $\angle ABC \cong \angle DEF$ or $\angle ABC < \angle DEF$.

It is an exercise of the reader to prove that $<$ provides a total¹ order on the set of angles in an congruence geometry.

Theorem 4.3.4 (Euclid's Fourth Postulate). *All right angles are congruent to each other.*

Proof. Let $\angle ABC$ and $\angle A'B'C'$ be right angles, that is, there exists points D and D' such that $A-B-D$ and $A'-B'-D'$ such that $\angle ABC \cong \angle CBD$ and $\angle A'B'C' \cong \angle C'B'D'$. We need to show that $\angle ABC \cong \angle A'B'C'$. By Angle Translation, let P be a point in $H^\circ(\overleftrightarrow{A'D'}, C')$ such that $\angle ABC \cong \angle A'B'P$. Since $\angle CBD$ and $\angle PB'D'$ are supplements to congruent angles, we have that $\angle CBD \cong \angle PB'D'$.

Now, P is on one side of $\overleftrightarrow{B'C'}$. Without the loss of generality, assume that $P \in H^\circ(\overleftrightarrow{B'C'}, A')$, that is, $P \in \angle^\circ A'B'C'$. Then $\overrightarrow{B'A'} - \overrightarrow{B'P} - \overrightarrow{B'C'}$, which implies that $\angle A'B'P \leq \angle A'B'C'$. On the other hand, since A' and C' are on the opposite side of $\overleftrightarrow{B'P}$ by the Crossbar Theorem and A' and D' are on opposite sides of $\overleftrightarrow{B'P}$ since $A'-B'-D'$, we conclude by Plane Separation that C' and D' are on the same side of $\overleftrightarrow{B'P}$. This, of course, gives us that $\overrightarrow{B'P} - \overrightarrow{B'C'} - \overrightarrow{B'D'}$. Thus, $\angle A'B'P \geq \angle C'B'D'$. Therefore, we have the following chain of relations:

$$\angle ABC \cong \angle A'B'P \leq \angle A'B'C' \cong \angle C'B'D' \leq \angle PB'D' \cong \angle CBD \cong \angle ABC.$$

Since the above chain shows that $\angle ABC \leq \angle ABC$ but, in fact, $\angle ABC \cong \angle ABC$, we must conclude that all of the $\leq$'s in the above chain are actually $\cong$'s. In particular, we have that $\angle ABC \cong \angle A'B'C'$ by transitivity of congruence. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be congruence geometries.

1. (Angle Subtraction) Let ray $\overrightarrow{BC}$ be between rays $\overrightarrow{BA}$ and $\overrightarrow{BD}$, and let ray $\overrightarrow{B'C'}$ be between rays $\overrightarrow{B'A'}$ and $\overrightarrow{B'D'}$. Prove if $\angle ABC \cong \angle A'B'C'$ and $\angle ABD \cong \angle A'B'D'$, then $\angle CBD \cong \angle C'B'D'$.

Definition 4.3.5. We say that angles $\angle ABC$ and $\angle CBD$ are **complementary** if $\angle ABD$ is a right angle and $\overrightarrow{BA} - \overrightarrow{BC} = \overrightarrow{BD}$. We also say the two angles are **complements** of each other.

2. (Complementary Angles) Prove complements of congruent angles are congruent.
3. (Total Order of Angles) Prove that $<$ forms a total strict order on the set of angles.

ⁱFor the totality axiom, if $\angle ABC \not\cong \angle DEF$ then either $\angle ABC < \angle DEF$ or $\angle ABC > \angle DEF$.

“Many individuals have, like uncut diamonds, shining qualities beneath a rough exterior.” – Juvenal

4.4 Interior and Exterior Angles

Definition 4.4.1. Let ℓ and m be two distinct lines transversed by a third line t in an ordered geometry. Let $P \in \ell \cap t$ and $Q \in m \cap t$. Let $A \in \ell$ and $B \in m$ such that A and B are on the same side of t . Let $A - P - A'$ and $B - Q - B'$. The four angles $\angle APQ$, $\angle A'PQ$, $\angle BQP$, and $\angle B'QP$ are called **interior angles** to the transversal t . The pairs $\angle APQ$, $\angle BQP$ and $\angle A'PQ$, $\angle B'QP$ are each called **consecutive interior angles**. The pairs $\angle APQ$, $\angle B'QP$ and $\angle A'PQ$, $\angle BQP$ are each called **alternate interior angles**.

Theorem 4.4.2 (Alternate Interior Angle Theorem). *If two distinct lines cut by a transversal have a pair of congruent alternate interior angles, then the two lines are parallel. (Progressive Geometry Proofs Slides)*

Proof. Use the same notion as introduced in the previous definition. Thus, we will assume that $\angle APQ \cong \angle B'QP$. Suppose for the sake of contradiction that $\ell \nparallel m$. Let $R \in \ell \cap m$. Without the loss of generality, we may assume that R is on the same side of t as A and B . By segment translation, we let R' be a point on $\overrightarrow{QB'}$ such that $\overline{QR'} \cong \overline{PR}$. Then $\triangle PQR \cong \triangle QPR'$ by SAS. Hence, $\angle PQB = \angle PQR \cong \angle QPR'$.

Since $\angle A'PQ$ is supplementary to $\angle APQ$ and $\angle B'QP$ is supplementary to $\angle BQP$, we conclude that $\angle A'PQ \cong \angle BQP$, since supplements of congruent angles are congruent. By transitivity of congruence, we have that $\angle A'PQ \cong \angle BQP \cong \angle QPR'$. By uniqueness of angle translation, we conclude that $\angle A'PQ = \angle R'PQ$, in particular, $\overline{PA'} = \overline{PR'}$. This shows that $R' \in \ell$. Hence, $R, R' \in \ell \cap m$. But $R \neq R'$ since they are on opposite sides of t . This contradiction line determination. Therefore, $\ell \parallel m$. $\square$

Theorem 4.4.3 (Consecutive Interior Angle Theorem). *If two distinct lines cut by a transversal have a pair of consecutive interior angles such that one interior angle is congruent to the others supplement, then the two lines are parallel.*

Proof. Let $Q - P - R$ such that $\angle APR$ is supplementary to $\angle APQ$. Suppose that $\angle BQP \cong \angle APR$. Since $\angle APR$ and $\angle A'PQ$ are vertical, we have $\angle APR \cong \angle A'PQ$. By transitivity, we have that $\angle BQP \cong \angle A'PQ$. The result then follows from the Alternate Interior Angle Theorem. $\square$

Definition 4.4.4. Let A , B , and C be three distinct, non-collinear points in an ordered geometry. Let $C - A - D$. Then the angle is called $\angle DAB$ is called an **exterior angle** to the triangle $\triangle ABC$. Note that $\angle DAB$ is a supplement to the triangle's angle $\angle CAB$. We say the other two angles of the triangle $\angle ABC$ and $\angle BCA$ are **remote interior angles** to $\angle DAB$.

Theorem 4.4.5 (Exterior Angle Theorem). *An exterior angle of a triangle is greater than each of its remote interior angles.*

Proof. Consider the triangle $\triangle ABC$. Let $C - A - D$ such that $\angle DAB$ is an exterior angle to the triangle. We will first show that $\angle ABC = \angle B < \angle DAB$. Suppose that $\angle B \cong \angle DAB$. Note that $\angle B$ and

$\angle DAB$ are alternate interior angles of the lines $\overleftrightarrow{AC}$ and $\overleftrightarrow{BC}$ with transversal $\overleftrightarrow{AB}$. By the Alternate Interior Angle Theorem, this implies that $\overleftrightarrow{AC} \parallel \overleftrightarrow{BC}$, which is false since $C \in \overleftrightarrow{AC} \cap \overleftrightarrow{BC}$. Thus, $\angle B \not\cong \angle DAB$.

Suppose that $\angle B > \angle DAB$. Then there exists a point $F \in \angle^\circ ABC$ such that $\angle ABF \cong \angle DAB$. Again, $\angle ABF$ and $\angle DAB$ are alternate interior angles and we conclude that $\overleftrightarrow{BF} \parallel \overleftrightarrow{AC}$. By the Crossbar Theorem, the ray $\overrightarrow{BF}$ intersects $\overleftrightarrow{AC}$ at some point G . But then $G \in \overleftrightarrow{BF} \cap \overleftrightarrow{AC}$, a contradiction. Therefore, we conclude that $\angle B < \angle DAB$.

Let $B - A - E$. Then by the above argument, we conclude that $\angle C < \angle EAC$. Since $\angle EAC$ and $\angle DAB$ are vertical angles, then they are congruent, which proves the result. $\square$

Exercise

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be congruence geometries.

1. (Erecting a Perpendicular from a Line) Let ℓ be a line and let P be a point on ℓ . Prove there exists a unique line m such that $P \in m$ and $m \perp \ell$.
2. Prove that two distinct lines perpendicular to a common line are parallel.
3. (Guaranteed Parallel Lines) Prove that through a point P not on a line ℓ there exists at least one line m such that $P \in m$ and $m \parallel \ell$. This line is called the **guaranteed parallel**.

ⁱThe progressive slides in this section were created by Hayley John.

“Only a man’s character is the real criterion of worth.” – Eleanor Roosevelt

4.5 Triangle Congruence

Theorem 4.5.1 (Angle-Side-Angle Criterion). *Let $\triangle ABC$ and $\triangle A'B'C'$ be two triangles such that $\angle BAC \cong \angle B'A'C'$, $\overline{AB} \cong \overline{A'B'}$, and $\angle ABC \cong \angle A'B'C'$. Then $\triangle ABC \cong \triangle A'B'C'$.*

Proof. By segment translation, let C'' be the unique point on the ray $\overrightarrow{A'C'}$ such that $\overline{AC} \cong \overline{A'C''}$. Then by SAS, we have $\triangle ABC \cong \triangle A'B'C''$. In particular, $\angle ABC \cong \angle A'B'C''$. By transitivity of congruence, this implies that $\angle A'B'C' \cong \angle A'B'C''$. By uniqueness of angle translation, we then conclude that $\angle A'B'C' = \angle A'B'C''$. In particular, $\overline{B'C'} = \overline{B'C''}$. This shows that B', C' , and C'' are collinear, that is, $C', C'' \in \overrightarrow{B'C'}$. By construction of C'' , we also have that $C', C'' \in \overrightarrow{A'C'}$. Since lines intersect at unique points, we conclude that $C' = C''$. Therefore, $\triangle ABC \cong \triangle A'B'C'$. $\square$

Theorem 4.5.2 (Side-Side-Side Criterion). *Let $\triangle ABC$ and $\triangle A'B'C'$ be two triangles such that $\overline{AB} \cong \overline{A'B'}$, $\overline{AC} \cong \overline{A'C'}$, and $\overline{BC} \cong \overline{B'C'}$. Then $\triangle ABC \cong \triangle A'B'C'$.*

Proof. By triangle translation, there exists a unique point C'' in the half plane opposite to C' with respect to $\overleftrightarrow{A'B'}$ such that $\triangle ABC \cong \triangle A'B'C''$. By transitivity of congruence, we have $\overline{A'C'} \cong \overline{A'C''}$ and $\overline{B'C'} \cong \overline{B'C''}$. So, $\triangle A'C'C''$ and $\triangle B'C'C''$ are isosceles. In particular, $\angle A'C'C'' \cong \angle A'C''C'$ and $\angle B'C'C'' \cong \angle B'C''C'$.

Since C' and C'' are on opposite sides of $\overleftrightarrow{A'B'}$, let $D' \in \overline{C'C''} \cap \overleftrightarrow{A'B'}$. If $A' - D' - B'$, then $D' \in \angle A'C'B'$ and $D' \in \angle A'C''B'$ by the Between-Cross Lemma. So, $\overline{C'A'} - \overline{C'D'} - \overline{C'B'}$ and $\overline{C''A'} - \overline{C''D'} - \overline{C''B'}$. By angle addition, we then have that $\angle A'C'B' \cong \angle A'C''B'$. Otherwise, we may without the loss of generality assume that $A' - B' - D'$ by trichotomy. Then $B' \in \angle D'C'A'$ and $B' \in \angle D'C''A'$, that is, $\overline{C'A'} - \overline{C'B'} - \overline{C'D'}$ and $\overline{C''A'} - \overline{C''B'} - \overline{C''D'}$. By angle subtraction, we again have that $\angle A'C'B' \cong \angle A'C''B'$. Therefore $\triangle A'B'C' \cong \triangle A'B'C'' \cong \triangle ABC$ by SAS. $\square$

Theorem 4.5.3 (Side-Angle-Angle Criterion). *Let $\triangle ABC$ and $\triangle A'B'C'$ be two triangles such that $\overline{AB} \cong \overline{A'B'}$, $\angle BAC \cong \angle B'A'C'$, and $\angle ACB \cong \angle A'C'B'$. Then $\triangle ABC \cong \triangle A'B'C'$.*

Please note that SAA does not immediately imply ASA. Often in Euclidean geometry, we use the notion that the angle sum of a triangle is 180° . Thus, if we know the measure of two angles then we can compute angle measure of the third angle. Unfortunately, the above angle sum property of triangles is actually equivalent to the Euclidean parallel postulate. As such, it does not hold in neutral geometry. Instead, we will show that the Exterior Angle Theorem implies that SAA can be converted to ASA.

Proof. Suppose for the sake of contradiction that $\angle ABC > \angle A'B'C'$. Then there exists some point $D \in \angle ABC$ such that $\angle ABD \cong \angle A'B'C'$. By the Crossbar Theorem, there exists some point of intersection $E \in \overline{AC} \cap \overline{BD}$. Then $\triangle ABE \cong \triangle A'B'C'$ by ASA. In particular, $\angle AEB \cong \angle A'C'B'$. By transitivity of congruence, we also have that $\angle AEB \cong \angle ACB$. But $\angle AEB$ is an exterior angle to the triangle $\triangle ECB$ with $\angle ECB$ as a remote interior angle. Thus, $\angle ACB = \angle ECB < \angle AEB$ by the Exterior Angle Theorem, a contradiction.

A similar contradiction is discovered if we assume that $\angle ABC < \angle A'B'C'$. Thus, we conclude that $\angle ABC \cong \angle A'B'C'$. Then $\triangle ABC \cong \triangle A'B'C'$ by ASA. $\square$

In closing, it is important to recognize certain congruence conditions which do not imply congruence of triangles. The first to note is Angle-Angle-Angle. In Euclidean geometry, AAA does not imply congruence of triangles. It, in fact, leads to the notion of *similar triangles*, triangles which are proportional sides but not necessarily congruent. Interestingly, in hyperbolic geometry, AAA does imply congruence of triangles. Therefore, in neutral geometry AAA does not necessarily imply congruence of triangles.

The last congruence condition Side-Side-Angle does not imply the congruence of triangles. In Euclidean geometry, SSA is often called the *ambiguous case* because given an angle measure and two segment measures there may exist exactly one congruence class of triangles, two congruence classes of triangles, or no congruence classes of triangles. In particular, since two triangles with the same SSA might be non-congruence, SSA is not a reliable congruence condition in neutral geometry.

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be congruence geometries.

Definition 4.5.4. Let A, B, C, D be four distinct points in a congruence geometry no three of which are collinear. Let $\square ABCD := \triangle ABD \cup \triangle CBD$ be called a **quadrilateral** if $\triangle ABD \cap \triangle CBD = \overline{BD}$.

We say that a quadrilateral $\square ABCD$ is a **kite**ⁱ if $\overline{AB} \cong \overline{BC}$ and $\overline{AD} \cong \overline{CD}$.

We say that a kite $\square ABCD$ is a **rhombus** if additionally $\overline{AB} \cong \overline{AD}$.

1. Prove the Side-Angle-Side-Angle-Side (SASAS) Congruence condition for quadrilaterals, that is, if the vertices of two quadrilaterals are in one-to-one correspondence such that three sides and the two included angles of one quadrilateral are congruent to the corresponding parts of the second quadrilateral, then the quadrilaterals are congruent.

2. Prove the Angle-Side-Angle-Side-Angle (ASASA) Congruence condition for quadrilaterals, that is, if the vertices of two quadrilaterals are in one-to-one correspondence such that three angles and the two included sides of one quadrilateral are congruent to the corresponding parts of the second quadrilateral, then the quadrilaterals are congruent.

Definition 4.5.5. We say that $\square ABCD$ is a **parallelogram**ⁱⁱ if $\overleftrightarrow{AC} \parallel \overleftrightarrow{BD}$ and $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$.

We say that $\square ABCD$ is a **trapezoid** if at least one of the previous pairs of lines are parallel.

3. Disprove that Side-Side-Side-Side (SSSS) Congruence condition for quadrilaterals, that is, if the vertices of two quadrilaterals are in one-to-one correspondence such that all four sides of one quadrilateral are congruent to the corresponding sides of the second quadrilateral, then the quadrilaterals are congruent. This shows that there is no analog of SSS for quadrilaterals.

4. (Converse of Isosceles Triangle Theorem) Prove the Converse of the Isosceles Triangle Theorem, that is, if two angles of a triangle are congruent, then the sides opposite those angles are congruent.

5. Given $\triangle ACD$ and $\triangle A'C'D'$, where $A - B - D$, $A' - B' - D'$, $\overline{AB} \cong \overline{A'B'}$, $\overline{BD} \cong \overline{B'D'}$, $\overline{AC} \cong \overline{A'C'}$, and $\overline{BC} \cong \overline{B'C'}$, prove $\overline{CD} \cong \overline{C'D'}$.

6. (Hypotenuse-Leg Criterion) Prove the Hypotenuse-Leg Triangle Congruence Criterion: if two right triangles have the hypotenuse and leg of one congruent, respectively, to the hypotenuse and leg of the other then the triangles are congruent.

ⁱWe note that $\square C'A'C'B''$ in Theorem 4.5.2 is a kite.

ⁱⁱWe will mention that in congruence geometry, there are two non-equivalent ways to define the notion of a parallelogram. The above definition of parallelograms is actually where its name comes from, which we adopt of the official definition. Alternatively, we could define a parallelogram as a quadrilateral $\square ABCD$ if $\triangle ABC \cong \triangle DCA$. In neutral geometry, these two notions can be different. In fact, the equivalence of these definitions is equivalent to the Euclidean Parallel Postulate.

“There are two sides to every issue: one side is right and the other is wrong, but the middle is always evil.”
– Ayn Rand

4.6 Midpoints and Bisectors

Definition 4.6.1. Let A, C be points in a congruence geometry. Let B be a point such that $A - B - C$ and $\overline{AB} \cong \overline{BC}$. Then the point B is called the **midpoint** of $\overline{AC}$. We say that a line ℓ is a **perpendicular bisector** of $\overline{AC}$ if $B \in \ell$ and $\ell \perp \overleftrightarrow{AC}$.

Theorem 4.6.2 (Midpoint Theorem). *Every segment has a unique midpoint. (Progressive Geometry Proof Slides)*

Proof. Consider the segment $\overline{BC}$. Let A be a point off the line $\overleftrightarrow{BC}$. Let D be a point such that $A - C - D$. Then $\angle DCB$ is an exterior angle to the triangle $\triangle ABC$ with remote interior angle $\angle B$. Then by the Exterior Angle Theorem, $\angle DCB > \angle B$. Thus, there exists a point $E \in \angle^\circ DCB$ such that $\angle B \cong \angle ECB$. By segment translation we may assume that $\overline{CE} \cong \overline{AB}$.

Since $E \in \angle^\circ DCB$, we know that $E \in H^\circ(\overleftrightarrow{BC}, D)$. On the other hand, A and D are on opposite sides of $\overleftrightarrow{BC}$ since $A - C - D$. So E and A are on opposite sides of $\overleftrightarrow{BC}$ by Plane Separation. Let $F \in \overline{AE} \cap \overleftrightarrow{BC}$. In particular, $A - F - E$.

We also have that $E \in H^\circ(\overleftrightarrow{CD}, B) = H^\circ(\overleftrightarrow{AC}, B)$, since $E \in \angle^\circ DCB$. Since $\angle ABC$ and $\angle ECB$ are congruent alternate interior angles to the pair of lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CE}$, we conclude that $\overleftrightarrow{AB} \parallel \overleftrightarrow{CE}$ by the Alternate Interior Angle Theorem. In particular $\overleftrightarrow{AB} \parallel \overleftrightarrow{CE}$. So, $E \in H^\circ(\overleftrightarrow{AB}, C)$, which shows that $E \in \angle^\circ BAC$. The Between-Cross Lemma then shows that $B - F - C$.

Finally, the angles $\angle AFB \cong \angle EFC$ since they are vertical angles (remember that $A - F - E$ and $B - F - C$). Therefore, $\triangle AFB \cong \triangle EFC$ by AAS. In particular, $\overline{BF} \cong \overline{FC}$. This provides the existence of midpoints.

Let F and F' be midpoints of $\overline{BC}$, and let $\preceq$ be an ordering of $\overleftrightarrow{BC}$ such that $B \preceq C$. Without the loss of generality, we may assume that $F' \preceq F$. In particular, $B - F' - F - C$. Then

$$\overline{BF'} \leq \overline{BF} \cong \overline{FC} \leq \overline{F'C} \cong \overline{BF'}.$$

Hence, $\overline{BF'} \cong \overline{BF}$. But by the uniqueness of Segment Translation, we conclude that $F = F'$, that is, midpoints are unique. $\square$

Definition 4.6.3. Let A, B, D be points in a congruence geometry. Let $C \in \angle^\circ ABD$ such that $\angle ABC \cong \angle CBD$. Then the ray $\overrightarrow{BC}$ is called the **bisector** of $\angle ABD$.

A similar existence and uniqueness theorem holds for bisectors but whose proof is left as an exercise for the reader.

Theorem 4.6.4 (Bisector Theorem). *(a) Every angle has a unique bisector. (b) Every segment has a unique perpendicular bisector.*

Proposition 4.6.5 (Angle-Opposite-Side Relation). *For any triangle $\triangle ABC$, we have $\angle A > \angle B$ if and only if $\overline{BC} > \overline{AC}$.*

Proof. Suppose $\overline{BC} > \overline{AC}$. Let $\overrightarrow{CP}$ be the bisector of $\angle C$, which will intersect $\overline{AB}$ at some point D by the Crossbar Theorem. By Segment Translation, let $E \in \overrightarrow{CB}$ such that $\overline{CE} \cong \overline{AC}$. By our original assumption, we also know that $B - E - C$. Then $\triangle ACD \cong \triangle ECD$ by SAS. In particular, $\angle A \cong \angle CED$. Since $B - E - C$, $\angle CED$ is exterior to $\triangle BED$. Note that $\angle CED > \angle B$ by the Exterior Angle Theorem with respect to the triangle $\triangle BED$. Hence, $\angle A > \angle B$.

Suppose $\angle A > \angle B$. Assume for the sake of contradiction that $\overline{BC} \cong \overline{AC}$. Then $\triangle ABC$ is isosceles, which implies that $\angle A \cong \angle B$, a contradiction. Suppose next that $\overline{BC} < \overline{AC}$. This implies $\angle A < \angle B$ by the first part of this proof, which is another contradiction. By trichotomy, we conclude that $\overline{BC} > \overline{AC}$. $\square$

Corollary 4.6.6 (The Best Approximation Theorem). *(a) If $\angle Q$ is a right angle for a triangle $\triangle PQR$, then $\angle P < \angle Q$ and $\angle R < \angle Q$. (b) Let P be a point off a line ℓ and let m be the unique perpendicular line m containing P and intersecting ℓ . Let $Q \in m \cap \ell$. Let $R \in \ell \setminus \{Q\}$. Then $\overline{PR} > \overline{PQ}$.*

Proof. Considering (a), let S be a point such that $\angle PQS$ is an exterior angle to $\triangle PQR$ (supplementary to $\angle PQR = \angle Q$). By the Exterior Angle Theorem, we have that $\angle P, \angle R < \angle PQS$. Since $\angle Q$ is right, we have $\angle Q \cong \angle PQS$. This proves (a). Part (b) follows immediately from (a) and AOS (Angle-Opposite-Side) on $\triangle PQR$. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be congruence geometries.

1. Prove Theorem 4.6.4.

Definition 4.6.7. A **scalene triangle** is a triangle for which no two sides are congruent.

2. Prove that the largest angle of a scalene triangle is opposite the largest side.
3. Prove that if two angles of a triangle are not congruent, then the sides opposite them are not congruent, and the larger side is opposite the larger angle.

ⁱThe progressive slides in this section were created by Hayley John.

“The worst form of inequality is to try to make unequal things equal.” – Aristotle

4.7 The Triangle Inequality

Definition 4.7.1. Let $\overline{AB}$ and $\overline{CD}$ be two segments in a congruence geometry. By Segment Translation and Extension, there exists a unique point E such that $A - B - E$ and $\overline{BE} \cong \overline{CD}$. We define the **sum** of $\overline{AB}$ and $\overline{CD}$ to be $\overline{AE}$, denoted $\overline{AB} + \overline{CD} = \overline{AE}$.

We should mention some properties of this notion of segment addition. First, if $\overline{PQ} \cong \overline{CD}$, then $\overline{AB} + \overline{CD} = \overline{AB} + \overline{PQ}$. Also, it should be mentioned that $\overline{AB} + \overline{CD} \neq \overline{BA} + \overline{CD}$ even though $\overline{AB} = \overline{BA}$. This is because the definition of addition depends on the direction since $\overline{AB}$ and $\overline{BA}$ are opposite. On the other hand, $\overline{AB} + \overline{CD} \cong \overline{BA} + \overline{CD}$ by the Segment Addition Axiom (this is why it got its name). Finally, it should be mentioned that $\overline{AB} + \overline{CD} \neq \overline{CD} + \overline{AB}$ because the first sum lies on $\overleftrightarrow{AB}$ and the second lies on $\overleftrightarrow{CD}$. But again,

$$\overline{AB} + \overline{CD} \cong \overline{CD} + \overline{AB}$$

by the Segment Addition Axiom. Therefore, the sum of two segments is well-defined up to congruence, and we will mostly just be interested in the congruence classes of these segments and not necessarily on the particular representatives.

We mention one last property of segment sums. Segment addition is associative, that is,

$$(\overline{AB} + \overline{CD}) + \overline{EF} \cong \overline{AB} + (\overline{CD} + \overline{EF}).$$

This result follows from the Segment Addition Axiom and the Betweenness Preservation Property. With associativity, we can define for each $n \in \mathbb{N}$ the **multiple** of the segment,

$$n \cdot \overline{AB} = \underbrace{\overline{AB} + \dots + \overline{AB}}_{n\text{-times}}$$

when $n > 0$ and we define $0 \cdot \overline{AB} = \overline{AA} = A$. We can also extend this notion to negative integer as $(-n) \cdot \overline{AB}$ is defined to be the unique segment on $\overleftrightarrow{AB}$ which is congruent to $n \cdot \overline{AB}$ by segment translation. This also allows use to construct the **difference** of segments.

Lastly, we define $\frac{1}{2} \cdot \overline{AC}$ to be the segment $\overline{AB}$ where B is the midpoint of $\overline{AC}$. This allows use to describe $\frac{a}{2^k} \cdot \overline{AB}$ where $a \in \mathbb{Z}$ and $k \in \mathbb{N}$. The number $\frac{a}{2^k}$ is known as a **dyadic rational number**. Without the presence of some Continuity Axiom, we cannot guarantee the existence of any other multiples in congruence geometry.

Definition 4.7.2. Given two angles $\angle ABC$ and $\angle DEF$, let P be the unique point P in the half plane of $\overleftrightarrow{BC}$ opposite of A such that $\angle CBP \cong \angle DEF$. If $P \in H(\overleftrightarrow{AB}, C)$, then we define the **sum** of $\angle ABC$ and $\angle DEF$ to be $\angle ABP$, denoted $\angle ABC + \angle DEF = \angle ABP$.

Analogously, we can extend all notions of segment sums to angles sums, like commutativity, associativity, and dyadic multiples. Unlike segment sums, it is possible that the sum of two angles is undefined, which would occur if the sum would have been greater than a flat angle.

Theorem 4.7.3 (The Triangle Inequality). *Let A , B , and C be three distinct, non-collinear points. Then*

$$\overline{AC} < \overline{AB} + \overline{BC}.$$

If A , B , and C are collinear, then without the loss of generality we may assume $A - B - C$ by trichotomy. Then $\overline{AB} + \overline{BC} = \overline{AC}$ by the Segment Addition Axiom. Therefore, the triangle inequality is sometimes

expressed as

$$\overline{AC} \leq \overline{AB} + \overline{BC}$$

where equality (really congruence) happens if and only if $A - B - C$.

Proof. Let D be the unique point on $\overrightarrow{AB}$ such that $\overline{AD} = \overline{AB} + \overline{BC}$. Thus, we need to show that $\overline{AD} > \overline{AC}$. Then by construction, we have that $\overline{BD} \cong \overline{BC}$ and $A - B - D$. Then $\triangle BCD$ is isosceles, which implies that $\angle BCD \cong \angle BDC$. Also, by the Between-Cross Lemma, $B \in \angle^\circ ACD$. This shows that $\angle ACD > \angle BCD \cong \angle BDC$. Finally, $\overline{AD} > \overline{AC}$ by AOS in the triangle $\triangle ACD$. $\square$

Theorem 4.7.4 (The Hinge Theorem). *Let $\triangle ABC$ and $\triangle A'B'C'$ be triangles with $\overline{AB} \cong \overline{A'B'}$ and $\overline{AC} \cong \overline{A'C'}$. Then $\angle A > \angle A'$ if and only if $\overline{BC} > \overline{B'C'}$.*

Proof. Let $\angle BAC > \angle B'A'C'$. Let $D \in \angle^\circ BAC$ such that $\angle BAD \cong \angle B'A'C'$. By segment translation, we may assume that $\overline{AD} \cong \overline{A'C'}$. Then $\triangle ABD \cong \triangle A'B'C'$ by SAS. This implies that $\overline{BD} \cong \overline{B'C'}$. Let $E \in \overline{BC} \cap \overrightarrow{AD}$, guaranteed by the Crossbar Theorem. In particular, $B - E - C$. Let $\overrightarrow{AF}$ be the bisector of $\angle CAD$. Again, by the Crossbar Theorem, we may assume that $E - F - C$. By SAS, we have $\triangle ACF \cong \triangle ADF$. Hence, $\overline{FC} \cong \overline{FD}$. Therefore, $\overline{B'C'} \cong \overline{BD} < \overline{BE} + \overline{ED} \cong \overline{BE} + \overline{FC} = \overline{BC}$, where the inequality follows from the Triangle Inequality. This proves the first direction.

Conversely, let $\overline{BC} > \overline{B'C'}$. If $\angle BAC \cong \angle B'A'C'$, then $\triangle ABC \cong \triangle A'B'C'$ by SAS. This implies that $\overline{BC} \cong \overline{B'C'}$, a contradiction. If $\angle BAC < \angle B'A'C'$, then $\overline{BC} < \overline{B'C'}$ by the first part of this proof, which is another contradiction. By trichotomy, we conclude that $\angle BAC > \angle B'A'C'$. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be congruence geometries.

1. Consider a polygon with vertices $A_1, A_2, \dots, A_n$. Prove that

$$\overline{A_1 A_k} \leq \sum_{i=1}^{k-1} \overline{A_i A_{i+1}}$$

for all $1 < k \leq n$. In other words, partial perimeters are necessarily longer than diagonals in an arbitrary polygon.

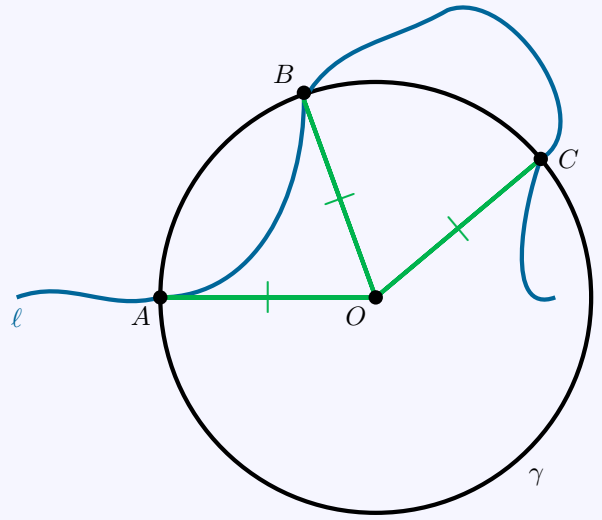
“The wheel is come full circle.” – William Shakespeare

4.8 Circles

Definition 4.8.1. Let O and R be two distinct points in a congruence geometry. Then we define a **circle** γ centered at O with radius $\overline{OR}$ to be the set of all points P in the geometry such that $\overline{OP} \cong \overline{OR}$. The **interior** γ° of a circle γ is the set of all points P such that $\overline{OP} < \overline{OR}$. Note that by definition, $\gamma^\circ \not\subseteq \gamma$. Instead, $\gamma \cup \gamma^\circ$ is called a **disc** with center O and radius $\overline{OR}$. The **exterior** of a circle γ is the set of all points P such that $\overline{OP} > \overline{OR}$.

Proposition 4.8.2 (Line-Circle Intersection). *A line intersects a circle in at most two distinct points.*

Proof. Let γ be a circle with center O and radius OR . Let ℓ be a line. Suppose for the sake of contradiction that ℓ intersects γ at three distinct points A , B , and C . Without the loss of generality, we may assume that $A - B - C$. Then $\overline{OA} \cong \overline{OB} \cong \overline{OC} \cong \overline{OR}$. Thus, the triangles $\triangle ABO$ and $\triangle CBO$ are isosceles, which implies $\angle OAB \cong \angle OBA$ and $\angle OBC \cong \angle OCB$. Since a triangle can have at most one non-acute angle by the Exterior Angle Theorem, we conclude that all four angles are acute. But $\angle OBA$ and $\angle OBC$ are supplementary, which implies that $\angle OBC$ is also obtuse. This contradicts the trichotomy of the ordering of angles. This proves the result. $\square$



Unfortunately, an important follow up to this proposition is the fact that three non-collinear points form a unique circle (circle determination), but this fact is not a theorem in congruence geometry. It is not even a theorem of neutral geometry! In particular, in hyperbolic geometry of collection of three non-collinear points might not determine a circle, but all sets of three non-collinear points in the Euclidean plane determine a circle. In other words, circle determination is independent of the neutral geometry axioms. Furthermore, we will later see that circle determination is equivalent to the Euclidean parallel postulate in neutral geometry.

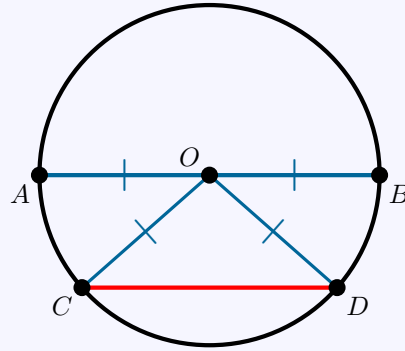
Definition 4.8.3. A **secant** of a circle is a line that intersects the circle in two distinct points. Similarly, a **chord** of a circle is a segment whose endpoints are two distinct points on the circle. A **diameter** of a circle is a chord which contains the center of the circle. A **tangent** of a circle is a line that intersects the circle in a unique point, called the **point of tangency**.

It is immediate that the segment sum of two radii of a circle is a diameter.

Proposition 4.8.4. *If $\overline{AB}$ is a diameter of a circle and if $\overline{CD}$ is another chord of the same circle that is not a diameter, then $\overline{AB} > \overline{CD}$.*

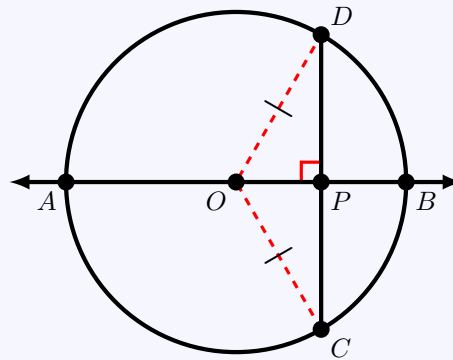
Proof. Let γ be a circle with center O . Let $\overline{AB}$ and $\overline{CD}$ be a diameter and a non-diametric chord of γ , respectively. Then by the Triangle Inequality applied to $\triangle COD$, we have

$$\overline{CD} < \overline{OC} + \overline{OD} \cong \overline{OA} + \overline{OB} = \overline{AB}. \quad \square$$



Proposition 4.8.5 (Diameter-Perpendicular-Bisector or DPB). *If a diameter of a circle is perpendicular to a chord of the circle, then the diameter bisects the chord.*

Proof. Let γ be a circle with center O . Let $\overline{AB}$ and $\overline{CD}$ be a diameter and a non-diametric chord of γ , respectively, such that $\overline{AB} \perp \overline{CD}$. Let $P \in \overline{AB} \cap \overline{CD}$. Now, $\overline{OD} \cong \overline{OC}$ since they are radii of γ . Likewise, $\angle OPD \cong \angle OPC$ as they are both right. Then $\triangle OPD \cong \triangle OPC$ by the hypotenuse-leg congruence condition of right triangles. In particular, $\overline{CP} \cong \overline{PD}$. $\square$



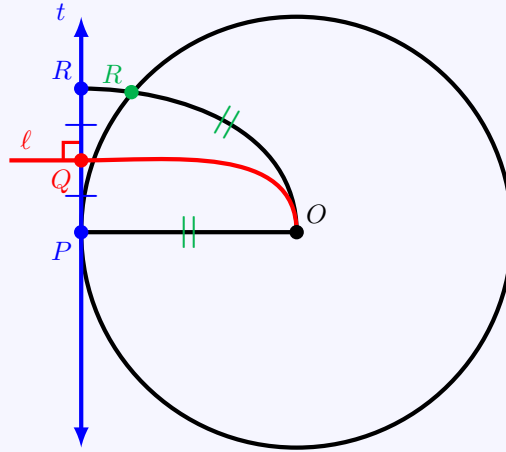
Proposition 4.8.6 (Diameter-Bisector-Perpendicular or DBP). *If a diameter of a circle bisects a non-diametric chord of the circle, then the diameter is perpendicular to the chord.*

Proposition 4.8.7 (Perpendicular-Bisector-Diameter or PBD). *The perpendicular bisector of a chord of a circle contains a diameter of the circle.*

The proofs of the previous two propositions are left as exercises to the reader.

Theorem 4.8.8 (Tangent-Radius Theorem). *A line is tangent to a circle if and only if it is perpendicular to the radius drawn to the point of tangency.*

Proof. Let γ be a circle with center at O and let t be a line tangent to γ at the point P . Suppose for the sake of contradiction that $\overline{OP} \not\perp t$. Then let ℓ be the unique line containing O that is perpendicular to t . Let $Q \in t \cap \ell$. By Extension and Segment Translation, let R be the unique point on t on the opposite ray of $\overrightarrow{QP}$ such that $\overline{QR} \cong \overline{QP}$. Note that $\angle PQO$ and $\angle RQO$ are both right. Then by SAS, $\triangle OQR \cong \triangle OQP$. This implies that $\overline{OR} \cong \overline{OP}$. But $\overline{OP}$ is a radius of γ , which means that $R \in \gamma$. This contradicts t being a tangent line. Therefore, $\overleftrightarrow{OP} \perp t$, which proves the first direction.



The other direction of this proof is left as an exercise for the reader. □

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be congruence geometries.

1. Prove Proposition 4.8.6.
2. Prove Proposition 4.8.7.
3. Consider a point P in the exterior of a circle γ centered at O with radius $\overline{OA}$. Prove that the two tangent segments drawn from P to γ are congruent.
4. Finish the proof of Theorem 4.8.8.

ⁱFigures in this section were created by Yaxin Lan.

Chapter 5

Neutral Geometry

“Continuous effort - not strength or intelligence - is the key to unlocking our potential.”
– Winston Churchill

5.1 The Continuity Axioms

The last family of axioms from Hilbert’s Axioms of Plane Geometry are the Continuity Axioms. This is the final piece which completes our notion of Euclidean geometry: incidence, betweenness, congruence, the Euclidean Parallel Postulate, and Continuity (or sometimes called the Completeness Axiom). The Completeness Axiom is a property coming from Topology and of up most importance in Real Analysis because the Completeness Axiom is what makes the Real Numbers $\mathbb{R}$ the Real Numbers as opposed to some other number system. In Analysis, the Completeness Axiom is often referred to as the *Least Upper Bound Property*: all non-empty sets of numbers with an upper bound must have a least upper bound, called the *supremum* of the set. The Real Numbers $\mathbb{R}$ as a set is then the unique ordered field which also satisfies the least upper bound property.

Now the least upper bound property is an axiom of $\mathbb{R}$, and it sets it apart from other ordered fields, like the rational numbers $\mathbb{Q}$. The Euclidean plane $\mathbb{R}^2$ is then deeply concerned with the properties of $\mathbb{R}$ and a completeness axiom is necessary to differentiate an affine, congruence geometry, like $\mathbb{Q}^2$, from what we mean as Euclidean geometry $\mathbb{R}^2$. The least upper bound property is often referred as the *Dedekind Axiom* of completeness/continuity because the model of $\mathbb{R}$ that best exemplifies the least upper bound property is the model of Dedekind cuts on $\mathbb{Q}$. As such, we will recast our continuity axiom of geometry after the least upper bound property and will refer to it as the Dedekind axiom.

Definition 5.1.1. Let ℓ be a line and let $\Sigma_1, \Sigma_2 \subseteq \ell$. We say that (Σ_1, Σ_2) is a **Dedekind cut** of ℓ if Σ_1, Σ_2 are two nonempty, convex subsets of ℓ such that $\ell = \Sigma_1 \cup \Sigma_2$ and $\Sigma_1 \cap \Sigma_2 = \emptyset$.

The Continuity Axiom

Dedekind’s Axiom: Given any Dedekind cut (Σ_1, Σ_2) of a line ℓ , there exists a unique point P on ℓ and an ordering of ℓ such that $\Sigma_1 = (-\infty, P]$ and $\Sigma_2 = (P, \infty)$ or $\Sigma_1 = (-\infty, P)$ and $\Sigma_2 = [P, \infty)$.

We say a geometry is **complete**ⁱ if it satisfies Dedekind’s axiom. A complete congruence geometry is called a **neutral geometry** (or **absolute geometry**).

Example 5.1.2. Both the real Euclidean plane $\mathbb{R}^2$ and the real hyperbolic plane $\mathbb{H}^2$ are examples of neutral geometries.

The real projective plane $\mathbb{RP}^2$ also satisfies Dedekind’s axiom. We note that although $\mathbb{RP}^2$ has a well-defined notion of betweenness, it fails the Trichotomy and Pasch’s Axioms, so it is

not even an ordered geometry and hence cannot be a neutral geometry. We also note that $\mathbb{RP}^2$ has a well-defined notion of congruences of segments and angles and that it satisfies all six congruence axioms. Therefore, $\mathbb{RP}^2$ is only two betweenness axioms short of being a neutral geometry.

The Euclidean sphere S^2 also satisfies Dedekind's axiom. Similar to $\mathbb{RP}^2$ has a well-defined notion of betweenness and congruence. Under these interpretations, S^2 satisfies all the neutral geometry axioms except the Line Determination and Trichotomy axioms. Thus, S^2 is only two axioms away from being neutral. The advantage of switching from S^2 to $\mathbb{RP}^2$ is that $\mathbb{RP}^2$ is an incidence geometry but at the cost of losing Plane Separation

Example 5.1.3. Consider the geometry $\mathbb{Q}^2$ as a subset geometry of $\mathbb{R}^2$. The geometry $\mathbb{Q}^2$ consists of only points whose both coordinate are rational points. Line in $\mathbb{Q}^2$ are equations of the form $Ax + By = C$ where $A, B, C \in \mathbb{Q}$ and A and B are not both zero. The notion of incidence, betweenness, and congruence take on their usual interpretations as in $\mathbb{R}^2$. Thus, $\mathbb{Q}^2$ is a congruence geometry but not a neutral geometry since the continuity axiom fails in $\mathbb{Q}^2$. Consider the line $\ell : y = 0$ (the x -axis) in $\mathbb{Q}^2$. Let $\Sigma_1 = \{(x, 0) \mid x^2 \geq 2\}$ and let $\Sigma_2 = \{(x, 0) \mid x^2 < 2\}$. It is simple to see that $\ell = \Sigma_1 \cup \Sigma_2$ and $\Sigma_1 \cap \Sigma_2 = \emptyset$. Suppose that $a, c \in \Sigma_1$. Then $a^2 \geq 2$ and $c^2 \geq 2$. Suppose without the loss of generality that $a \geq c$. Let b be between a and c , that is, $a - b = c$. Then there exists a rational number t such that $b = ta + (1 - t)c$. Then

$$b^2 = (ta + (1 - t)c)^2 \geq (tc + (1 - t)c)^2 = c^2 \geq 2.$$

Thus, $b \in \Sigma_1$, which shows that Σ_1 is convex. A similar argument shows that Σ_2 is convex. Thus, (Σ_1, Σ_2) is a Dedekind cut of ℓ , but there is no rational number p such that Σ_1 or Σ_2 are intervals one of which contains p . This follows from the fact that $\sqrt{2}$ is not a rational number. Therefore, $\mathbb{Q}^2$ fails the continuity axiom.

The Continuity Axiom guarantees the existence of many important points in neutral geometry, such as points of intersection between various geometric objects. A few point existence principles will be listed below, without proof. Although Dedekind's Axiom is very important in neutral geometry, the following theorems, especially Theorems 5.1.7 and 5.1.8 are very difficult and technical to prove. In fact, their proofs belong more to the study of Analysis than Geometry, which is why they are omitted. All of the remaining theorems below are consequences of Dedekind's Axiom and do not hold for general congruence geometries.

Theorem 5.1.4 (The Archimedean Principle). *In neutral geometry, given a segment $\overline{AB}$ and a ray r with vertex C , for each point $D \neq C$ on r , there exists an integer n and a point E on r such that $\overline{CE} \cong n \cdot \overline{AB}$ where either $E = D$ or $C - D - E$.*

Theorem 5.1.5 (Circular Continuity Principle). *In neutral geometry, if each of two circles has one point inside the other and one point outside the other, then the two circles intersect at two distinct points.*

Theorem 5.1.6 (Elementary Continuity Principle). *In neutral geometry, if one endpoint of a segment is inside a circle and the other endpoint of the same segment is outside the same circle, then the segment intersects the circle.*

Theorem 5.1.7 (Segment Measure). *Fix a segment $\overline{OI}$ in a neutral geometry, called the **unit segment**. There exists a unique mapping from the set of all segments of the geometry to the set of positive real numbers, $\overline{AB} \mapsto AB \in \mathbb{R}$, where AB is called the **measure** of $\overline{AB}$, satisfying the following properties:*

- (a) $OI = 1$;
- (b) $AB = CD$ if and only if $\overline{AB} \cong \overline{CD}$;
- (c) $A - B - C$ if and only if $AC = AB + BC$;
- (d) $AB < CD$ if and only if $\overline{AB} < \overline{CD}$;
- (e) For each positive real number x , there exists points A, B such that $AB = x$.

In the proof of the above theorem, the Continuity Axiom is only needed to prove (e).

Theorem 5.1.8 (Angle Measure). *There exists a unique mapping from the set of all angles of a neutral geometry to the interval of positive real numbers $[0, 180]$, $\angle ABC \mapsto m\angle ABC \in [0, 180]$, where $m\angle ABC$ is called the **measure** of $\angle ABC$, satisfying the following properties:*

- (a) $m\angle ABC = 0^\circ$ if and only if $C \in \overrightarrow{BA}$, that is, $\angle ABC$ is a null angle;
- (b) $m\angle ABC = 180^\circ$ if and only if $A - B - C$, that is, $\angle ABC$ is a flat angle;
- (c) $m\angle ABC = 90^\circ$ if and only if $\angle ABC$ is a right angle;
- (d) $m\angle ABC = m\angle DEF$ if and only if $\angle ABC \cong \angle DEF$;
- (e) $\overrightarrow{BA} - \overrightarrow{BC} - \overrightarrow{BD}$ if and only if $m\angle ABD = m\angle ABC + m\angle CBD$;
- (f) $m\angle ABC < m\angle DEF$ if and only if $\angle ABC < \angle DEF$;
- (g) For each positive real number $x \in [0, 180]$, there exists points A, B, C such that $m\angle ABC = x$.

In the proof of the above theorem, the Continuity Axiom is only needed to prove (g).

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be neutral geometries.

Definition 5.1.9. A triangle $\triangle ABC$ is **equilateral** if $\overline{AB} \cong \overline{BC} \cong \overline{AC}$.

1. (Euclid's Equilateral Triangle Proposition) For pair of distinct points A and B , prove there exists a point C such that the triangle $\triangle ABC$ is equilateral.
2. Prove that $\angle ABC$ and $\angle CBD$ are supplementary angles if and only if $m\angle ABC + m\angle CBD = 180^\circ$.
3. Given any angle $\angle ABC$, prove that the angle can be trisected, that is, there exists some point $D \in \angle ABC$ such that $m\angle ABD = \frac{1}{3}m\angle ABC$.

ⁱNote that some authors define an *ordered geometry* as a geometry satisfying the incidence, betweenness, and Dedekind's axioms. We will call such a geometry a *complete ordered geometry*.

“If you want to see the true measure of a man, watch how he treats his inferiors, not his equals.”
– J. K. Rowling

5.2 The Saccheri-Legendre Theorem

Definition 5.2.1. In a congruence geometry, we say that an angle is **acute** if it is less than a right angle. Likewise, we say that an angle is **obtuse** if it is greater than a right angle.

In congruence geometry, it can be proven that the supplement of an acute angle is obtuse and vice versa. Likewise, the complement of an acute angle is acute, while an obtuse angle has no complement. Finally, we have shown in congruence geometry that a triangle cannot have two right angles or a right angle and an obtuse angle. A modification of this theorem can also be used to show that no triangle can have more than one obtuse angle. Of course, such triangles with multiple right or obtuse angles can occur in elliptic geometry since they fail the betweenness axioms.

In neutral geometry, we can simply say that an angle is acute if it is less than 90° and obtuse if it is more than 90° . The above observations are much easier to prove in neutral geometry as they will be immediate consequences of the Saccheri-Legendre Theorem.

Lemma 5.2.2. *In neutral geometry, the sum of measures of any two angles of a triangle is less than 180° .*

Proof. Let $\triangle ABC$ be a triangle in neutral geometry and consider the angles $\angle A$ and $\angle B$. By Extension, let D be a point such that $C - B - D$. Since the angle $\angle DBA$ is exterior to $\triangle ABC$, we have $\angle ABD > \angle A$ by the Exterior Angle Theorem. Also, angles $\angle ABC$ and $\angle ABD$ are supplementary angles, which implies that $m\angle ABC + m\angle ABD = 180^\circ$. Therefore,

$$180^\circ = m\angle ABC + m\angle ABD > m\angle ABC + m\angle A = m\angle B + m\angle A.$$

The other two pairings of angles are handled similarly. □

Lemma 5.2.3. *In neutral geometry, for any $\triangle ABC$ there exists $\triangle A_1B_1C_1$ such that the two triangles have the same angle sums but $m\angle A_1 \leq \frac{1}{2}(m\angle A)$. ([Progressive Geometry Proofs Slides](#))*

Proof. Let D be the midpoint of $\overline{BC}$. Let E be the point such that $A - D - E$ and $\overline{AD} \cong \overline{DE}$. Since $\angle ADC \cong \angle EDB$ as vertical angles, we conclude that $\triangle ADC \cong \triangle EDB$ by SAS. In particular, $\angle CAD \cong \angle BED$, $\angle EBD \cong \angle ACD$. Let $\triangle A_1B_1C_1 = \triangle ABE$. Also,

$$\begin{aligned} m\angle A + m\angle B + m\angle C &= m\angle BAC + m\angle ABC + m\angle BCA \\ &= (m\angle CAD + m\angle BAD) + m\angle ABC + m\angle BCA \\ &= m\angle BED + m\angle BAD + m\angle ABC + m\angle EBD \\ &= m\angle BAD + (m\angle ABC + m\angle EBD) + m\angle BED \\ &= m\angle BAD + m\angle ABE + m\angle BED \\ &= m\angle A_1 + m\angle B_1 + m\angle C_1 \end{aligned}$$

Therefore, $\triangle ABC$ and $\triangle A_1B_1C_1$ have the same angle sum. Since $m\angle A = m\angle A_1 + m\angle C_1$, where angle measures are nonnegative real numbers, we have that $m\angle A_1 \leq \frac{1}{2}(m\angle A)$ or $m\angle C_1 \leq \frac{1}{2}(m\angle A)$. Switching the labels A_1 and C_1 if necessary, this finishes the proof. □

Theorem 5.2.4 (The Saccheri-Legendre Theorem). *In neutral geometry, the angle sum of any triangle is less than or equal to 180° .*

Proof. For the sake of contradiction, suppose there exists a triangle $\triangle ABC$ with angle sum equal to $180^\circ + x$, where $x > 0$. By Lemma 5.2.3, there exists a triangle $\triangle A_1B_1C_1$ with angle sum $180^\circ + x$ but $m\angle A_1 \leq \frac{1}{2}(m\angle A)$. Applying the Lemma again, we also have a triangle $\triangle A_2B_2C_2$ with angle sum $180^\circ + x$ but $m\angle A_2 \leq \frac{1}{2}(m\angle A_1) \leq \frac{1}{4}(m\angle A)$. By induction, there exists a triangle $\triangle A_nB_nC_n$ with angle sum $180^\circ + x$ but $m\angle A_n \leq \frac{1}{2^n}(m\angle A)$. By the Archimedean Principle, there exists a sufficiently large N such that $m\angle A_N \leq \frac{1}{2^N}(m\angle A) < x$. Therefore,

$$\begin{aligned} m\angle A_N + m\angle B_N + m\angle C_N &= 180^\circ + x \\ m\angle B_N + m\angle C_N &= 180^\circ + (x - m\angle A_N) > 180^\circ, \end{aligned}$$

which is a contradiction to Lemma 5.2.2. The result then follows. $\square$

We will see soon that the angle sum of a triangle equaling 180° is equivalent to the Euclidean parallel postulate and the angle sum being less than 180° is equivalent to the Hyperbolic parallel postulate. As might be guessed, in Elliptic geometry the angle sum of a triangle is greater than 180° and the two notions are equivalent. The above proof falling apart in Elliptic geometry since the Exterior Angle Theorem does not hold.

Definition 5.2.5. In ordered geometry, a **quadrilateral** $\square ABCD$ is the union of two triangles which share a common side, that is, $\square ABCD = \triangle ABD \cup \triangle CBD$ such that $\triangle ABD \cap \triangle CBD = \overline{BD}$. When addressing angles in quadrilaterals, we use the notation $\angle A = \angle BAD$, $\angle B = \angle ABD \cup \angle CBD$, $\angle C = \angle BCD$, and $\angle D = \angle ADB \cup \angle CDB$. Thus, in a quadrilateral, we allow angles to have measure greater than 180° .

Corollary 5.2.6. *The angle sum of any quadrilateral is less than or equal to 360° .*

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be neutral geometries.

1. (Saccheri-Legendre Theorem for Quadrilaterals) Prove that the angle sum of any quadrilateral is less than or equal to 360° .
2. (Exterior Angle Theorem) Prove that the measure of an exterior angle of a triangle is greater than or equal to the sum of the measures of the two remote interior angles.
3. (Converse of Euclid's 5th) Prove the converse of Euclid's fifth postulate, that is, prove that if two lines intersect then the sum of the interior angles with respect to some transversal on the same side as this intersection will be less than 180° .

ⁱThe progressive slides in this section were created by Hayley John.

“A man travels the world over in search of what he needs and returns home to find it.” – George A. Moore

5.3 Quadrilaterals

Geometers for centuries believed that Euclid’s fifth postulate, which is equivalent what is now known as the Euclidean parallel postulate which is actually due to Playfair, could be proven as a theorem of what is now known as neutral geometry. Many attempts to prove this have been attempted, but all have failed since the Euclidean parallel postulate is independent to the neutral axioms. Its independence was finally proven by the discovery of Hyperbolic geometry by Beltrami.

Because of these many failed attempts to prove the Euclidean parallel postulate, many equivalents of the Euclidean parallel postulate were found. Several of these will be mentioned and proven in the next few lectures. One attempt to prove the Euclidean parallel postulate that we will explore in this section is the existence of rectangle.

Definition 5.3.1. In congruence geometry, a **Saccheri quadrilateral** $\square ABCD$ is a quadrilateral such that $\angle A$ and $\angle B$ are right and $\overline{AD} \cong \overline{BC}$. The side $\overline{AB}$ is called the **base** of the quadrilateral, the side $\overline{CD}$ is called the **summit**, and the sides $\overline{AD}$ and $\overline{BC}$ are called the **legs**. The angle $\angle C$ and $\angle D$ are called the **summit angles**.

Since the legs of a Saccheri quadrilateral have a common perpendicular line, they are necessarily parallel to each other.

Proposition 5.3.2. *In congruence geometry, the diagonals of a Saccheri quadrilateral are congruent. Likewise, the summit angles of a Saccheri quadrilateral are congruent.*

Proof. Let $\square ABCD$ be a Saccheri quadrilateral with diagonals $\overline{AC}$ and $\overline{BD}$. By SAS, $\triangle ABD \cong \triangle BAC$. This implies that $\overline{BD} \cong \overline{AC}$. Then $\triangle ACD \cong \triangle BDC$ by SSS. In particular, $\angle D \cong \angle C$. $\square$

Proposition 5.3.3. *The segment joining the midpoints of both the summit and base of a Saccheri quadrilateral is called the **altitude** of the quadrilateral. The altitude is perpendicular to both the base and summit of the quadrilateral. In particular, this implies that the base and summit of Saccheri are parallel.*

The proof of this proposition is left as an exercise for the reader. Since base and summit of a Saccheri quadrilateral have a common perpendicular line, they are necessarily parallel to each other. The above proposition shows that a Saccheri quadrilateral is a parallelogram.

Theorem 5.3.4. *In neutral geometry, the summit angles of a Saccheri quadrilateral are not obtuse.*

Proof. We have already seen that the angle sum of a quadrilateral cannot exceed 360° by the Saccheri-Legendre Theorem. Let x be the measure of a summit angle of a Saccheri quadrilateral. Then the angle sum of the quadrilateral is $2x + 180^\circ \leq 360^\circ$. This implies that $x \leq 90^\circ$. $\square$

Of course, then the summit angles of a Saccheri quadrilateral are right or acute. As we will see, the summit angles being right is equivalent to the Euclidean parallel postulate and the angles being acute is equivalent to the Hyperbolic parallel postulate.

Theorem 5.3.5. *In neutral geometry, the base in a Saccheri quadrilateral is less than or equal to the summit.*

Proof. Let $\square ABCD$ be a Saccheri quadrilateral with diagonal $\overline{BD}$. If $\angle ADB \cong \angle DBC$ then $\triangle ABD \cong \triangle CDB$ by SAS. This implies that $\overline{AB} \cong \overline{CD}$. If $\angle ADB < \angle DBC$ then $\overline{AB} < \overline{CD}$ by the Hinge Theorem. Lastly, we must consider the case that $\angle ADB > \angle DBC$. We consider next the angle sum of $\triangle ABD$:

$$m\angle ADB + m\angle A + m\angle DBA = m\angle ADB + 90^\circ + (90^\circ - m\angle DBC) > m\angle DBC + 180^\circ - m\angle DBC = 180^\circ,$$

where the first equality follows since $\angle DBA$ and $\angle DBC$ are complementary. But this contradicts the Saccheri-Legendre Theorem, which disallows this last case. The result then follows. $\square$

Again, equality is equivalent to the Euclidean parallel postulate and less than is equivalent to the Hyperbolic parallel postulate. In particular, in neutral geometry it is not necessary that opposite sides of parallelograms are congruent. Such a congruence requirement is also equivalent to the Euclidean parallel postulate. In Hyperbolic geometry, parallelograms can have non-congruent opposite sides.

Lambert took an alternative approach to prove the existence of rectangles.

Definition 5.3.6. In congruence geometry, a **Lambert quadrilateral** $\square ABCD$ is a quadrilateral such that $\angle A$, $\angle B$, and $\angle C$ are right.

Likewise, a Lambert quadrilateral is also a parallelogram.

Proposition 5.3.7. *In neutral geometry, the fourth angle of a Lambert quadrilateral is not obtuse.*

Proposition 5.3.8. *In neutral geometry, the side of a Lambert quadrilateral between two right angles is less than or equal to the opposite side.*

The proofs of both of these theorems are left as an exercise to the reader.

As we will see, the fourth angle being right is equivalent to the Euclidean parallel postulate and the angle being acute is equivalent to the Hyperbolic parallel postulate. Again, equality of lengths is equivalent to the Euclidean parallel postulate and less than is equivalent to the Hyperbolic parallel postulate.

Theorem 5.3.9. *The altitude of a Saccheri quadrilateral is less than or equal to either leg.*

Proof. Let M and N be the midpoints of $\overline{AB}$ and $\overline{CD}$, respectively, in the Saccheri quadrilateral $\square ABCD$. By Proposition 5.3.3, the altitude $\overline{MN}$ is perpendicular to $\overline{AB}$ and $\overline{CD}$. Thus, $\square AMND$ and $\square BMNC$ are Lambert quadrilaterals. By the previous proposition, we conclude that $\overline{MN} < \overline{AD}, \overline{BC}$, which proves the result. $\square$

As to be expected, equality is equivalent to the Euclidean geometry and less than is equivalent to Hyperbolic geometry. Interesting enough, this shows that in Hyperbolic geometry parallel lines are not equidistant. In fact, equidistance of parallel lines is another equivalent of the Euclidean parallel postulate.

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be neutral geometries.

1. Prove that the line joining the midpoints of both the summit and the base of a Saccheri quadrilateral is perpendicular to both.
2. The summit and the base of a Saccheri quadrilateral are parallel.
3. Prove Proposition 5.3.7.
4. Prove Proposition 5.3.7.

“I always believe that the sky is the beginning of the limit.” – MC Hammer

5.4 Angle of Parallelism

While studying Saccheri and Lambert quadrilaterals we have seen that the summit and fourth angles must be less than or equal to 90° , but how small can they get? The initial goal was to prove that the summit or fourth angles are necessarily right in order to prove the Euclidean parallel postulate, but why does the presence of acute angles in Saccheri or Lambert quadrilaterals negate the Euclidean parallel postulate? The answers to both of these questions have a lot to do with the line $\overleftrightarrow{AB}$ and segment $\overline{AD}$.

Let $\ell = \overleftrightarrow{AB}$ be a line. Let t be the unique perpendicular line which passes through A on ℓ . Let D be some point on t not equal to A . Finally, let C be some point not on t and not equal to B . Let $m = \overleftrightarrow{CD}$. We then ask the question, “Are m and ℓ parallel?” Let

$$\mathcal{M} = \{m\angle ADC \mid C \in \angle BAD, \overleftrightarrow{AB} \nparallel \overleftrightarrow{CD}\} \subseteq \mathbb{R}.$$

This is a nonempty set of real numbers, since if $C = B$, $B \in \ell \cap m$. Also, if $m \perp t$, that is, $m\angle ADC = 90^\circ$, then $m \parallel \ell$ since they have a common perpendicular. Thus, $\mathcal{M}$ is bounded above since 90° is an upper bound for $\mathcal{M}$. Then the Least Upper Bound Property of the Real Numbers applies and says that $\mathcal{M}$ has a least upper bound, d_0 . We say that this value d_0 is the **angle of parallelism** for the line ℓ and point D . It should also be mentioned that the segment $\overline{AD}$ is significant in determining the angle of parallelism, not just the line ℓ .

Since 90° is an upper bound for any line and point by the Alternate Interior Angle Theorem, it holds that $d_0 \leq 90^\circ$ for any line and point. We should also note that if $m\angle ADC < d_0$, then $m\angle ADC \in \mathcal{M}$, which implies that $\ell \nparallel m$. Thus, any angle measure less than d_0 implies that the lines intersect. Similarly, if $m\angle ADC > d_0$, then $m\angle ADC \notin \mathcal{M}$, which implies that $\ell \parallel m$. Thus, any angle measure greater than d_0 implies that the lines are parallel.

Proposition 5.4.1. Suppose the angle of parallelism for line $\overleftrightarrow{AB}$ and point D is d_0 . Suppose $m\angle ADC = d_0$. Then $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$. (*Progressive Geometry Proofs Slides*)

Proof. Suppose to the contrary that $\overleftrightarrow{AB} \nparallel \overleftrightarrow{CD}$. Without the loss of generality, we may assume that $B \in \overleftrightarrow{AB} \cap \overleftrightarrow{CD}$. Let E be a point given by Extension that $A - B - E$. Since $A - B - E$, we know that $\angle ADC = \angle ADB < \angle ADE$. Then $d_0 = m\angle ADC < m\angle ADE$ but $\overleftrightarrow{DE} \nparallel \overleftrightarrow{AB}$, which is a contradiction to d_0 being the least upper bound. $\square$

Definition 5.4.2. A line $\overleftrightarrow{CD}$ parallel to line $\overleftrightarrow{AB}$ is called a **limiting parallel** (or **asymptotic parallel**) if $m\angle ADC = d_0$, the angle of parallelism for $\overleftrightarrow{AB}$ and D .

We need to also mention that as we defined it, the angle of parallelism was found using rays, not line. Fortunately, d_0 is well-defined, that is, if $C - D - C'$, then the angle of parallelism with respect to the ray $\overrightarrow{DC}$ is the same as the ray $\overrightarrow{DC'}$. Suppose to the contrary that they are different, where the angles of parallelism are d_0 and d'_0 for the sides containing C and C' , respectively. Without the loss of generality we may assume $d_0 > d'_0$. Then let $E \in H(\overleftrightarrow{AB}, C)$ such that $m\angle ADE = d'_0$. This implies that $\overleftrightarrow{DE} \nparallel \overleftrightarrow{AB}$, that is, $\triangle ADE$ is a triangle with $m\angle ADE = d'_0$. Translate this triangle into $H(\overleftrightarrow{AD}, C')$, that is, let E' be a point on the opposite side of $\overleftrightarrow{AD}$ as E such that $\triangle ADE \cong \triangle ADE'$. Thus, $m\angle ADE' = d'_0$ but $\overleftrightarrow{DE'} \parallel \overleftrightarrow{AB}$, which contradicts d'_0 being the angle of parallelism on this side of the line.

For a Saccheri quadrilateral, the summit angles are necessarily greater or equal to the angle of parallelism. Likewise, the fourth angle of a Lambert quadrilateral is greater than or equal to the angle of parallelism.

Theorem 5.4.3. *If the angle of parallelism for line ℓ and point P is acute, then there exists more than one line through P which is parallel to ℓ , that is, the Hyperbolic parallel postulate holds.*

The proof of the above extremely important theorem is simple and left as an exercise for the reader.

Theorem 5.4.4. *In neutral geometry, the Euclidean parallel postulate is equivalent to the statement: The measure of the angle of parallelism is 90° for all lines and points.*

Proof. Assume the Euclidean parallel postulate. Let ℓ be a line and let $P \notin \ell$. Let t be the unique perpendicular line to ℓ through P . Let d_0 be the angle of parallelism for line ℓ and point P . Let m be a limiting parallel through P that obtains this angle d_0 , which exist by Dedekind's Axiom. But by the Euclidean parallel postulate, m is the only line parallel to ℓ through P . On the other hand, by the Alternate Interior Angle Theorem, the unique line perpendicular to t through P is parallel to ℓ . Thus, m is this line, which implies that $d_0 = 90^\circ$ since $m \perp t$. This proves the first direction.

For the reverse direction, suppose angle of parallelism is always $d_0 = 90^\circ$. Let ℓ be a line and $P \notin \ell$. Let t be the transversal such that $P \in t$ and $t \perp \ell$. We know from the Alternate Interior Angle Theorem that at least one parallel line exists, specifically the line perpendicular to t and contains P . This line, of course, forms a right angle with t , and hence is a limiting parallel since the angle measure is $d_0 = 90^\circ$. Therefore, no other parallel line can form an angle with t less than 90° . But the two angles formed by such a parallel line on either side of t are supplementary, that is, either one of the angles is acute or they are both right. As the first case is impossible, every parallel line to ℓ is perpendicular to t . As perpendicular lines are unique, the Euclidean parallel postulate follows. $\square$

The proof of the previous theorem actually shows a more powerful result: the number of lines parallel to ℓ through P is at most one if and only if the angle of parallelism is equal to 90° .

Theorem 5.4.5. *In neutral geometry, the hyperbolic parallel postulate is equivalent to the statement: The measure of the angle of parallelism is acute for all lines and points.*

Proof. One direction is handled by Theorem 5.4.3. Let the geometry be hyperbolic, that is, there exist more than one line parallel to ℓ through P . Thus, the angle of parallelism is not 90° . Since the angle of parallelism is less than or equal to 90° , we conclude the angle must be acute. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be neutral geometries.

1. Prove that if the angle of parallelism for line ℓ and point P is acute, then there exists more than one line through P which is parallel to ℓ , that is, the Hyperbolic parallel postulate holds.
2. Prove that if $\angle RPQ$ is the angle of parallelism for P and $\overleftrightarrow{RS}$, $\angle R'P'Q'$ is the angle of parallelism for P' and $\overleftrightarrow{R'S'}$, and $\overline{PR} \cong \overline{P'R'}$, then $\angle RPQ \cong \angle R'P'Q'$.

ⁱThe progressive slides in this section were created by Hayley John.

Chapter 6

Euclidean Geometry

“The noblest search is the search for excellence.” – Lyndon B. Johnson

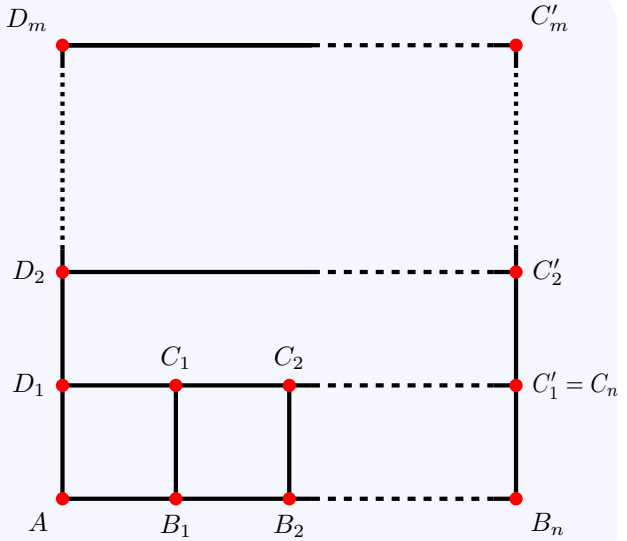
6.1 Rectangles

Definition 6.1.1. We say that a quadrilateral $\square ABCD$ is a **rectangle**ⁱ if all its angles are right.

Lemma 6.1.2. In neutral geometry, if one rectangle exists, then there exists a rectangle with two arbitrary large side measure.

Proof. Let $\square ABCD = \square AB_1C_1D$ be a rectangle. D_m Using segment translation and extension, there exist points B_2 and C_2 such that $A-B_1-B_2$, $D-C_1-C_2$, $\overline{AB_1} \cong \overline{B_1B_2}$, and $\overline{DC_1} \cong \overline{C_1C_2}$. Of course, $m\angle C_2C_1B_1 = m\angle B_2B_1C_1 = 90^\circ$ since they are supplements to right angles. Therefore, $\square ABCD \cong \square B_1B_2C_2C_1$ by SASAS. Thus, $\square B_1B_2C_2C_1$ is a rectangle, which implies that $\square AB_2C_2D$ is also a rectangle with side length $AB_2 = 2(AB)$. By induction, there exists a rectangle $\square AB_nC_nD$ such that $AB_n = n(AB)$. Repeating this process in the other direction, we can construct a rectangle $\square AB_nC'_mD_m$ with sides $AB_n = n(AB)$ and $AD_m = m(AD)$.

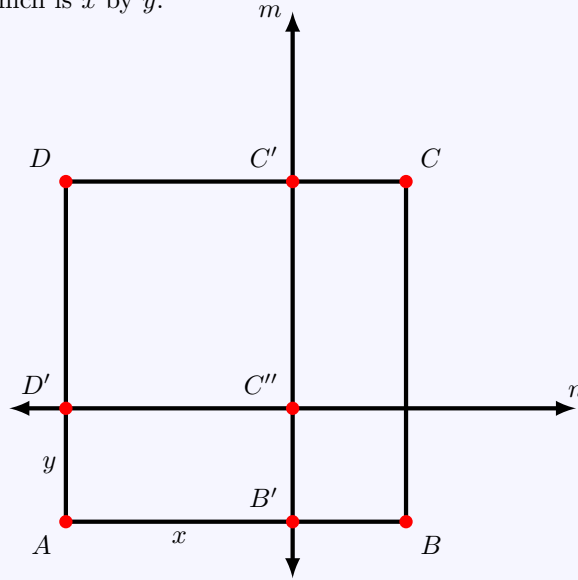
By the Archimedean property, we can choose n and m sufficiently large so that $AB_n > x$ and $AD_m > y$ for any real numbers x and y . $\square$



Lemma 6.1.3. In neutral geometry, if one rectangle exists, then there exists a rectangle with sides measures equal to x and y for any two positive real numbers x and y .

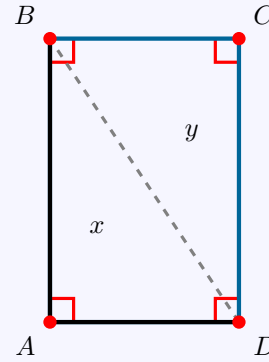
Proof. Let $\square ABCD$ be a rectangle such that $AB > x$ and $AD > y$. Then there exists a point B' such that $A-B'-B$ and $AB' = x$. Let m be the unique line perpendicular to $\overleftrightarrow{CD}$ and contains B' . Let $C' \in m \cap \overleftrightarrow{CD}$. Necessarily $D-C'-C$, otherwise we could construct a triangle with angle sum greater than 180° . Then $\square AB'C'D$ and $\square B'BCC'$ are Lambert quadrilaterals, where $\angle AB'C'$ and $\angle BB'C'$ are

the fourth angles, respectively. If $\angle AB'C'$ is acute, then $\angle BB'C$ is obtuse, a contradiction. Therefore, $\angle AB'C'$ is right, that is, $\square AB'C'D$ is a rectangle with one side measure equal to x . A similar construction can produce a rectangle which is x by y . $\square$



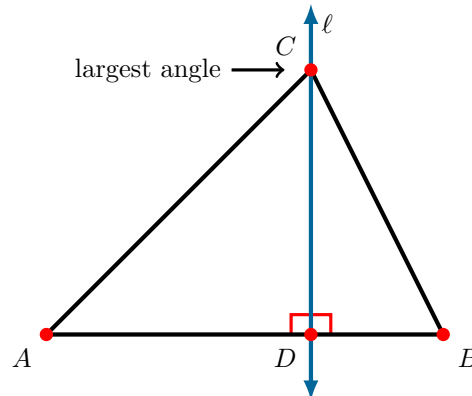
Lemma 6.1.4. *In neutral geometry, if one rectangle exists, then every right triangle has an angle sum of 180° .*

Proof. Let $\triangle PQR$ be a right triangle with $m\angle Q = 90^\circ$. Let x be the angle sum of $\triangle PQR$. By the previous lemma, there exists a rectangle $\square ABCD$ such that $AB = PQ$ and $AD = QR$. Thus, $\triangle ABD \cong \triangle PQR$ by SAS. By triangle translation, we may assume $P = B$, $Q = A$, and $R = D$. Thus, the angle sum of $\triangle ABD$ is x . Let y be the angle sum of the right triangle $\triangle CBD$. Then $x + y = 360^\circ$. If $x < 180^\circ$, then $y > 180^\circ$, which contradicts the Saccheri-Legendre Theorem. Since $x \not> 180^\circ$ by the same theorem, we conclude that $x = 180^\circ$. $\square$



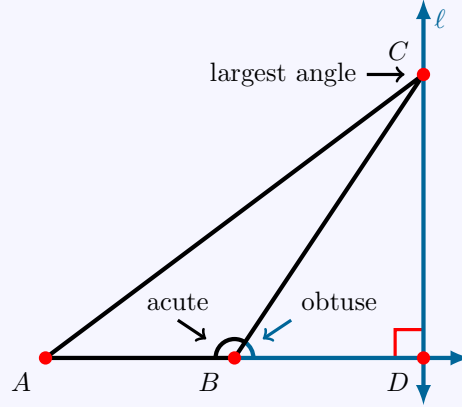
Theorem 6.1.5. *In neutral geometry, if one rectangle exists, then every triangle has an angle sum of 180° .*

If an **altitude** of a triangle is the line which is perpendicular to a side of the triangle and passes through the vertex on the opposite side, then the above theorem can be proven using the altitude of the largest angle and the above lemma. The details of the proof are left as an exercise to the reader.



Theorem 6.1.6 (The All-Or-Nothing Theorem). *In neutral geometry, if one triangle has an angle sum equal to 180° , then a rectangle exists.*

Proof. Let $\triangle ABC$ be a triangle with angle sum equal to 180° . Let $\angle C$ be the largest angle and let ℓ be an altitude of $\triangle ABC$. Thus, $\ell \perp \overleftrightarrow{AB}$ and let $D \in \ell \cap \overleftrightarrow{AB}$. We claim that $A - D - B$. If not, we may without the loss of generality assume that $A - B - D$. Then the triangle $\triangle CDB$ is right, with right angle $\angle CDB$. Also, since $\angle C$ is the largest angle of $\triangle ABC$, the angle $\angle ABC$ must be acute. Thus, its supplement $\angle DBC$ must be obtuse. This implies that the angle sum of $\triangle CDB$ exceeds 180° , a contradiction. Thus, $A - D - B$.



Let x and y be the angle sums of $\triangle ADC$ and $\triangle BDC$, respectively. Then we have

$$\begin{aligned} x + y &= (m\angle A + m\angle ADC + m\angle DCA) + (m\angle B + m\angle BDC + m\angle DCB) \\ &= (m\angle A + m\angle DCA + m\angle DCB + \angle B) + m\angle ADC + m\angle BDC \\ &= 180^\circ + 90^\circ + 90^\circ = 360^\circ. \end{aligned}$$

As we have seen before, both x and y are positive real numbers which cannot exceed 180° . Thus, we conclude that $x = y = 180^\circ$.

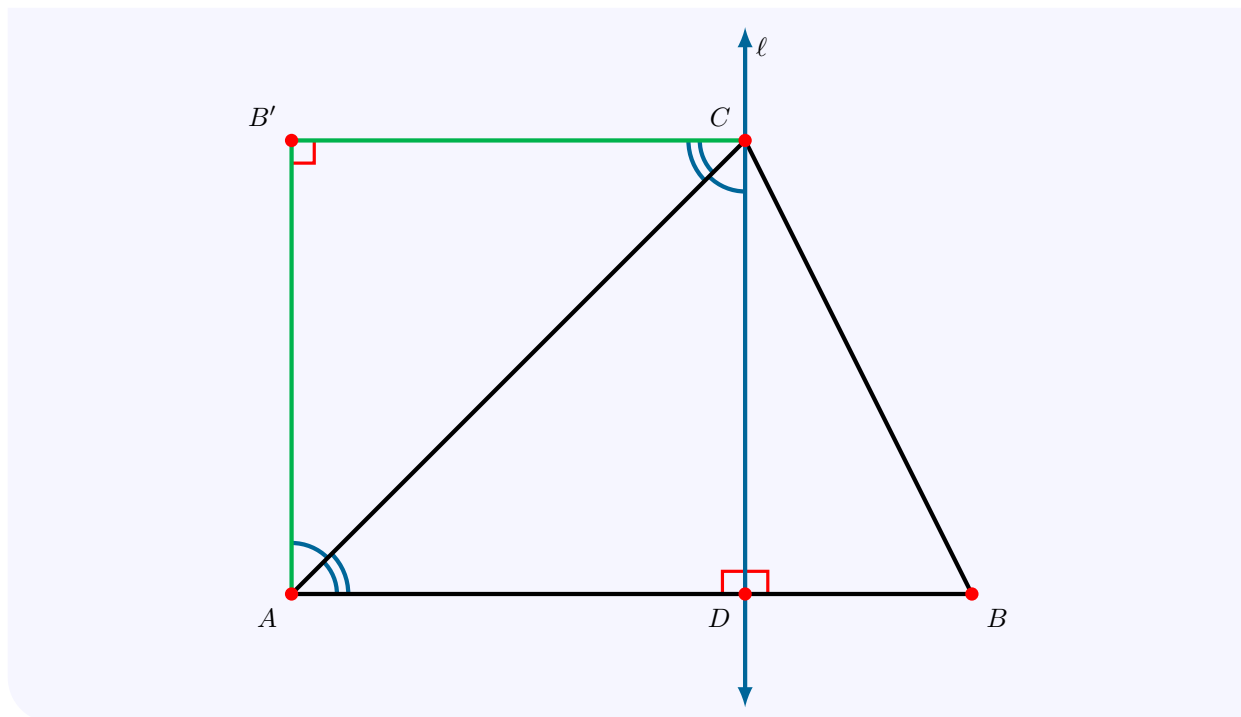
Next, we will translate the right triangle $\triangle ACD$ onto the segment $\overline{AC}$ on the opposite of D using the point B' such that $\triangle ACD \cong \triangle CAB'$. Thus, $\angle ACD \cong \angle CAB'$ and $\angle CAD \cong \angle ACB'$. Then the quadrilateral $\square AB'CD$ has the angle sum:

$$\begin{aligned} m\angle A + m\angle B' + m\angle C + m\angle D &= (m\angle CAD + m\angle CAB') + m\angle AB'C + (m\angle ACD + m\angle ACB') + m\angle ADC \\ &= (m\angle ADC + m\angle CAD + m\angle ACD) + (m\angle AB'C + m\angle CAB' + m\angle ACB') \\ &= 180^\circ + 180^\circ = 360^\circ \end{aligned}$$

But on the other hand,

$$\begin{aligned} m\angle A + m\angle B' + m\angle C + m\angle D &= (m\angle CAD + m\angle CAB') + m\angle AB'C + (m\angle ACD + m\angle ACB') + m\angle ADC \\ &= (m\angle CAD + m\angle CAB') + 90^\circ + (m\angle CAB' + m\angle CAD) + 90^\circ \\ &= 180^\circ + 2(m\angle CAD + m\angle CAB') = 180^\circ + 2(m\angle A) \end{aligned}$$

Thus, $180^\circ + 2(m\angle A) = 360^\circ \Rightarrow m\angle A = 90^\circ$. Similarly, we have $m\angle C = 90^\circ$. Therefore, $\square AB'CD$ is a rectangle. $\square$



This theorem has two immediate corollaries whose proofs are left as exercises to the reader.

Corollary 6.1.7. *In neutral geometry, if one triangle has an angle sum equal to 180° , then all triangles have angle sums equal to 180° .*

Corollary 6.1.8. *In neutral geometry, if one triangle has an angle sum less than 180° , then all triangles have angle sums less than 180° .*

The collection of the statements from the above theorem and its corollaries are known as the All-Or-None Theorem.

Exercises

*Unless explicitly stated otherwise, all geometries in these exercises are assumed to be **neutral** geometries.*

1. Prove Theorem 6.1.5.
2. Prove Corollary 6.1.7.
3. Prove Corollary 6.1.8.

ⁱAn alternative way to define a rectangle is to define them to be quadrilaterals which are equiangular, that is, all angles are congruent. Of course, in Euclidean geometry the two notions are equivalent since the angle sum of a quadrilateral must be 360° . In hyperbolic geometry, quadrilaterals can be equiangular but no quadrilateral can have four right angles since the angle sum is less than 360° .

ⁱⁱFigures in this section were created by Ethan Olcott.

“Distinctions drawn by the mind are not necessarily equivalent to distinctions in reality.”
 – Thomas Aquinas

6.2 Euclidean Geometry

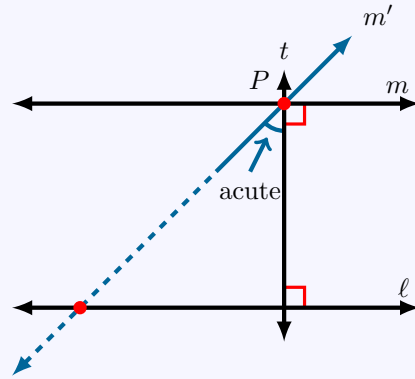
Definition 6.2.1. We say that a neutral geometry is **Euclidean** if in addition it satisfies the Euclidean parallel postulate.

Geometers extensively tried to prove that the Euclidean parallel postulate was a theorem of neutral geometry, but failed. This is due to the fact the the Euclidean parallel postulate is independent of the neutral axioms, evidence of this being consistent models of hyperbolic geometry. Although the Euclidean parallel postulate could not be proven from the neutral axioms, many important equivalents of the Euclidean parallel postulate were discovered which are very useful theorems of Euclidean (and neutral) geometry!

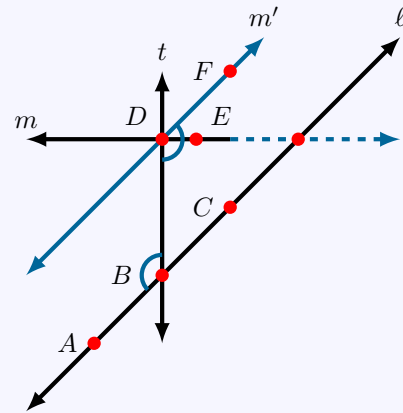
Throughout, we will never need to prove that at least one parallel exists, because this has already been proven as a consequence of the Alternate Interior Angle theorem using perpendicular lines. We will refer to this parallel line as the **guaranteed parallel line**. There is also a unique line which is perpendicular to the original line and the guaranteed parallel intersecting the guaranteed parallel at the specified point. We will call this line the **perpendicular transversal**. Therefore, to prove that some statement implies the Euclidean parallel postulate, we will only need to prove that any other parallel line is equal to the guaranteed one.

Theorem 6.2.2. *In neutral geometry,¹ the Euclidean parallel postulate is equivalent to Euclid’s fifth postulate, namely, if two distinct lines ℓ , m intersect a transversal t in such a way that the sum of the consecutive interior angles is less than a flat angle, then ℓ , m meet on the side of t having the these two consecutive interior angles.*

Proof. Assume first Euclid’s fifth postulate. Let ℓ be a line and let $P \notin \ell$ be a point. Let m be the guaranteed parallel line through P and let t be the perpendicular transversal. Suppose that m' is any other line passing through P . Since $m' \not\perp t$, one of the vertical angle pairs formed by m' and t must be acute. Thus, one side of t contains an acute angle from m' and a right angle from ℓ . By Euclid’s fifth postulate, $m' \not\parallel \ell$ since they intersect on this side of t . Therefore, m is the unique line parallel to ℓ containing P , which proves the Euclidean parallel postulate.



Assume next the Euclidean parallel postulate. Let ℓ, m be lines with transversal t such that on one side of t the consecutive interior angles add to less than a flat angle. In particular, there exists points $A, B, C \in \ell$, $D, E \in m$ such that $A - B - C$, $B \in \ell \cap t$, $D \in m \cap t$, $E \in H^\circ(t, C)$, and $m\angle CBD + m\angle EDB < 180^\circ$. Since $\angle ABD$ is supplementary to $\angle CBD$, we have that $m\angle ABD + m\angle CBD = 180^\circ$, which implies that $\angle EDB < \angle ABD$. By angle translation, there exists a point $F \in H^\circ(t, E)$ such that $\angle BDF \cong \angle ABD$. By the Alternate Interior Angle Theorem, $m' = \overleftrightarrow{DF} \parallel \ell$. Also, $m' \neq m$ since $\angle EDB < \angle BDF$.

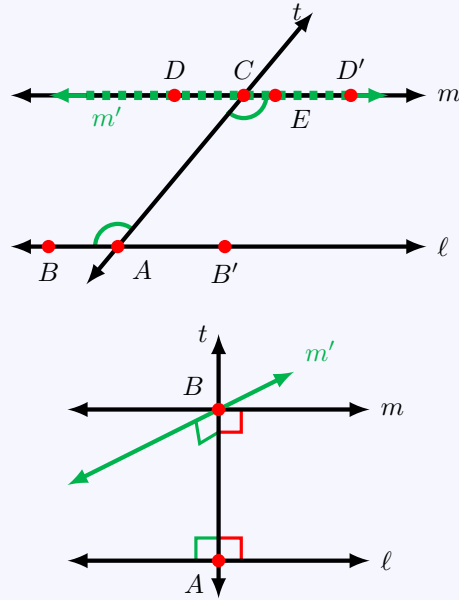


Thus by the Euclidean parallel postulate, $m \nparallel \ell$, and the intersection must be in $H^\circ(t, C)$ since the sum of the supplements of these consecutive interior angles is greater than 180° . This proves Euclid's fifth postulate. $\square$

Theorem 6.2.3. *In neutral geometry, the Euclidean parallel postulate is equivalent to the converse of the Alternate Interior Angle Theorem, that is, if two lines are parallel then their alternate interior angles are congruent.*

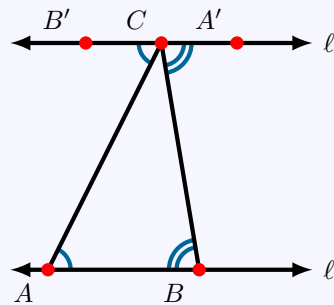
Proof. Assume first the Euclidean parallel postulate. Let $\ell \parallel m$ and let t be a transversal of these lines such that $A \in \ell \cap t$ and $C \in m \cap t$. Let $B, B' \in \ell$ and $D, D' \in m$ such that $B - A - B'$, $D - C - D'$ and $D \in H^\circ(t, B)$. By angle translation, there exists a point $E \in \angle D'CA$ such that $\angle BAC \cong \angle ECA$. Let $m' = \overleftrightarrow{CE}$. Since the alternate interior angles are congruent, we conclude that $\ell \parallel m'$ by the Alternate Interior Angle Theorem. But by the Euclidean parallel postulate, there is only one line passing through C that is parallel to ℓ . Therefore, $m = m'$ and alternate interior angles of ℓ and m with respect to t are congruent. This proves the Converse of the Alternate Interior Angle Theorem.

The remaining implication is left as an exercise to the reader. An illustration that, well, illustrates the basic idea of the proof is provided. $\square$



Theorem 6.2.4. *In neutral geometry, the Euclidean parallel postulate is equivalent to the angle sum of a triangle equaling 180° .*

Proof. Assume first the Euclidean parallel postulate, and let $\triangle ABC$ be a triangle. Let ℓ be the unique line parallel to $\overleftrightarrow{AB}$ containing C . Let A' and B' be points on ℓ such that $A' - C - B'$ and $A' \in H^\circ(\overleftrightarrow{AC}, B)$. A simple application of Plane Separation and the Crossbar Theorems will show that $B' \in H^\circ(\overleftrightarrow{BC}, A)$. Then $m\angle ACB' + m\angle ACB + m\angle A'CB = m\angle ACB' + m\angle ACA' = 180^\circ$, since A and B are on the same side of ℓ . By the converse of the Alternate Interior Angle Theorem, we have that $\angle A = \angle BAC \cong \angle ACB'$ and $\angle B = \angle ABC \cong \angle BCA'$. Therefore, $m\angle A + m\angle C + m\angle B = m\angle BAC + m\angle ACB + m\angle ABC = 180^\circ$.



The other half of the proof will be provided in the next section. $\square$

Corollary 6.2.5. *In neutral geometry, the Euclidean parallel postulate is equivalent to the existence of rectangles.*

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be **neutral** geometries.

1. Finish the proof of Theorem 6.2.3, that is, prove that the converse of the Alternate Interior Angle Theorem implies the Euclidean Parallel Postulate.

Lemma (Proclus' Lemma). *Suppose that ℓ , m , and n are lines such that ℓ is parallel to m and m intersects n . Then ℓ intersects n also.*

2. Prove that the Euclidean parallel postulate is equivalent to Proclus' lemma.

Lemma 6.2.6 (Right Angle Proclus' Lemma). *Suppose that ℓ , m , and n are lines such that $\ell \parallel m$ and $m \perp n$. Then $\ell \perp n$ also.*

3. Prove that the Euclidean parallel postulate is equivalent to the Right Angle Proclus' lemma.

ⁱThis equivalence actually holds in congruence geometry. On the other hand, the equivalence of the Euclidean parallel postulate to right angles of parallelism requires continuity and neutral geometry is the setting for that equivalence.

ⁱⁱFigures in this section were created by Ethan Olcott.

“Too much may be the equivalent of none at all.” – Lee Loevinger

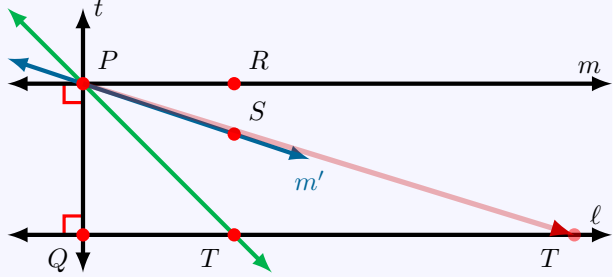
6.3 Equivalents to the Euclidean Parallel Postulate

Remainder of the Proof of Theorem 6.2.4.

Assume next that the angle sum of each triangle is 180° . Let ℓ be a line, let P be a point with $P \notin \ell$, let m be the guaranteed parallel, and let t be the perpendicular transversal. For the sake of contradiction let $m' \neq m$ be a line parallel to ℓ through P . Let $Q \in \ell \cap t$, let $S \in m'$ be a point not equal to P such that $\angle SPQ$ is acute (such a point exists since $m' \not\perp t$), and let $R \in m \cap H^\circ(t, S)$. Then $S \in \angle RPQ$ since $\angle SPQ < \angle RPQ$. By the Archimedean property, there exists a point $T \in \ell \cap H^\circ(t, S)$ such that $m\angle QTP < m\angle SPR$. By trichotomy on the betweenness of rays, either $\overrightarrow{PQ} - \overrightarrow{PT} - \overrightarrow{PS}$ or $\overrightarrow{PQ} - \overrightarrow{PS} - \overrightarrow{PT}$. If the latter happens, then $S \in \angle QPT$ and the Crossbar Theorem would say that $\overrightarrow{PS}$ intersects $\overline{QT}$, which contradicts $m' \parallel \ell$. Thus, we may assume $T \in \angle QPS$. Then $m\angle TPQ < m\angle SPQ$. Therefore,

$$m\angle QTP + m\angle TPQ < m\angle SPR + m\angle SPQ = 90^\circ,$$

since $\angle SPR$ and $\angle SPQ$ are complementary. But since the third angle of $\triangle PQT$ is right, this above inequality shows that $\triangle PQT$ has an angle sum less than 180° , which is a contradiction. Therefore, parallel lines are unique. $\square$



We summarize next a few of equivalent statements to the Euclidean parallel postulate most of which have already been proven (even if the proof was only implied):

- 1) Uniqueness of parallel lines through a point (this is Playfair's and our version of the axiom).
- 2) Proclus' Lemma (We proved one direction in affine geometry. The other direction is immediate in congruence geometry since parallels exist).
- 3) Transitivity of Parallelism (Ditto).
- 4) Angles of Parallelism are always right.
- 5) Euclid Fifth postulate.
- 6) The Converse to the Alternate Interior Angle Theorem.
- 7) Triangles have angle sum equal to 180° (by the All-or-None Theorem, if one triangle does then they all do, that is, the quantifier on the triangle does not matter).
- 8) Rectangles exist.
- 9) The Right Angle Hypothesis (Summit angles of Saccheri and the Fourth angle of Lambert quadrilaterals are right).
- 10) Parallel lines are equidistant. (This follows from the right angle hypothesis for Saccheri quadrilaterals).
- 11) The Pythagorean Theorem or its converse (a right triangle is dissected using multiple altitudes to form a rectangle).

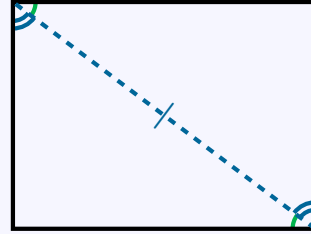
More examples are listed in the text.

Although there are numerous equivalents to the Euclidean parallel postulate, we will not prove the equivalence of all of these. Instead we will be more interested in proving theorems of Euclidean geometry.

Many of these theorems will be genuine equivalents but we will not bother to prove the converses, for example, the next two theorems are actually equivalents but we will not provide the converse. The only need to know if two statements are logically equivalent is if we are trying to prove a specific interpretation of undefined terms is a model for a certain axiomatic system and we believe it to be easier to prove an equivalent to an axiom instead of the original or if we want to study a more broad or more narrow axiomatic system, that is, we want to create independent axioms.

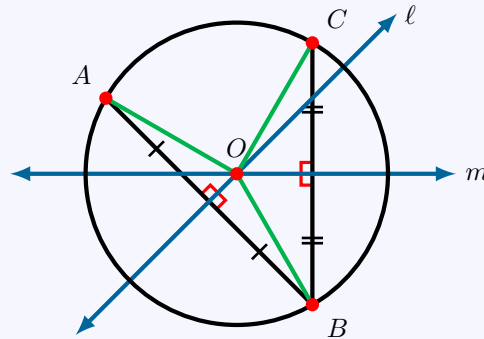
Theorem 6.3.1. *In Euclidean geometry, the opposite sides of parallelograms are congruent.*

Proof. Let $\square ABCD$ be a parallelogram, that is, $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ and $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$. Consider the triangles $\triangle ABD$ and $\triangle CDB$. By the converse to the Alternate Interior Angle Theorem, we have $\angle ABD \cong \angle CDB$ and $\angle ADB \cong \angle CBD$. Obviously, $\overline{BD} \cong \overline{BD}$. Thus, $\triangle ABD \cong \triangle CDB$ by ASA. In particular, $\overline{AB} \cong \overline{CD}$ and $\overline{AD} \cong \overline{BC}$, which proves the theorem. $\square$



Theorem 6.3.2. *In Euclidean geometry, three non-collinear points determine a unique circle. (Progressive Geometry Proofs Slides)*

Proof. Let A , B , and C be three non-collinear points. Let ℓ and m be the two perpendicular bisectors of $\overline{AB}$ and $\overline{BC}$. We claim that ℓ and m intersect each other. If not, that is, if $\ell \parallel m$, then m intersects $\overleftrightarrow{AB}$ by Proclus' lemma. By the converse of the Alternate Interior Angle theorem, $m \perp \overleftrightarrow{AB}$ since $\ell \perp \overleftrightarrow{AB}$. But then either $\overleftrightarrow{AB} \parallel \overleftrightarrow{BC}$ since $\overleftrightarrow{AB} \neq \overleftrightarrow{BC}$ and by the Alternate Interior Angle theorem with m as a common perpendicular. This contradicts $B \in \overleftrightarrow{AB} \cap \overleftrightarrow{BC}$. Thus, ℓ and m intersect at some point O .



By a basic SAS argument, we see that $\overline{OA} \cong \overline{OB} \cong \overline{OC}$. Thus, A , B , C are all incidence to the circle centered at O and with radius $\overline{OA}$. This circle is unique, since by a basic SSS argument shows that $\overline{PA} \cong \overline{PB} \cong \overline{PC}$ implies that $P \in \ell \cap m = \{O\}$. $\square$

Exercises

*Unless explicitly stated otherwise, all geometries in these exercises are assumed to be **neutral** geometries.*

1. Prove that if all parallelograms containing one right angle are rectangles, then the geometry is Euclidean.
2. Prove that the Euclidean parallel postulate is equivalent to the congruence of opposite sides of parallelograms.
3. In **Euclidean** geometry, prove that the diagonals of a parallelogram bisect each other.

ⁱFigures in this section were created by Ethan Olcott. The progressive slides in this section were created by Hayley John.

Chapter 7

Hyperbolic Geometry

“Achieving life is not the equivalent of avoiding death.” – Ayn Rand

7.1 The Hyperbolic Parallel Postulate

It is not the intention of this book to provide proofs of all the well-known theorems of Euclidean geometry nor even a subset of these theorems. With few exceptions, the reader has certainly had great exposure and familiarity with these theorems through past courses in (high-school) geometry, trigonometry, calculus, and linear algebra. As such, for the remainder of this book, we will study non-Euclidean geometries, namely hyperbolic and elliptic, and any future reference to Euclidean geometry will only serve as a comparison to non-Euclidean geometries. Any reader interested in learning more about the development of Euclidean geometry from its axioms is encouraged to study the one of the many good textbooks concerning Euclidean geometry. Our rigorous treatment of axiomatic geometry should have well prepared the reader to be able to study the remaining topics of Euclidean geometry individually. Believe in yourself and give it a try! Life long learning will not take place in a classroom with an instructor. It rests in your hands.

Definition 7.1.1. We say that a neutral geometry is **hyperbolic** if in addition it satisfies the hyperbolic parallel postulate.

There exists many equivalents to the hyperbolic parallel postulate are listed in the text. Essentially the negation of any equivalent of the Euclidean parallel postulate within the realm of the Saccheri-Legendre theorem is an equivalent to the hyperbolic parallel postulate. For example, the Acute Angle Hypothesis is equivalent to the hyperbolic parallel postulate since the negation of the Euclidean parallel postulate says the summit angles of a Saccheri quadrilateral are not right. They cannot be obtuse, by which we conclude that they must be acute.

But what about the negation of the converse of the Alternate Interior Angle theorem? The negation would simply say that there exists a pair of parallel lines and a transversal such that alternate interior angles are not congruent. It does not say that alternate interior angles are never congruent. In fact, the existence of parallel lines in congruence geometry follows from the Alternate Interior Angle theorem because the parallel lines in question have a common perpendicular. Thus, a natural question to ask is when do parallel lines satisfy the converse of the Alternate Interior Angle Theorem. The answer is simple: hyperparallels (to be defined later) where the transversal intersects the midpoint of the common perpendicular.

Theorem 7.1.2. *Two parallel lines that are crossed by a transversal have congruent alternate interior angles if and only if the transversal contains the midpoint of a segment that is perpendicular to both lines.*

Proof. Let ℓ and m be two parallel lines crossed by transversal t such that alternate interior angles are congruent. Let $C \in \ell \cap t$, $D \in m \cap t$, let M be the midpoint of $\overline{CD}$, and let A be the foot of the

perpendicular dropped from M onto ℓ . Let B be a point on the line m such that B and A are on opposite side of t and $\overline{BD} \cong \overline{AC}$. By assumption $\angle ACM \cong \angle BDM$, the alternate interior angles. Also, $\overline{CM} \cong \overline{DM}$ as M is the midpoint of $\overline{CD}$. Then $\triangle ACM \cong \triangle BDM$ by SAS. Thus, $\angle AMC \cong \angle BMD$ as corresponding parts of congruent triangles are congruent. Let $B' \in \overleftrightarrow{AM}$ be a point such that $A - M - B'$. Then $\angle AMC \cong \angle B'MD$, as they are vertical angles. Thus, $\angle BMD \cong \angle B'MD$. Note that B' and A are on opposite sides of t since $A - M - B'$. Thus, B and B' are on the same side of t . Therefore, by uniqueness of angle translation, we see that $\overrightarrow{MB} = \overrightarrow{MB'}$. This shows that A , M , and B are all collinear and $B \in \overleftrightarrow{AM} \cap m$. Additionally, $\angle BMD \cong \angle CAM$, which implies they are both right angles. Thus, $\overleftrightarrow{AB}$ is a common perpendicular to ℓ and m . Also, $\overline{AM} \cong \overline{MB}$, that is, M is the midpoint for the common perpendicular. This proves the first direction. The other direction is left as an exercise for the reader. $\square$

Definition 7.1.3. Let ℓ and ℓ' be two lines in a congruence geometry. Let $A, B, C, \dots \in \ell$, and let $A', B', C', \dots \in \ell'$ be the feet of the perpendicular lines dropped from $A, B, C, \dots$, respectively, onto ℓ' . We say that $A, B, C, \dots$ are **equidistant** from ℓ' if $\overline{AA'} \cong \overline{BB'} \cong \overline{CC'} \cong \dots$

Theorem 7.1.4 (At most two points equidistant). *Given two distinct lines in hyperbolic geometry, no equidistant set contains more than two points.*

Proof. Suppose to the contrary that $A, B, C \in \ell$ are equidistant from ℓ' . Without the loss of generality, we may assume that $A - B - C$. Let $A', B', C' \in \ell'$ be the respective feet of the perpendiculars dropped from A, B, C onto ℓ' . We can then conclude $A' - B' - C'$, otherwise B and B' would be on opposite sides of one of the other perpendiculars thus causing parallel lines to intersect. Then the quadrilaterals $\square A'B'BA$, $\square B'C'CB$, and $\square A'C'CA$ are Saccheri. Then the summit angles are congruent, namely:

$$\angle A'AB \cong \angle B'BA, \quad \angle B'BC \cong \angle C'CB, \quad \angle A'AB \cong \angle C'CB$$

By transitivity, $\angle B'BA \cong \angle B'BC$. As these two angles are supplements, they must both be right. Therefore, rectangles exist, a contradiction to the hyperbolic parallel postulate. $\square$

Corollary 7.1.5. *The length of altitude of a Saccheri quadrilateral is strictly less than the length of either leg in hyperbolic geometry.*

In particular, in hyperbolic geometry, there are now two cases to consider for parallel lines. What type of parallels have an equidistant pair and which do not? In the latter case, by Dedekind's axiom the Intermediate Value Theorem applies to the distance between the parallel lines. Thus, two parallel lines are nowhere equidistant if and only if the distance between the lines is monotonic, that is, the distance is monotonically increasing or decreasing, depending on the ordering of the lines. The next result follows after a bit of calculus.

Theorem 7.1.6. *If ℓ and ℓ' be two parallel lines in hyperbolic geometry, then ℓ' is an asymptotic parallel to ℓ (a parallel line obtaining the angle of parallelism on at least one side) if and only if ℓ and ℓ' are nowhere equidistant.*

Definition 7.1.7. We say that two parallel lines are **hyperparallel** if there exists a pair of equidistant points.

Hyperparallel lines are the types of parallels involved in creating Saccheri quadrilaterals, which provides the proof of the following theorem and is left to the reader.

Theorem 7.1.8. *In hyperbolic geometry, parallel lines are hyperparallel if and only if they have a common perpendicular.*

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be hyperbolic geometry.

1. Prove that if two lines share a common perpendicular the distance between the lines is least on that segment.
2. Prove that if a transversal crossing two parallel lines contains the midpoint of a common perpendicular to those lines, then the alternate interior angles are congruent.

“To realize that you do not understand is a virtue; Not to realize that you do not understand is a defect.”
– Lao Tzu

7.2 Defect of Triangles

We have seen previously that in neutral geometry, if one triangle has an angle sum of 180° then they all do and this is equivalent to Euclidean geometry. Therefore, in hyperbolic geometry all triangle have an angle sum less than 180° . The angle amount that a triangle’s angle sum is less than 180° is called the **defect** of the triangle. For triangle $\triangle ABC$, we will denote the defect as $d(\triangle ABC)$.

It is important to mention that the defects of triangles are not all the equal in hyperbolic geometry. In fact, if the defect was constant, that is, all triangles have the same defect, then we can argue that every defect is 0° , that is, all angle sums are 180° , which implies Euclidean. Therefore, there exist triangles in hyperbolic geometry with different defects. At least we can say by the All-Or-Nothing and Saccheri-Legendre theorems that $d(\triangle ABC) > 0$ for all triangles in hyperbolic geometry. Also, if $\triangle ABC \cong \triangle A'B'C'$ then $d(\triangle ABC) = d(\triangle A'B'C')$ since the two triangles have identical angle sums.

Theorem 7.2.1 (Additivity of Defects). *Let $\triangle ABC$ be a hyperbolic triangle and let $A - D - B$. Then*

$$d(\triangle ABC) = d(\triangle ADC) + d(\triangle BDC).$$

Proof. We note that $m\angle ADC + m\angle BDC = 180^\circ$ since the angles are supplementary and that from the definition of defect we have

$$\begin{aligned} d(\triangle ADC) &= 180^\circ - (m\angle DAC + m\angle ADC + m\angle ACD) \\ d(\triangle BDC) &= 180^\circ - (m\angle BDC + m\angle DBC + m\angle DCB) \\ d(\triangle ABC) &= 180^\circ - (m\angle BAC + m\angle ABC + m\angle ACB) \\ &= 180^\circ - (m\angle DAC + m\angle DBC + (m\angle ACD + m\angle BCD)) \\ &= 180^\circ - (m\angle DAC + m\angle DBC + m\angle ACD + m\angle BCD) + 180^\circ - (m\angle ADC + m\angle BDC) \\ &= 180^\circ - (m\angle DAC + m\angle ACD + m\angle ADC) + 180^\circ - (m\angle DBC + m\angle BCD + m\angle BDC) \\ &= d(\triangle ADC) + d(\triangle BDC), \end{aligned}$$

which proves the result. □

Note that using the previous theorem as a base case, if a triangle $\triangle ABC$ is dissected into n non-overlapping triangles $\triangle 1, \triangle 2, \dots, \triangle n$, then

$$d(\triangle ABC) = \sum_{i=1}^n d(\triangle i).$$

Because of the previous theorem, we can show that triangles can have arbitrarily small defects, always positive, of course. To see this, take any triangle $\triangle ABC$ and dissect it into two triangles along a common vertex. Since defects are additive and positive, at least one of these two triangles have defect less than or equal to $\triangle ABC$. Call this triangle $\triangle A_1 B_1 C_1$. Thus, $d(\triangle A_1 B_1 C_1) \leq \frac{1}{2} d(\triangle ABC)$. Inductively, we can construct a triangle $\triangle A_n B_n C_n$ such that $d(\triangle A_n B_n C_n) \leq \frac{1}{2^n} d(\triangle ABC)$. Because of the Archimedian property of the real numbers, for any $\varepsilon > 0$ there exists a number N sufficiently large such that $d(\triangle A_N B_N C_N) < \varepsilon$. In particular, the “smaller” the triangle is the smaller the defect will be.

Theorem 7.2.2 (Angle-Angle-Angle). *Two hyperbolic triangles are congruent if all corresponding angles are congruent. ([Progressive Geometry Proofs Slides](#))*

Proof. Let $\triangle ABC$, $\triangle A'B'C'$ be triangles in hyperbolic geometry such that $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$. In particular, $d(\triangle ABC) = d(\triangle A'B'C')$. If any pair of sides of these two triangles are congruent, namely $\overline{AB} \cong \overline{A'B'}$, $\overline{AC} \cong \overline{A'C'}$, or $\overline{BC} \cong \overline{B'C'}$, then $\triangle ABC \cong \triangle A'B'C'$ by ASA. For the sake of contradiction, we will suppose that no such pair of sides of $\triangle ABC$ and $\triangle A'B'C'$ are congruent. Then one of the triangles has at least two sides longer than their correspondents in the other triangle. Without the loss of generality, we may suppose that $\overline{AB} > \overline{A'B'}$ and $\overline{AC} > \overline{A'C'}$. Then there exists points $A - B'' - B$ and $A - C'' - C$ such that $\overline{A'B'} \cong \overline{AB''}$ and $\overline{A'C'} \cong \overline{AC''}$. Therefore, $\triangle AB''C'' \cong \triangle A'B'C'$ by SAS. In particular, $d(\triangle AB''C'') = d(\triangle A'B'C') = d(\triangle ABC)$.

By the convexity of $\triangle ABC$, the triangle can be partitioned into the union of three triangles, namely

$$\triangle ABC = \triangle AB''C'' \cup \triangle B''C''B \cup \triangle BCC''.$$

The additivity of defects implies that

$$\begin{aligned} d(\triangle ABC) &= d(\triangle AB''C'') + d(\triangle B''C''C) + d(\triangle BB''C) = d(\triangle ABC) + d(\triangle B''C''C) + d(\triangle BB''C) \\ &\Rightarrow 0 = d(\triangle B''C''C) + d(\triangle BB''C). \end{aligned}$$

Given that defects are always positive, the above equation leads to a contradiction. $\square$

Furthermore, we can define a **polygon** as a union of adjacent, non-overlapping triangles (meaning that the intersection of two triangles' boundaries is a side of both triangles but the intersection of their interiors is empty). For example, $\square ABCD = \triangle ABD \cup \triangle CBD$ such that $\triangle^\circ ABD \cap \triangle^\circ CBD = \emptyset$. Then we can extend the notion of defect to any polygon by defining it as the sum of the defects of the respective triangles in the union. Then certainly the additive property of defects is extended, but now we must worry if this definition is well-defined. That is, if a polygon can be partitioned into a union of triangles in two different ways then will the defects with respect to these two different partitions agree?

Theorem 7.2.3. *If a polygon is partitioned into triangles in any manner, the defect of the polygon is equal to the sum of defects of the component triangles. ([Progressive Geometry Proofs Slides](#))*

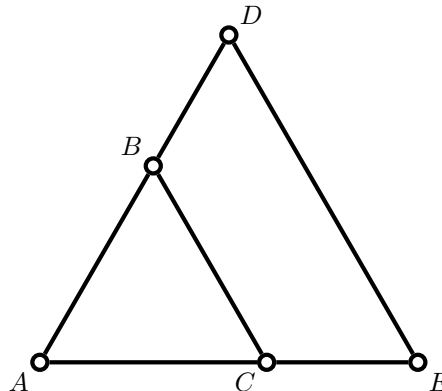
Proof. The proof relies on finding a common refinement of each partition, that is, a **refinement** $\mathcal{R}$ of a partition $\mathcal{P}$ of triangles is a partition of the original polygon such that each triangle in $\mathcal{P}$ is a union of adjacent, non-overlapping triangles of $\mathcal{R}$. In other words, the refinement $\mathcal{R}$ is a union of partitions of triangles for each triangle in $\mathcal{P}$. It is a combinatorially argument that any two partitions of triangles of a common polygon have a common refinement. The details rely primarily upon corresponding all the possible intersections and symmetric differences of two overlapping triangles and why each intersection or symmetric difference can be partitioned into triangles. A double induction argument then follows. The complete details are left to the reader. When a common refinement is obtained, the additivity of defects of triangles implies that the two original partitions of the polygon will have the same defect as the common refinement. Therefore, the defect of a polygon is well-defined. $\square$

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be hyperbolic geometry.

1. Consider the hyperbolic points such that $A-B-D$, $A-C-E$, $\{A, B, C\}$ is non-collinear, and $\overline{AB} \cong \overline{AC}$.

- (i) Which triangle, $\triangle ABC$ or $\triangle ADE$, has the greater defect? Why?
- (ii) If $d(\triangle ABC) = k_1^\circ$ and $d(\triangle ADE) = k_2^\circ$, find $m\angle BCED$ (the sum of the angle measures of $\angle BCED$).



2. Suppose an equilateral triangle $\triangle ABC$ satisfies $m\angle A = 59^\circ$. Find $d(\triangle ABC)$.
3. Suppose a Lambert quadrilateral $\square ABCD$ satisfies $m\angle D = 59^\circ$ (the fourth angle). Find $d(\square ABCD)$.
4. Suppose a Saccheri quadrilateral $\square ABCD$ satisfies $m\angle D = 59^\circ$ (a summit angle). Find $d(\square ABCD)$.

ⁱThe progressive slides in this section were created by Hayley John.

“A man’s country is not a certain area of land, of mountains, rivers, and woods, but it is a principle and patriotism is loyalty to that principle.” – George William Curtis

7.3 Axioms of Area

Before we introduce the notion of area to hyperbolic geometry, we first need to introduce the notion of *area* to geometry. As might be expected, **area** is an undefined term for geometry but axioms are provided to give it the most specific, yet still general meaning. Area is a function A assigning to each triangle $\triangle ABC$ a unique real number $A(\triangle ABC)$ satisfying the following three axioms:

Definition 7.3.1. Area Axioms

Axiom A.1 (Positive): The area of each triangle is positive.

Axiom A.2 (Congruence Preservation): If two triangles are congruent then they have equal areas.

Axiom A.3 (Additivity): If $A - D - B$ then $A(\triangle ABC) = A(\triangle ADC) + A(\triangle DBC)$.

We will also extend the notion of area to any polygon using partitions of triangles. Since any two partitions of a polygon have a common refinement, the area of a polygon is well-defined. The proof of these follows the same reasoning that showed the defect of hyperbolic polygons is well-defined. After all, we have already seen that defect satisfies the area axioms!

Definition 7.3.2. We say that two polygons are **equivalent** if there exists a partition of adjacent, non-overlapping triangles for each polygon such that there exists a congruence-preserving one-to-one correspondence between the triangles among the two partitions.

Theorem 7.3.3. *Polygons are equivalent if and only if they have equal area.*

The proof that equivalent polygons have equal area is a straight forward application of the area axioms and is left as an exercise of the reader. The converse of this statement, although equally true, is much more difficult to prove and will be omitted.

Definition 7.3.4. We say that a congruence geometry that satisfies the three area axioms is an **area geometry**.

We will see that every neutral geometry is an area geometry. This follows from the fact that the All-or-Nothing theorem implies that each neutral geometry is Euclidean or hyperbolic, and we will see that both of these have a well-defined notion of area. In fact, we will see that these area functions are essentially unique, determined by the parallel postulate. We will begin with the notion of area in Euclidean geometry.

In Euclidean geometry, rectangles of arbitrarily large and small sizes exist. Let $\square ABCD$ be a rectangle and let n be a positive integer. Then there exists points B_n, C_n such that $\square AB_n C_n D$ is a rectangle with $AB_n = \frac{1}{n}AB$. As we can tile $\square ABCD$ with translated copies of $\square AB_n C_n D$, the additivity axiom of area shows $A(\square AB_n C_n D) = \frac{1}{n}A(\square ABCD)$. In particular, the length of a rectangle is directly proportional to the area of the rectangle. A similar argument shows that the area of a rectangle is also directly proportional to the width of the rectangle. Therefore, in Euclidean geometry area of a rectangle is jointly proportional to the length and the width. Thus, there exists some positive real number k such that

$$A(\square ABCD) = k(AB)(AD) = k(\ell w).$$

The proportionality coefficient k is arbitrary and, in effect, defines a measurement scale. For example, if acres is our desired measurement for area and miles for distance, our area of a rectangle would be $A = 640\ell w$, since there are 640 acres per square mile. Most often though in practice, area is measured using the square

unit of distance. In this case, $k = 1$ and we recapture the usual area formula for rectangles $A = \ell w$. Without the loss of generality, we may assume our choice of measurements sets $k = 1$.

From this important formula, we can recapture the remaining area formulas of polygons. We will provide the proof of the parallelogram area formula. It is an exercise of the reader to provide the rest.

Theorem 7.3.5. *In Euclidean geometry, the area of a parallelogram is the product of the lengths of its base and its height.*

Proof. Let $\square ABCD$ be a parallelogram. By the term *base*, we mean the length of an side of the parallelogram, for example AB . By the term *height*, we mean the distance between the base side and the opposite side of the parallelogram. Since the geometry is Euclidean, the two parallel lines are equidistant and the notion of height of the parallelogram with respect to a side is well-defined.

If $\square ABCD$ is a rectangle, then we are done. So suppose that the parallelogram is not a rectangle. Since the opposite angles of a Euclidean parallelogram are congruent and the angle sum is necessarily 360° , we know that one pair of congruence angles is obtuse. Without the loss of generality, suppose that $\angle B, \angle D$ are obtuse. Let E and F be the feet of the perpendicular dropped from C and D , respectively, onto $\overleftrightarrow{AB}$. Under our assumption, we have $A - F - B - E$ (otherwise there exists a Euclidean triangle with angle sum greater than 180°). By the Euclidean parallel postulate, opposite sides of parallelograms are congruent and parallel lines are equidistant. Therefore, $\triangle AFD \cong \triangle BEC$ by hypotenuse-leg. Thus, $\square FECD$ is a Saccheri quadrilateral, which must be a rectangle in Euclidean geometry. Furthermore, $\square ABCD$ and $\square FECD$ are equivalent polygons and $A(\square ABCD) = A(\square FECD) = (EF)(CE) = (AB)(CE)$. $\square$

Definition 7.3.6. In plane geometry, a **shear mapping** (or **transvection**) is a map that displaces each point in fixed direction, by an amount proportional to its signed distance from a line that is parallel to that direction.

In other words (similar to how Euclid explained it), a parallelogram is a shear of another parallelogram if they share a common base and have the sides opposite this base contained in the same parallel line.

Theorem 7.3.7. *In Euclidean geometry, area is an invariant of any shearing mapping.*

Proof. It suffices to prove the claim for the unit square $\square ABCD$. Let $\square EBCF$ be a shearing of the unit square along the line $\overleftrightarrow{BC}$. Since opposite sides of parallelograms are congruent, we know that $\overline{AD} \cong \overline{BC} \cong \overline{EF}$, $\overline{AB} \cong \overline{DC}$, and $\overline{BE} \cong \overline{CF}$. Up to re-labeling of vertices, there are two cases to consider: $A - D - E - F$ and $A - E - D - F$. If $A - D - E - F$, it must be that $\overline{CD}$ intersects $\overline{BE}$ at some point G . Then $\triangle AFD \cong \triangle BEC$ by SSS. Then $\square ABCD$ and $\square EBCF$ are equivalent ($\square ABCD = \triangle ABE \cup \triangle BCG \setminus \triangle DGE$ and $\square EBCF = \triangle DCF \cup \triangle BCG \setminus \triangle DGE$). If $A - E - D - F$, $\triangle AEB \cong \triangle DFC$ by SSS. Again, we have $\square ABCD$ and $\square ABEF$ are equivalent. $\square$

It is interesting to see that the existence of arbitrary rectangles, an equivalence of the Euclidean parallel postulate, in conjunction with the area axioms is sufficient to provide the traditional area formula. Our usual notion of area can thus be described as being Euclidean area. The converse is also true, that is, an area geometry is Euclidean if and only if area of a rectangle is proportional to its length and width.

Exercises

*Unless explicitly stated otherwise, all geometries in these exercises are assumed to be **Euclidean** geometry.*

1. Prove the area of a right triangle is one-half the product of the length of its legs.
2. Prove the area of a triangle is one-half the product of its base and height.
3. Prove the area of a trapezoid is the product of its height and the arithmetic mean of its bases.
4. Prove the area of a rhombus is one-half the product of the lengths of its diagonals.

“There are no lines in nature, only areas of colour, one against another.” – Edouard Manet

7.4 Hyperbolic Area and Distance

Now that we understand the axioms of area and how they, when combined with the Euclidean axioms, produce the usual notion of area ($A = \ell w$), we are ready to explore the interpretation of area in hyperbolic geometry. Although we cannot use the Euclidean area for rectangle in hyperbolic geometry (since rectangles do not exist), it still might be possible to use the Euclidean triangle area formula to determine hyperbolic area.

To explore this idea, imagine two hyperbolic lines ℓ and m such that $\ell \perp m$. Let $A \in \ell \cap m$. Let $C \in m \setminus \ell$ such that $AC = 1$, let $B \in \ell \setminus m$ such that $AB = 1$, and let $B_n \in H^\circ(\ell, B)$ such that $AB_n = n$. Then, if hyperbolic triangles have area given as $\frac{1}{2}bh$, then each of the triangles $\triangle B_{n-1}B_nC$ has the same area as $\triangle ABC$ which is $\frac{1}{2}$. On the other hand, $\triangle AB_nC = \bigcup_{i=1}^n \triangle B_{i-1}B_iC$, which implies that

$$d(\triangle AB_nC) = \sum_{i=1}^n d(\triangle B_{i-1}B_iC).$$

We know that each of the triangles $\triangle B_{i-1}B_iC$ have equal area, so they must be equivalent. On the other hand, equivalent polygons must have equal defects, a simple fact that follows immediately from the additivity of defects. Let $d = d(\triangle B_{i-1}B_iC)$. Then $d(\triangle AB_nC) = nd < 180^\circ$ for all n . Of course, since d is a constant, there exists a sufficiently large N such that $Nd \geq 180^\circ$, a contradiction. Therefore, $\frac{1}{2}bh$ is not a consistent interpretation of area for hyperbolic triangles.

If not the Euclidean area formula for triangles, how do we interpret area for hyperbolic triangles? Honestly, the answer is already before us. The conflict in the above argument is that although the triangles had equal area, they had non-equal defects. We have already seen that the defect function satisfies the three axioms of area. Therefore, defect is an area function for hyperbolic triangles. We are fighting against the curve if we expect to find some superior area formula for hyperbolic geometry.

Definition 7.4.1. Let $\triangle ABC$ be a hyperbolic triangle. Let k be a positive real number. Then the **area** of $\triangle ABC$, denoted $A(\triangle ABC)$, is defined as

$$A(\triangle ABC) = k \cdot d(\triangle ABC).$$

Just like the Euclidean area formula, the coefficient k should be understood as a unit conversion factor. To take $k = 1$ implies that we are measure area using degrees.

We see that in hyperbolic geometry, $0 < A(\triangle ABC) < 180$. Thus, in hyperbolic geometry, triangles cannot be arbitrarily large in terms of their area. On the other hand, the lengths of their sides can get arbitrarily large.

Interpreting distances in hyperbolic geometry is much more difficult.

Definition 7.4.2. Let d be a function from pairs of distinct points to the real numbers. We call d a **distance** (or **metric**) functions if it satisfies the three following metric axioms:

Metric Axioms

Axiom A.1 (Positive): If A and B are two distinct points, then $d(A, B) > 0$.

Axiom A.2 (Symmetry): If A and B are two distinct points, then $d(A, B) = d(B, A)$.

Axiom A.3 (Additivity): If A , B , and C are three non-collinear points, then $d(A, C) < d(A, B) + d(B, C)$.

We say that an incidence geometry that satisfies the three metric axioms is a **metric geometry**.

Example 7.4.3. The Segment Measure Theorem guarantees that any neutral geometry is a metric geometry since segment congruence satisfies the metric axioms. In fact, this theorem can be extended to any Archimedean congruence geometry, such as $\mathbb{Q}^2$.

Of course, the details of the Segment Measure Theorem are obscure to us (remember that we did not prove it). Even with the proof, we would only know the existence of a distance function, not necessarily the specific implementation. Honestly, the same geometry can have different distance functions.

We will specify a metric for the disk model of hyperbolic geometry. Let A and B be two points in the disk. Let ℓ be the hyperbolic line determined by A and B . Let P and Q be the intersection between ℓ and the unit circle in $\mathbb{R}^2$. Then define the **hyperbolic distance** between A and B by the metric function

$$d(A, B) = \left| \ln \left(\frac{AQ}{BQ} \times \frac{BP}{AP} \right) \right|,$$

where AQ , BQ , BP , and AP are understood to be the Euclidean distance between the points A , B , P , and Q .

It takes some work (the details are in the book) but it can be shown that this function satisfies the metric axioms. This distance formula also allows us to put coordinates on a hyperbolic line, similar to how we do this on a Euclidean line. Let O be the midpoint between P and Q . This will take the role of the origin. Then if $a = d(A, O)$ then the point A would have the coordinate $(\pm a, 0)$ (the sign depends on the ordering of the line), as if the line were the x -axis.

The real problem with this interpretation of distance is that it requires finding a specific equation for the hyperbolic line in Euclidean coordinates, that is, we need the equation of the circle passing through the two points A and B which is perpendicular to the unit circle at the points of intersection. To find this equation, we need to find an endpoint (either one works) of the perpendicular bisector of A (or B) on the unit circle. Call such a point A'' . Then we would need to find the intersection between the tangent line of the unit circle at A'' with the ray $\overrightarrow{OA}$. Call this point A' , called the **inversion** of A . Then the hyperbolic line $\overleftrightarrow{AB}$ is the circle ℓ determined by the three points A , B , and A' (or B'). After this equation is found, the ideal points P and Q are found as the two intersection points between ℓ and the unit circle. Then four Euclidean distances are calculated, followed by the natural logarithms. You can imagine this is a very difficult calculation. Technology is very useful in speeding up this process.

By the way, angle measure in this hyperbolic model is simply the Euclidean angle measure of the tangent lines of the curves.

Exercises

Definition 7.4.4. Consider the ordered geometry $\mathbb{R}^2$. Define the **taxicab metric** (or $\mathcal{L}^1$ metric) on $\mathbb{R}^2$ as following, for points $P = (x_1, y_1), Q = (x_2, y_2) \in \mathbb{R}^2$:

$$PQ = |x_2 - x_1| + |y_2 - y_1|.$$

The plane $\mathbb{R}^2$ equipped with the taxicab metric is called **Taxicab geometry**.ⁱ In terms of ordered geometry, taxicab $\mathbb{R}^2$ and Euclidean $\mathbb{R}^2$ are identical. In particular, they have the same meaning of points, lines, incidence, and betweenness. Likewise, since they have the same meaning of lines and betweenness, taxicab geometry satisfies the Euclidean parallel postulate and Dedekind's axiom. Therefore, taxicab geometry is a complete, affine, ordered geometry, identical to the Euclidean plane in those regards. Obviously, they disagree on their notions of segment measure having different, non-equivalent metrics. Therefore, segment congruence means different things between the two geometries. Interpret angle measure the same in taxicab geometry as we do in Euclidean geometry. Therefore, taxicab geometry is a geometry the same as the Euclidean plane but segment measure has been modified.

*Unless explicitly stated otherwise, all geometries in these exercises are assumed to be **taxicab** geometry.*

1. Let $O = (0, 0)$, $A = (1, 2)$, $B = (-3, 5)$, $C = (-2, -2)$, $D = (2, -3)$. Find all ten distances between these five points.

2. Let $O = (0, 0)$ and $R = (3, 0)$. Draw the circle centered at O with a radius OR .

It is given by the equation $|x| + |y| = 3$.

Definition 7.4.5. In an affine metric space, we define the number π to be the ratio between the circumference of a circle by its diameter.

3. Prove that $\pi = 4$.

Definition 7.4.6. A **parabola** with **focus** F and **directrix** ℓ , in a metric space, is the set of all points equidistant between F and ℓ .ⁱⁱ

4. If $F = (0, 1)$ and ℓ is the line $y = -1$, draw the parabola associated to F and ℓ in taxicab geometry.

It is given by the equation $|x| + |y - 1| = y + 1$.

5. Prove that taxicab geometry violates SAS, that is, provide two triangles $\triangle ABC$ and $\triangle A'B'C'$ in taxicab such that $\overline{AB} \cong \overline{A'B'}$, $\angle B \cong \angle B'$, and $\overline{BC} \cong \overline{B'C'}$ but $\triangle ABC \not\cong \triangle A'B'C'$.

ⁱIt receives this name because if someone in a large town, such as NYC, can only travel along roads going North-South or East-West to arrive upon their destination, then this metric would accurately measure the distance as the taxicab drives. Conversely, the standard Euclidean metric (also called the $\mathcal{L}^2$ metric) would agree with how the pigeon flies.

ⁱⁱRecall that the distance between a point P and a line ℓ is PR where R is the foot of the perpendicular dropped from P onto ℓ .

Chapter 8

Elliptic Geometry

“Projective geometry is all geometry.” – Arthur Cayley

8.1 Projective Geometry

We can formalize projective geometry into an axiomatic system. We will see that projective geometry generalizing the properties of Fano geometry.

Definition 8.1.1. Projective Geometry

Axiom I.1 (Line Determination): For each two distinct points there exists a unique line containing both of them.

Axiom I.2 (Tricancy^a): For every line there exist at least three distinct points on it.

Axiom I.3 (Point Existence): There exists at least three distinct points.

Axiom I.4 (Non-Collinearity): Not all points lie on the same line.

Axiom P.1 (Elliptic Parallel Postulate): All lines intersect, that is, there exists no parallel lines.

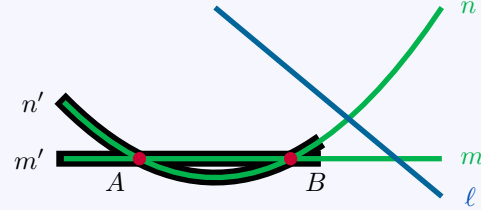
^aProjective geometries are incidence geometries since axioms I, II, and IV are identical to axioms of incidence. The difference here is the secancy axiom of incidence is strengthened to the tricancy axioms because all lines contain at least three points instead of two. The reason for this upgraded axiom is revealed in the proof of Theorem 8.1.3.

Example 8.1.2. The Fano plane is a finite model of projective geometry. It is, in fact, the smallest model. The real projective plane $\mathbb{RP}^2$ is a model of an infinite projective plane.

Theorem 8.1.3. Let $(\mathcal{P}, \mathcal{L})$ be a projective geometry where $\mathcal{P}$ is the set of points and $\mathcal{L}$ is the set of lines. Let $\ell \in \mathcal{L}$ be a fixed line. Let $\mathcal{P}' = \mathcal{P} \setminus \ell$ and $\mathcal{L}' = \{m \setminus \ell \mid m \in \mathcal{L}, m \neq \ell\}$. Then $(\mathcal{P}', \mathcal{L}')$ is an affine geometry, whose points are $\mathcal{P}'$ and whose lines are $\mathcal{L}'$.

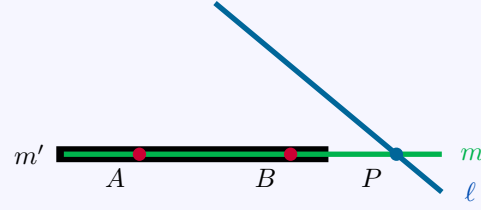
This demonstrates that a projective geometry is an affine geometry adjoined by a **line at infinity** or **ideal line**. The points on the ideal line are called **ideal points** or **points at infinity**. The other points and lines in the affine geometry are referred to as **ordinary**. We should also mention that in the projective geometry, distinct lines intersection at exactly one point. Thus, $m \setminus \ell$ is a line with one less point. In particular, every ordinary line in the projective geometry contains exactly one ideal point and all the rest are ordinary.

Proof. Let $A, B \in \mathcal{P}'$ be two distinct points in the ordinary geometry. Then $A, B \in \mathcal{P}$ are two distinct ordinary points in the projective geometry and there exists a unique line m containing both points by Line Determination. Since m contains two ordinary points, it must be that $m \neq \ell$, since ℓ contains only ideal points by construction. Then $m' = m \setminus \ell$ is a non-empty line in the ordinary geometry. If m'

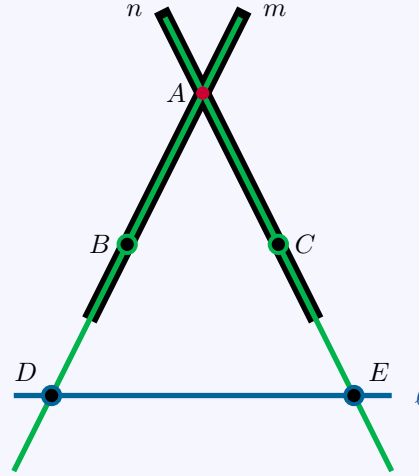


and n' are two line in the ordinary geometry containing A and B , then there exists lines m and n in the projective geometry containing A and B . But Line Determination guarantees that $m = n$. Thus, $m' = m \setminus \ell = n \setminus \ell = n'$. This proves Axiom I.1 for the ordinary geometry.

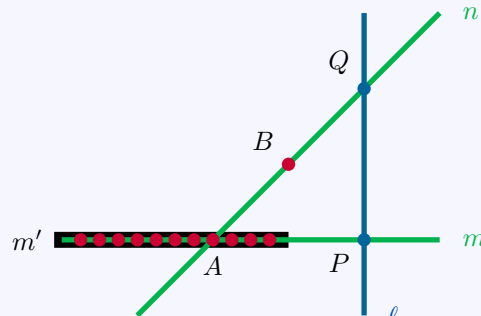
Let m' be a line in the ordinary geometry. Then $m' = m \setminus \ell$ for some line in the projective geometry. Let $P \in m \cap \ell$, a point guaranteed by the Elliptic Parallel Postulate. Hence, $m' = m \setminus \{P\}$. By Tricancy, m contains at least three points. Therefore, $m' = m \setminus \{P\}$ contains at least two points. This proves Axiom I.2 for the ordinary geometry.



We know that the projective geometry contains at least three non-collinear points since it is an incidence geometry, call them A, B, C . We know that the projective geometry has exactly one ideal line, which cannot contain all three of these points. Thus, we know that at least one point is ordinary. Without the loss of generality, assume A is ordinary. Thus, the lines determined by A and B and A and C , call them m and n respectively, are ordinary. By Tricancy, m and n must have a third point D and E , respectively. Since an ordinary line can have at most one ideal point, either B or D is ordinary and either C or E is ordinary. In all cases, we have found three ordinary points. This proves Axiom I.3 for ordinary geometry.

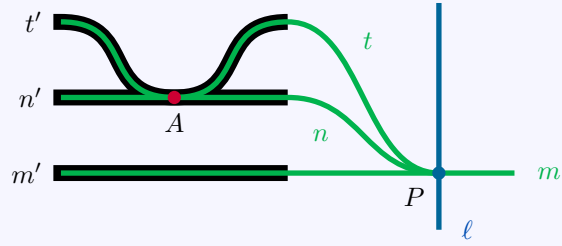


Suppose all points of the ordinary geometry are contained on the single line m' , that is, $\mathcal{P}' \subseteq m'$. Then $m' = m \setminus \{P\}$ for some projective line m and some ideal points P . By the Non-collinearity axiom, there exists some projective point Q not on m . If Q is an ordinary point, then Q is a point not on m' in the ordinary geometry. If Q is an ideal point, then consider the line n determined by Q and some ordinary points on m , say A (the existence of n follows from Line Determination and the existence of A follows from Tricancy).



The line n contains a third point B by Tricancy. Now $B \notin m$, otherwise $A, B \in m \cap n$, a contradiction. Since Q is ideal and on n , B must be ordinary. Therefore, B is an ordinary point not on m' . This proves Axiom I.4 for the ordinary geometry. Therefore, the ordinary geometry is an incidence geometry.

Let m' be a line in the ordinary geometry and let A be a point not on m' . Then $m' = m \setminus \{P\}$ for some projective line m and some ideal point P . Let n be the projective line determined by P and A . Let $n' = n \setminus \{P\}$ be the corresponding ordinary line. Since $m \cap n = \{P\}$, we note that $m' \cap n' = \emptyset$, that is, $m' \parallel n'$. Of course, $A \in n'$ since $A \in n$. If t' is any other ordinary line containing A and parallel to m' , then there exists a projective line t containing



By the Elliptic Parallel Postulate, this projective line intersects m . The intersection cannot be an ordinary point, otherwise $m' \nparallel n''$. Thus, the intersection is ideal. As m contains only a single ideal point, the point of intersection must be P . Since the projective extension of n'' contains A and P , this projective line must be n by Linear Determination. Thus, $n'' = n \setminus \{P\} = n'$. This proves Axiom P.2. Therefore, the ordinary geometry is an affine geometry. $\square$

The take away of the previous theorem is that removing a single line from projective geometry creates an affine geometry. The process is, of course, reversible. In affine geometry, when considering a family of parallel lines, we can make them intersect by adjoining a single ideal point to each line in the parallel family. We then would define the ideal line as the set of all ideal points adjoined to the geometry in this fashion. The projective geometry created from this affine geometry is called the **projective cover** (or **projective completion**) of the affine geometry.

Example 8.1.4. Removing a line from the Fano geometry, such as the odd circular line in the model, created an affine geometry. In fact, this is the Four-Point geometry we have already seen to be affine. Removing a line from $\mathbb{RP}^2$ will also create an affine geometry, namely $\mathbb{R}^2$. To see this, consider the hemisphere model of $\mathbb{RP}^2$ (as discussed previously in Section 2.4), we can remove the bounding circle to get a geometry identical to the open unit disk. Using the arctangent function, we can expand this open disk to the whole plane.

The following theorem is an immediate consequence of Theorem 8.1.3 and the previous theorems we have seen of affine geometry.

Theorem 8.1.5. *Consider an affine geometry of order n , that is, all lines have n points. Then its projective cover is a projective geometry with $n^2 + n + 1$ points, $n^2 + n + 1$ lines, each line contains $n + 1$ points, and each point is incident to $n + 1$ lines.*

As every projective geometry is the cover of some affine geometry, the above theorem applies to all projective geometries. We say that the **order** of a projective geometry is n if its associated affine geometry has order n .

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be projective geometries.

1. Prove every two distinct lines ℓ and m determine a unique point.
2. Prove there exist four distinct points no three of which are on the same line.
3. Prove every point is contained by at least three lines.
4. Prove there exists four lines no three of which are concurrent with the same point.

ⁱFigures in this section were created by Ethan Olcott.

“Distance not only gives nostalgia, but perspective, and maybe objectivity.” – Robert Morgan

8.2 The Elliptic Parallel Postulate

We have now studied extensively two important geometric systems: Euclidean geometry (real geometry with unique parallels) and hyperbolic geometry (real geometry with multiple parallels). It would then make sense to want to study the last possibility called elliptic geometry (real geometry with no parallels), but is such a geometry even possible? By this we mean, it was proven as a theorem of congruence/neutral geometry that parallel lines exist through a given point and with respect to a given line. Therefore, if we are interested in developing an axiomatic system for geometries without parallels then it cannot happen inside congruence/neutral geometry.

Thus, in order to construct elliptic geometry, we will need to discover axioms which closely resemble the neutral axioms as much as possible but avoid the existence of parallel lines. In order to create such an axiomatic system, we need to investigate why does neutral geometry have parallel lines. In particular, we want to preserve the notion of measure so that the geometry is still “real.” The proof of the existence of parallel lines roughly goes as follows: given a line ℓ and a point P off the line; there exists a perpendicular line m to ℓ incident to P ; there exists a perpendicular line n to m incident to P ; as right angles are congruent, the lines n and ℓ form congruent alternate interior angles with respect to m ; then n and ℓ are parallel by the Alternate Interior Angle Theorem.

There we have it! Parallel lines exist in neutral geometry because perpendicular lines exist, right angles are congruent, and the Alternate Interior Angle Theorem. Thus, to form elliptic geometry, at least one of these facts must go. Now, we want elliptic geometry to resemble neutral geometry as much as it can with respect to the notion of congruence. In particular, we want our measure of angles and segments to resemble neutral geometry as much as possible. As such, we very much would still want the existence of perpendicular lines and congruence of right angles, otherwise the Angle Measure theorem would not guarantee the existence of 90° . Thus, the Alternate Interior Angle theorem must be the theorem to be removed from the axiomatic system. After all, the Alternate Interior Angle theorem guarantees the existence of parallel lines under certain circumstances, so it should be seen as the least compatible theorem of the three with the Elliptic Parallel Postulate.

Next, we must ask how to get rid of the Alternate Interior Angle Theorem. As it is a theorem and not an axiom, we must look at its proof too. The proof went roughly as following: assume that ℓ and m are lines with congruent alternate interior angles with respect to the transversal t but $\ell \nparallel m$; this forms a triangle on one side of t ; using the congruent alternate interior angles translate this triangle to the other side of t ; this translated triangle provides a second intersection to ℓ and m , which is a contradiction. In summary, the proof relies on triangle translation (which is proven using segment translation, angle translation, and SAS), plane separation (an equivalent of Pasch’s axiom), and uniqueness of line intersection (an immediate consequence of line determination). As mentioned above, we want to preserve, the best we can, the notion of measure. Thus, we want the congruence axioms to be intact, if possible. Therefore, elliptic geometry must not satisfy Pasch’s axiom or the Line Determination axiom.

If Line Determination is removed, then our elliptic geometry will not even be an incidence geometry. Spherical geometry is an example of such an elliptic geometry, because there is not a unique “line” determined by antipodes of the sphere. In fact, every “line” containing a point will also contain its antipode. This version of elliptic geometry has the advantage that it can be embedded into $\mathbb{R}^3$, which means it can be envisioned in the reader’s mind¹ concretely. One disadvantage is that this model fails the Trichotomy axiom. Is this model specific or do all elliptic geometries lose Trichotomy? Perhaps the strongest disadvantage is that this interpretation of elliptic geometry is not even an incidence geometry! This choice of axiomatic removal would be the most difficult to manage as nearly all theorems of neutral geometry rely on the incidence axioms, either explicitly or implicitly. We might have too many other casualties by losing Line Determination. This interpretation of elliptic geometry is often called **double (point) elliptic geometry**.

¹Well, at least it can be visualized by the author’s mind concretely. The author should be cautious about supposing the reader is able to think 3-dimensionally and not able to think 4-dimensionally.

The real projective plane $\mathbb{RP}^2$ serves as our second model for elliptic geometry. The projective plane is an incidence geometry (it is, in fact, a projective geometry). This should be seen as a correction of the flaw in Spherical Geometry because antipodal points are identified together. The disadvantage of $\mathbb{RP}^2$, other than being much more difficult to visualize (it does not embed into $\mathbb{R}^3$) is that it loses plane separation/Pasch's axiom. Consider the hemisphere model of $\mathbb{RP}^2$ and a line ℓ that “separates” the hemisphere into two halves. At first glance, this “separation” may appear to create two half planes, but, in fact, the wrap-around feature of the line at infinity allows for a path between two points in “separate” half-planes to be connected without transversing ℓ . Thus, plane separation is lost in elliptic geometry. This interpretation of elliptic geometry is often called **single (point) elliptic geometry**.

Also, because there is essentially two straight paths between two given points, the direct path and the path through infinity, Trichotomy also fails in $\mathbb{RP}^2$. Our experimentation leads us to believe that in order to adjoin the Elliptic Parallel Postulate to the neutral axioms we must forsake Trichotomy and we must forsake either Line Determination or Pasch's Axiom. We will choose the latter for our interpretation of elliptic geometry. The advantage here is that elliptic geometry is an incidence geometry with infinitely many points on a line (by the Extension axiom) and satisfies the Elliptic Parallel Postulate. Therefore, elliptic geometry will always be projective. Considering our previous lecture about projective covers, we are now in the correct setting to axiomatize elliptic geometry.

Definition 8.2.1. We say that a geometry is **elliptic** if it is isomorphic to the projective cover of a Euclidean geometry.

By our definition of elliptic geometry, only the notion of betweenness is weakened to remove the guaranteed parallel line. But notions such as segment and angle, which are used in the congruence axioms and Dedekind's axiom, need to still be valid in elliptic geometry. Without Trichotomy (we will see later when our notion of Trichotomy is lost) and Plane Separation, we will only have a relative notion of betweenness, that is, we will say $A - B - C$ relative to the line ℓ which will mean that the point B is between A and C in the Euclidean geometry formed viewing ℓ as a line at infinity. A different choice of ideal line will change the notion of betweenness for all points off this line. Using **relative betweenness**, we define segments, rays, angles, half-planes, interior, exterior, etc. but only relative to an ideal line. Be cautious though! Although we can pick up betweenness from the Euclidean geometry relative to a line, we cannot pick up measure, that is, the measure in the elliptic geometry is dramatically different than the measure for the Euclidean geometry it covers.

Exercises

1. Determine which of the axioms and theorems proved in Neutral geometry that are not true in elliptic geometry (need only look at theorems in sections, not in exercises).

- (i) Incidence Geometry
- (ii) Ordered Geometry (using Relative Betweenness)
- (iii) Congruence Geometry
- (iv) Neutral Geometry

“He who exercises government by means of his virtue may be compared to the north polar star, which keeps its place and all the stars turn towards it.” – Confucius

8.3 Poles and Duality

The first main theorem of elliptic geometry is known as the Polar Property Theorem, which says that all lines in elliptic geometry have a pointⁱ/ (specific to the line) which is equidistant from all points on the line. This is called the **pole** of the line. Essentially, this theorem says that all lines in elliptic geometry are circles, although there may exist circles which are not lines. Lines are, in fact, circles of maximum radius.

Theorem 8.3.1 (The Polar Property Theorem). *If ℓ is a line in elliptic geometry, then there exists exactly one point P such that P is equidistant to all points on ℓ .*

Proof. Let A and B be two distinct points on ℓ , such that $AB = 1$. The two perpendicular lines erected from A and B must intersect, say at P . The triangle $\triangle ABP$ has two right angles, namely $\angle A$ and $\angle B$. Therefore, $\triangle ABP$ is isosceles and $\overline{AP} \cong \overline{BP}$ (the converse of the isosceles triangle theorem follows from ASA which follows from SAS).

Let M be the midpoint of $\overline{AB}$ (relative to some line at infinity). Then $\triangle APM \cong \triangle BPM$ by SAS, which implies that $\angle AMP \cong \angle BMP$. As these angles are supplementary, then they both must be right (this reproves the typical altitude argument of isosceles triangles). As $\triangle BMP$ is a double-right triangle, it is also isosceles, which implies that $\overline{AP} \cong \overline{MP}$.

Next, let C be a point such that B is the midpoint of $\overline{AC}$ (relative to some line at infinity). Then $\triangle ACP$ is also a double right triangle and hence isosceles, so that $\overline{CP} \cong \overline{AP}$. In particular, if D is any other point on ℓ such that AD is dyadic rational number (that is, some combinations of doubling and halving can reach the point D , e.g. $5.25 \cdot AB = 2(2 \cdot AB) + \frac{1}{2}(\frac{1}{2} \cdot AB)$), then $\overline{DP} \cong \overline{AP}$.

If $AD = x \in \mathbb{R}^+$, then there exists a sequence $\{x_i\}_i$ of dyadic rational numbers that converge to x . Each term x_i in this sequence corresponds to some point $B_i \in \ell$ such that $AB_i = x_i$. As above, $B_iP = AP$ for all i . Then the sequence of points $\{B_i\}_i$ converges toward D and by continuity, we have that $\overline{AB_i} \rightarrow \overline{AD}$ and, more importantly, $\overline{B_iP} \rightarrow \overline{DP}$, which proves the result as this last sequence has constant length.

To see uniqueness of the points, this follows from the uniqueness of translation. Let P and P' be two poles of ℓ . Let A and B be two distinct points on ℓ . Then $\triangle ABP \cong \triangle ABP'$ by ASA. Angle Translation guarantees that rays are actually equal when translated onto the same half plane. In elliptic geometry, there is only one “half-plane,” which implies that $\angle ABP = \angle ABP'$, that is, $\overrightarrow{BP} = \overrightarrow{BP'}$. The uniqueness of intersection, which follows from Line Determination then implies that $P = P'$. $\square$

Definition 8.3.2. Let P be the pole of an elliptic line. Then the segment $\overline{AP}$, where A is a point on the line, is called the **polar distance** of ℓ .

Corollary 8.3.3. *The polar distance of any elliptic line is constant, that is, distinct lines have congruent polar distances.*

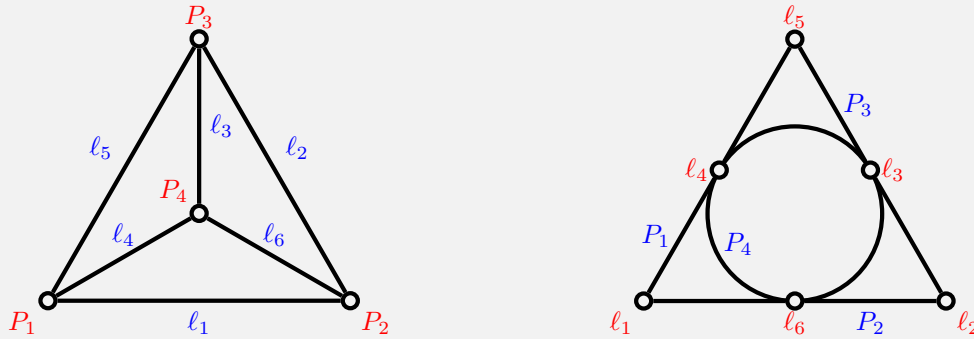
Proof. Let ℓ and ℓ' be two distinct elliptic lines. Let $A, B \in \ell$ be distinct points. Let $A', B' \in \ell'$ be points such that $\overline{AB} \cong \overline{A'B'}$. We know that such points exist by the Tricancy and Segment Translation axioms. Let P and P' be the poles of ℓ and ℓ' , respectively. Then $\triangle ABP \cong \triangle A'B'P'$ by ASA. Thus, $\overline{AP} \cong \overline{A'P'}$. $\square$

Since polar distances are congruent (and equal when a specific measure is applied), we speak instead of THE polar distance of the elliptic geometry. The specific measure of this distance is an artifact of the specific measure of the geometry. Thinking of elliptic geometry using the hemisphere model, a hemisphere with a larger radius will have a larger polar distance. If the boundary, wrap-around line at the equator of the hemisphere is our line ℓ , then its pole P is the point at the very top of the hemisphere, often called the north pole of the sphere. This geographic analogy is where the term gets its name.

We have seen that every line has a pole, but it is also true that every point is a pole of a line. Starting with any point P , we can construct two segments of polar length emanating from P , namely $\overrightarrow{AP}$ and $\overrightarrow{BP}$. By an argument similar to the proof above, if $\ell = \overleftrightarrow{AB}$, then P is the pole for ℓ .

This, in fact, demonstrates a *duality* between points and lines in the elliptic geometry. Given an incidence geometry, its **dual geometry** is formed when the undefined terms *points* and *lines* are interchanged and the relation of *incidence* is preserved. This is a construction that makes sense for any incidence geometry. This often leads to a new geometry.

Example 8.3.4. Recall the order 2 affine geometry (aka Four-Point Geometry), which is illustrated below. It has four points and six lines, where incidence can be read from the diagram. Its dual would be a six point, four line geometry whose incidence relations are illustrated below.



Notice that if we overlay the two geometries, we would get a diagram resembling the Fano geometry, which is a finite projective geometry.

In the case of projective geometry, the dual of the projective geometry is isomorphic to the original geometry, that is, projective geometry is **self-dual**.

Definition. Dual Projective Geometry

Axiom DI.1 (Point Determination): For each two distinct lines there exists a unique point incident to both.

Axiom DI.2 (Dual-Tricancy): For every point there exist at least three lines concurrent to it.

Axiom DI.3 (Line Existence): There exists at least three distinct lines.

Axiom DI.4 (Non-Concurrence): Not all lines are concurrent to the same point.

Axiom DP.1 (Dual-Elliptic Parallel Postulate): All pairs of points are incident to a common line.

The Dual Projective Geometry axioms are theorems of projective geometry. Thus, any theorem that can be proven from the dual axioms can ALSO be proven from the original axioms. This implies that for every incidence theorem in projective geometry its dual is ALSO a theorem of that projective geometry.

In elliptic geometry, this notion of duality can be extended to betweenness and congruences. The reader should try to convince one's self that point betweenness and ray betweenness are dual notions and that segment congruence and angle congruence are dual notions. In fact, the polar distance of an elliptic geometry is a tangible piece of duality. In elliptic geometry, we can form an explicit duality function that interchanges every line with its pole. As we saw above, this is a one-to-one correspondence. Furthermore, we can show that this correspondence is incidence, betweenness, and congruence preserving too. Therefore, elliptic geometry has a strong sense of self-duality, a property which is missing in Euclidean and hyperbolic geometry.

Exercises

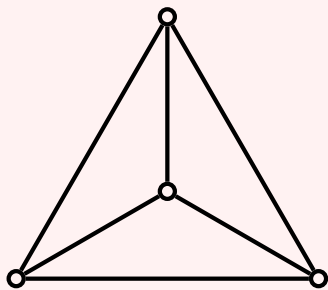
1. State the dual of each statement in projective geometry.
 - (i) There exist four distinct points no three of which are on the same line.
 - (ii) Every point is contained by at least three lines.
 - (iii) There exists four lines no three of which are concurrent with the same point.
 - (iv) If P is any point, then there exists a line that does not contain P .
 - (v) If P is any point and ℓ is any line that does not contain P , then there is a one-to-one correspondence between the points on ℓ and the lines that contain P .
 - (vi) The points of any line can be placed in one-to-one correspondence with the points of any other line.
2. The axioms of **Four-Line Geometry**, the duals of the axioms of *Four-Point Geometry*, are provided below.

Definition. Four-Point Geometry

Axiom 1: There exists exactly four distinct points.

Axiom 2: For each two distinct points there exists a unique line containing both of them.

Axiom 3: Each line is on exactly two points.

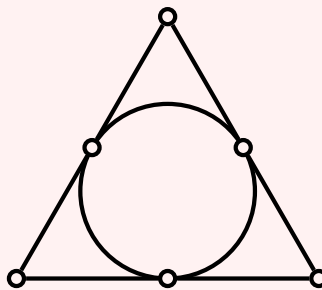


Definition. Four-Line Geometry

Axiom 1: There exists exactly four distinct lines.

Axiom 2: For each two distinct lines there exists a unique point contained by both of them.

Axiom 3: Each point is on exactly two lines.



Prove the following theorems of four-line geometry.

The theorems and proofs of Section 1.3 might be relevant.

- (i) Prove that if two distinct points intersect, then the intersection is exactly one line.
- (ii) Prove there are exactly six points.
- (iii) Prove each line has exactly three points incident to it.
- (iv) Prove each distinct point has exactly one point parallel to it. In fact, if P is a point and ℓ is a line not on P , then there exists exactly one parallel point to P which contains ℓ .

ⁱIn double point elliptic geometry, there are two such points. We say that two points are **antipodal** if they are poles for the same line.

“What we call the beginning is often the end. And to make an end is to make a beginning. The end is where we start from.” – T. S. Eliot

8.4 Elliptic Polygons

The following is an elliptic-version of the Angle-Opposite-Side property, a theorem lost to elliptic geometry because of the lack of Exterior Angle theorem.

Theorem 8.4.1. *In any right, elliptic triangle, each of the other two angles has a measure less than, equal to, or greater than a right angle dependent on whether the side opposite it has a measure that is less than, equal to, or greater than the polar distance.*

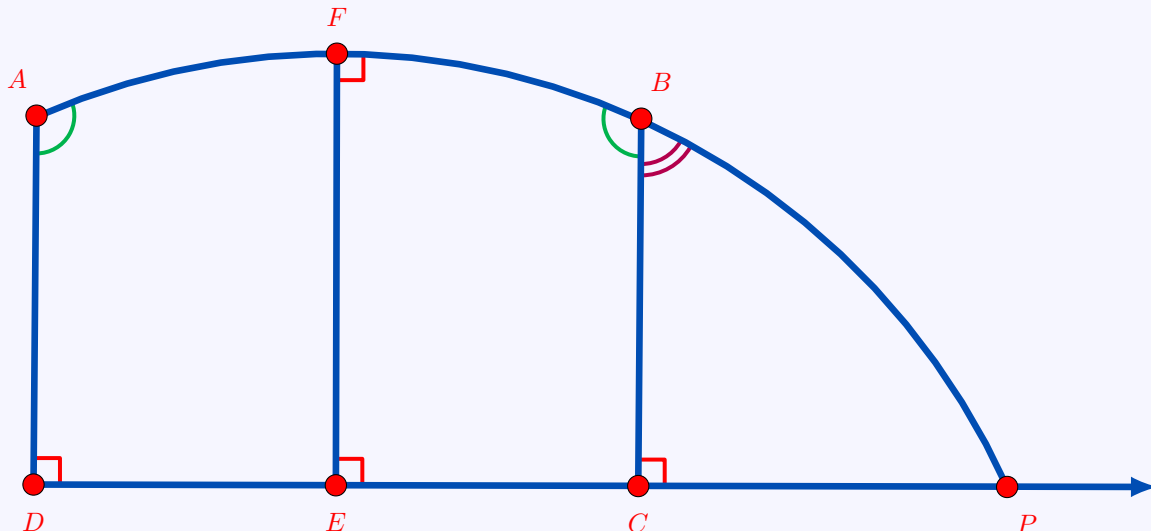
Proof. Let $\triangle ABC$ be a right triangle where $\angle C$ is right. Let P be the pole of the line $\overleftrightarrow{AC}$. Thus, $\overleftrightarrow{AP} \perp \overleftrightarrow{AC}$. Thus, if $B = P$, then $\angle CAB$ is right. If $C - B - P$ (relative to some appropriately chosen ideal line not intersecting $\triangle ABC$ nor $\overleftrightarrow{CP}$), then $\angle CAB < \angle CAP$, which implies that $\angle A$ is acute. Conversely, if $C - P - B$, then $\angle CAB > \angle CAP$, which implies that $\angle A$ is obtuse. $\square$

Let us consider quadrilaterals in elliptic geometry. The proofs of congruence of diagonals and summit angles and the existence of altitudes depend only upon the congruence axioms. Therefore, they are still valid. Unfortunately, the proofs regarding the measure of the summit angles and the comparisons of the measures of summit/base and leg/altitudes require betweenness axioms. These proofs are no longer valid. In fact, we will prove different results about these quadrilaterals.

Theorem 8.4.2 (The Obtuse Angle Hypothesis). *In elliptic geometry, the summit angles of a Saccheri quadrilateral are obtuse. ([Progressive Geometry Proofs Slides](#))*

Proof. Let $\square DCBA$ be a Saccheri quadrilateral with altitude $\overline{EF}$ with $F \in \overleftrightarrow{AB}$. By the elliptic parallel postulate, the lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{DC}$ must intersect at some point, say P . Relative to the ideal line $\overleftrightarrow{AD}$, we may say that $F - B - P$ and $E - C - P$.

Since $\triangle EFP$ is a double-right triangle, Theorem 8.4.1 shows that $\overline{EP}$ and $\overline{FP}$ are polar lengths. Also, $\triangle BPC$ is a right, with $m\angle C = 90^\circ$. By Theorem 8.4.1, $\angle CBP$ is acute since $\overline{CP} < \overline{EP}$ (relative to $\overleftrightarrow{AD}$). Then its supplement, $\angle CBF$ is obtuse. $\square$



The next theorem, an alternate version of the obtuse angle hypothesis, is proven similarly and left to the reader.

Theorem 8.4.3 (The Obtuse Angle Hypothesis). *In elliptic geometry, the fourth angle of a Lambert quadrilateral are obtuse.*

Corollary 8.4.4. *Rectangles do not exist in elliptic geometry.*

Theorem 8.4.5. *In elliptic geometry, the angle sum of a triangle is greater than 180° .*

Proof. We will prove the special case that the triangle is right. It is left to the reader to prove the general case using altitudes and partitions of triangles.

Let $\triangle ABC$ be a right triangle where $\angle C$ is right. By angle translation, there exists a point X on the “same” side of $\overleftrightarrow{AB}$ as C such that $\angle XAB \cong \angle ABC$. Let Q be the midpoint of $\overline{AB}$ and let R be the foot of the perpendicular dropped from Q onto $\overleftrightarrow{BC}$. Let $P = \overleftrightarrow{QR} \cap \overleftrightarrow{AX}$. Then $\triangle BQR \cong \triangle AQP$ by ASA. Thus, $\angle QPA$ is right. This implies that $\square CRPA$ is a Lambert quadrilateral. Therefore, $m\angle CAB + m\angle ABC = m\angle CAB + m\angle BAP = \angle CAP > 90^\circ$. As the third angle is right, the angle sum exceeds 180° . $\square$

Corollary 8.4.6. *The angle sum of any elliptic quadrilateral is greater than 360° .*

The angle amount that a triangle’s angle sum is greater than 180° is called the **excess** of the triangle. For triangle $\triangle ABC$, we will denote the defect as $e(\triangle ABC)$. Modifying the proofs of the corresponding hyperbolic results, we can prove the following two theorems by switching from defect to excess.

Theorem 8.4.7. *The excess function is an area function for elliptic geometry. In fact, if A is an area function for elliptic geometry, then $A(\triangle ABC) = k * e(\triangle ABC)$ for some constant k .*

Theorem 8.4.8 (Angle-Angle-Angle). *Two elliptic triangles are congruent if all corresponding angles are congruent.*

On the other hand, it should be mentioned that AAS is not a valid triangle congruence in elliptic geometry. For example, consider a double-right triangle. By Theorem 8.4.1, we know that the two sides opposite the two right angles must be polar lengths. On the other hand, the third side/angle could be any measure. Thus, AAS does not hold in elliptic geometry even though it holds in neutral geometry. This essentially shows that AAS cannot be proven without something equivalent to the Alternate Interior Angle theorem (we used the Exterior Angle theorem in our proof).

The above counterexample also shows that SSA is still not valid in elliptic nor neutral geometry. Sadly though, HL (a special case of SSA) does not hold in elliptic geometry even though it holds in neutral.

In hyperbolic geometry, triangle’s angle sums could range anywhere on the interval $(0^\circ, 180^\circ)$, where 0° could only be obtained by ideal triangles, triangles whose three vertices were points at infinity. In elliptic geometry, triangle’s angles sum can range anywhere on the interval $(180^\circ, 360^\circ)$. Can the 360° limit be

obtained by some generalized triangle? Since ideal points were already included to form elliptic geometry, ideal triangles offer nothing new.

Instead, imagine the limit of an elliptic triangle as the angle measure one angle approaches 180° . This limit is called a **lune** (or **biangle**). More precisely, a lune $\mathfrak{L} ABC = H(\overleftrightarrow{AB}, C) \cap H(\overleftrightarrow{BC}, A)$ relative to the line polar to B . These can be easily visualized in double-point elliptic geometry as the region bounded by two arcs connecting antipodes on the sphere. This is where it gets its name, as the region resembles the phases of the moon. Essentially a lune is a triangle where one of the angles is flat. Thus, the excess/area of a lune is the sum of its angles, which are necessarily congruent (they are vertical). Thus, the upper limit of triangle angle sum is 360° , which is realized by a right lune. Of course, lunes themselves have an angle sum between $(180^\circ, 540^\circ)$, where the upper limit is approached as the lunar angle approaches 180° .

Exercises

Unless explicitly stated otherwise, all geometries in these exercises are assumed to be elliptic geometry.

1. Prove the fourth angle of a Lambert quadrilateral are obtuse.
2. Finish the proof of Theorem 8.4.5 by proving if all right triangles have angle sums greater than 180° , then all triangles have angle sum greater than 180° .
3. Prove that in a Saccheri quadrilateral the length of the summit is less than the length of the base.
4. Prove that in a Saccheri quadrilateral the length of each leg is less than the length of the altitude.

ⁱFigures in this section were created by Ethan Olcott. The progressive slides in this section were created by Hayley John.